

Pathophysiology of Acute Kidney Injury

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CHAPTER OUTLINE

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PATHOPHYSIOLOGY OF CLINICAL ACUTE KIDNEY INJURY

The three major pathophysiologic categories—namely, prerenal, intrinsic, and postrenal (obstructive)—provide a framework for understanding the mechanisms of acute kidney injury (AKI).

PRERENAL ACUTE KIDNEY INJURY

Prerenal azotemia is the most common cause of AKI, accounting for approximately 40% to 55% of all cases.^{1–3} It results from kidney hypoperfusion due to reductions in actual or effective arterial blood volume (EABV—the volume of blood effectively perfusing the body organs). Common conditions causing true hypovolemia include hemorrhage (traumatic, gastrointestinal, surgical), gastrointestinal (GI) losses (vomiting, diarrhea, nasogastric suction), renal losses (overdiuresis, diabetes insipidus), and third spacing (pancreatitis, hypoalbuminemia). In addition, cardiogenic shock, septic shock, cirrhosis, hypoalbuminemia, and anaphylaxis all are pathophysiologic conditions that decrease EABV, independent of total body volume status, resulting in reduced renal blood flow. Prerenal azotemia reverses rapidly if renal perfusion is restored because, by definition, the integrity of the renal parenchyma has remained intact. However, severe and prolonged hypoperfusion may result in tissue ischemia, leading to acute tubular necrosis (ATN). Therefore, prerenal azotemia and ischemic ATN are part of a spectrum of manifestations of renal hypoperfusion.

Prerenal azotemia has also been divided into volume-responsive and volume-nonresponsive forms. The former is easy to comprehend, but the latter is less straightforward. In volume-nonresponsive forms, additional intravenous volume is of no help in restoring kidney perfusion and function. Disease processes such as congestive heart failure, liver failure, and sepsis may not respond to intravenous fluids because markedly reduced cardiac output or total vascular resistance, respectively, prevent improved kidney function.

True or effective hypovolemia causes a decrease in mean arterial pressure that activates baroreceptors and initiates a cascade of neural and humoral responses, leading to activation of the sympathetic nervous system and increased production of catecholamines, especially norepinephrine. There is increased release of antidiuretic hormone, mediated primarily by hypovolemia, resulting in vasoconstriction, water retention, and urea back diffusion into the papillary interstitium. In response to volume depletion or states of decreased EABV, there is increased intrarenal angiotensin II (Ang II) activity via activation of the renin-angiotensin-aldosterone system (RAAS). Ang II is a very potent vasoconstrictor that preferentially increases efferent arteriolar resistance, preserving glomerular filtration rate (GFR) in the setting of decreased renal perfusion through the maintenance of glomerular hydrostatic pressure. In addition, Ang II increases proximal tubular sodium absorption through a combination of alterations in hydrostatic forces in the peritubular capillaries and through direct activation of sodium-hydrogen exchangers. During severe volume depletion, Ang II activity is even greater, leading to afferent arteriolar constriction, which reduces renal plasma flow, GFR, and the filtration fraction, and markedly augments proximal tubular sodium reabsorption in an effort to restore plasma volume.⁴ Ang II has also been shown to have direct effects on transport in the proximal tubule through receptors located in the proximal tubule. It has also been postulated that the proximal tubule can produce Ang II locally. Hence, under conditions of volume depletion, Ang II stimulates a larger fraction of tubular transport, whereas volume expansion will blunt this response.^{5–9}

Renal sympathetic nerve activity is significantly increased in prerenal azotemia. Studies have shown that in the setting of hypovolemia, adrenergic activity independently constricts the afferent arteriole and changes the efferent arteriolar resistance through Ang II. α_1 -Adrenergic activity primarily influences kidney vascular resistance, whereas renal nerve activity is linked to renin release through β -adrenergic receptors on renin-containing cells. In contrast, α_2 -adrenergic agonists primarily decrease the glomerular ultrafiltration

coefficient via Ang II. Although vasodilation might be expected as a result of acute removal of adrenergic activity, a transient increase in Ang II is actually seen, maintaining GFR and renal blood flow. Even after subacute renal denervation, renal vascular sensitivity increases to Ang II as a result of major upregulation of Ang II receptors. Hence complex effects on renin-angiotensin activity occur within the kidney secondary to increased renal adrenergic activity during prerenal azotemia.¹⁰

All these systems work together and stimulate vasoconstriction in musculocutaneous and splanchnic circulations, inhibit salt loss through sweat, and stimulate thirst, thereby causing retention of salt and water to maintain blood pressure and preserve cardiac output and cerebral perfusion. Concomitantly, there are various compensatory mechanisms to preserve glomerular perfusion.¹¹ Autoregulation is achieved by stretch receptors in afferent arterioles that cause vasodilation in response to reduced perfusion pressure. Under physiologic conditions, autoregulation works only to a mean systemic arterial blood pressure of 75 to 80 mm Hg. Below this level, the glomerular ultrafiltration pressure and GFR decline abruptly. Renal production of prostaglandins, kallikrein, and kinins, as well as nitric oxide (NO), is increased, contributing to the vasodilation.^{12,13} Nonsteroidal antiinflammatory drugs (NSAIDs), by inhibiting prostaglandin production, worsen kidney perfusion in patients with hypoperfusion. Selective efferent arteriolar constriction, a result of Ang II, helps preserve the intraglomerular pressure and hence the GFR. Angiotensin-converting enzyme (ACE) inhibitors inhibit synthesis of Ang II and therefore disturb this delicate balance in patients with severe reductions in EABV, such as severe congestive heart failure or bilateral renal artery stenosis and, in these settings, can worsen prerenal azotemia. On the other hand, very high levels of Ang II, as seen in circulatory shock, cause constriction of both afferent and efferent arterioles, negating its protective effect.

Although these compensatory mechanisms minimize the progression toward AKI, they too are overcome in states of severe hypoperfusion. Renovascular disease, hypertensive nephrosclerosis, diabetic nephropathy, and older age predispose patients to prerenal azotemia¹⁴ at lesser degrees of hypotension.¹⁴ Prerenal azotemia also predisposes patients to radiocontrast media-induced AKI and events such as anesthesia and surgery, which are known to result in further decreases in renal blood flow. Therefore it is imperative to diagnose prerenal azotemia promptly and initiate effective treatment because it is a potentially reversible condition that can lead to ischemic ATN and/or nephrotoxic AKI if therapy is delayed, or the severity of the condition increases. In patients with advanced liver disease and portal hypertension, the hepatorenal syndrome (HRS) represents an extreme form of prerenal disease, characterized by peripheral and splanchnic vasodilation, with intense intrarenal vasoconstriction unresponsive to volume resuscitation.^{15–17} AKI can also result from abdominal compartment syndrome (ACS), characterized by a marked elevation in intraabdominal pressure, resulting in a clinical presentation with features similar to those of prerenal AKI.^{18,19} A recent study in animal models of volume-responsive and intrinsic AKI used laser microdissection to isolate specific domains of the kidney, followed by RNA sequencing of the microdissected kidney tissue. Based on the gene expression profile, the investigators found dif-

ferent signal transduction pathways in the two models. Volume-responsive genes rapidly reverse with volume resuscitation, whereas intrinsic AKI genes did not change. These results suggest that volume-dependent AKI is not an attenuated form of intrinsic AKI.²⁰

INTRINSIC ACUTE KIDNEY INJURY

DISEASES OF LARGE VESSELS AND MICROVASCULATURE

Total occlusion of the renal artery or vein is an uncommon event but can be seen in certain scenarios such as trauma, instrumentation, thromboemboli, thrombosis, and dissection of an aortic aneurysm. Stenosis of the renal artery is a slow chronic process, with or without evidence of declining GFR, and rarely presents as an acute event. Renal vein thrombosis has classically and frequently been associated with hypercoagulable states, including nephrotic syndrome, particularly when associated with membranous nephropathy. An atheroembolic source should be considered in patients who present with AKI after instrumentation with angiography, arteriography, or aortic surgery or after blunt trauma or acceleration-deceleration injury.²¹ Cholesterol-laden atheroembolic plaques in the aorta or other larger arteries may become disrupted, and fragments may become trapped in smaller renal arteries, leading to hypoperfusion and an intense inflammatory reaction, akin to a vasculitis. Other organs may also be affected, leading to gastrointestinal ischemia, peripheral gangrene, livedo reticularis, and acute pancreatitis. Patients frequently develop fevers and exhibit eosinophilia, an elevated erythrocyte sedimentation rate, and hypocomplementemia, which sometimes help in differentiating this condition from other simultaneous insults (e.g., transient hypotension and/or radiocontrast media administration).

Renal artery thrombosis is usually a posttraumatic or postsurgical complication, especially in the transplantation setting, but can also occur in other hypercoagulable states, such as antiphospholipid antibody syndrome.^{22–24} Diseases affecting the small vessels, generally termed *vasculitides*, include polyarteritis nodosa, necrotizing granulomatous vasculitis, hemolytic-uremic syndrome, thrombotic thrombocytopenic purpura, and malignant hypertension; they tend to occlude vessels by fibrin deposition, along with platelets. Endothelial cell damage leads to an inflammatory response in the renal microvasculature (and in other organs), leading to reduced microvascular blood flow and tissue ischemia, sometimes giving rise to superimposed ATN. One should keep in mind the intricate relationship among these inflammatory vasculitides and subsequent ischemic injury because even though the origin of these disease processes is located at a site distant from the tubules, the final result is often ATN if not treated early. Hence, virtually any disease that compromises blood flow within the renal microvasculature can induce AKI.

DISEASES OF THE TUBULOINTERSTITIUM

Ischemic and septic ATN are the most common causes of intrinsic AKI. These are discussed extensively in later sections of the chapter on ATN. Other disorders of the tubulointerstitium causing AKI, such as acute allergic interstitial nephritis, drug-induced tubular toxicity, and endogenous toxins, are presented in the following sections.

Interstitial Disease

Acute interstitial nephritis (AIN) results from an idiosyncratic allergic response to different pharmacologic agents, most commonly to antibiotics (e.g., methicillin and other penicillins, cephalosporins, sulfonamides, quinolones), NSAIDs (e.g., ibuprofen, naproxen),²⁵ chemotherapeutic agents,²⁶ and proton pump inhibitors.²⁷ Although evidence supports a role of vancomycin as a tubule toxin, most biopsy reports have described AIN.²⁸ The expanding list of newer targeted agents has led to more cases of acute tubulointerstitial injury. Serine-threonine protein kinase inhibitors such as vemurafenib and dabrafenib target a protein called B-Raf and have been associated with acute and chronic tubulointerstitial damage.²⁹ Other agents that cause acute tubulointerstitial injury include checkpoint inhibitors such as ipilimumab, nivolumab, and pembrolizumab. Programmed cell death protein 1 (PD-1), a checkpoint protein, is expressed on T cells, and PD-L1 is expressed on normal as well as cancer cells. Binding of PD-1 ligand to its receptor (PD-L1) blocks immune elimination by T cells. Cancer cells evade immune surveillance in part by this mechanism. Similarly, CTLA-4 is another checkpoint protein expressed on T cells that leads to self-tolerance and various forms of autoimmune injury, including acute interstitial injury.³⁰

Other conditions such as leukemia, lymphoma, sarcoidosis, bacterial infections (e.g., *Escherichia coli*), and viral infections (e.g., cytomegalovirus) can also cause AIN, leading to AKI. Systemic allergic signs such as fever, rash, and eosinophilia are often present in antibiotic-associated AIN but are not usually present in NSAID-related AIN, in which lymphocytes tend to predominate.³¹ The presence of inflammatory infiltrates within the interstitium is the key hallmark of AIN. These inflammatory infiltrates are often patchy and present most commonly in the deep cortex and outer medulla. Interstitial edema is typically seen with the infiltrates, and sometimes patchy tubular necrosis may be present in close proximity to areas with extensive inflammatory infiltrates.²⁵ The composition of cells in the interstitial infiltrate can suggest the cause of AIN. The presence of frequent eosinophils suggests a diagnosis of drug-induced AIN, whereas a high number of neutrophils favors bacterial infection, and a high number of plasma cells is dominant in immunoglobulin G4 (IgG4)-related tubulointerstitial nephritis and in renal allografts infected with polyomavirus.³² Most cases of AIN are probably induced by extrarenal antigens being produced by drugs or infectious agents that may be able to induce AIN by the following: (1) binding to kidney structures; (2) modifying immunogenetics of native renal proteins; (3) mimicking renal antigens; or (4) precipitating as immune complexes and hence serving as the site of antibody- or cellular-mediated injury.³³ This reaction is triggered by many events, including activation of complement and release of inflammatory cytokines by T cells and phagocytes. In other cases, loss of tolerance from checkpoint inhibitors leads to immune-mediated inflammation and injury.²⁶

Tubular Disease—Exogenous Nephrotoxins

Nephrotoxic ATN is the second most common cause of intrinsic AKI. We shall briefly review the common drug nephrotoxicities in the context of AKI. The kidneys are vulnerable to toxicity due to the high blood flow, and they are the major elimination and metabolizing routes of many

of these nephrotoxins. Furthermore, because of the medullary tonicity, the concentration of drugs within the tubular lumen increases along the nephron, exposing the tubules to toxic levels for a more prolonged exposure time. Several other well-known therapeutic agents, such as amphotericin B, vancomycin, acyclovir, indinavir, cidofovir, foscarnet, pentamidine, and ifosfamide, can all directly cause acute tubular injury and associated AKI.

Radiocontrast Media-Induced Nephropathy. Iodinated radiocontrast medium-induced nephropathy (CIN) is a common complication of radiologic or angiographic procedures. The incidence varies from 3% to 7% in patients without any risk factors but can be as high as 50% in patients with moderate to advanced chronic kidney disease (CKD). Other risk factors include diabetes mellitus, intravascular volume depletion, use of high-osmolality contrast media, advanced age, proteinuria, and anemia.^{34,35}

Unlike many other forms of intrinsic tubular injury, radiocontrast medium-induced AKI is usually associated with urinary sodium retention and a fractional excretion of sodium (FE_{Na}) of less than 1%. AKI resulting from iodinated contrast media is typically nonoliguric and rarely requires dialysis. However, requirements for renal support, prolonged hospitalization, and increased mortality are associated with (although not necessarily caused by) this condition.

The pathophysiology of CIN likely consists of combined hypoxic and toxic renal tubular damage associated with renal endothelial dysfunction and altered microcirculation.^{36,37} The administration of radiocontrast media causes vasoconstriction and markedly affects renal parenchymal oxygenation, especially in the outer medulla, as documented in various studies where the cortical PO_2 declined from 40 to 25 mm Hg and the medullary PO_2 fell from 26 to 30 mm Hg to 9 to 15 mm Hg.^{37–39} Radiocontrast media injection leads to an abrupt but transient increase in renal plasma flow, GFR, and urinary output.⁴⁰ This effect is due to the hyperosmolar radiocontrast medium-enhancing solute delivery to the distal nephron and leads to increased oxygen consumption by enhanced tubular sodium reabsorption. Using video microscopy, it has also been documented that radiocontrast media markedly reduce inner medullary papillary blood flow, even to the extent of near-cessation of red blood cell (RBC) movement in papillary vessels, associated with RBC aggregation within the papillary vasa recta.⁴¹ In isolated vasa recta from rats and humans, contrast medium applied to the lumen has led to constriction and enhanced vasa recta responses to Ang II.^{42,43} However, it should be noted that there may be different patterns of response possibly related to the type, volume, and route of radiocontrast medium administration. Numerous neurohumoral mediators may contribute to the changes in renal microcirculation caused by radiocontrast medium injection. Intrarenal NO synthase activity, NO concentration, plasma endothelin, adenosine, prostaglandins, and vasopressin are all thought to play a role in altering the cortical and medullary microcirculation after radiocontrast medium injection. Mechanical factors may also play a role because radiocontrast media increase blood viscosity and may affect the flow in the complex, low-pressure medullary microcirculation.³⁹ An increased plasma viscosity after radiocontrast medium administration can interfere with blood flow, particularly under the hypertonic conditions of the

(inner) renal medulla, where the plasma viscosity is already increased as a result of hemoconcentration. Indeed, there are several animal studies that have shown a correlation between experimental CIN and viscosity of the radiopaque compound.^{44,45}

Evidence has also suggested direct tubular toxicity from radiocontrast media. Early studies on isolated renal tubules in vitro have shown direct toxic effects of radiocontrast media on proximal tubular cells (PTCs).⁴⁶ Radiocontrast media (e.g., diatrizoate, iopamidol) induced a decline in tubule K^+ , adenosine triphosphate (ATP), and total adenine nucleotide contents. At the same time, there was a decrease in the respiratory rate of the tubules and an increase in Ca^{2+} content. These changes were more pronounced with the very high-osmolality ionic compound diatrizoate than with the lower osmolality nonionic iopamidol. Importantly, the cytotoxic effects were aggravated by hypoxia, indicating interactions between direct cellular mechanisms and vasoconstriction-mediated hypoxia.⁴⁷ Andersen and coworkers have demonstrated the concentration-dependent radiocontrast media-mediated release of tubular marker enzymes, ultrastructural changes, and cell death in both Madin-Darby canine kidney (MDCK) and porcine kidney proximal tubule epithelial (LLC-PK₁) cells.⁴⁸ Radiocontrast medium-induced critical medullary hypoxia may lead to the formation of reactive oxygen species (ROS), with subsequent membrane and DNA damage. A vicious cycle of hypoxia, free radical formation, and further hypoxic injury may be activated after radiocontrast medium exposure. Clinically, CIN presents with an acute decline in the GFR within 24 to 48 hours of administration, with a peak serum creatinine concentration usually occurring in 3 to 5 days and return to baseline within 1 week, although patients with moderate to advanced CKD may take somewhat longer to return to the baseline serum creatinine concentration. Existing CKD, diabetic nephropathy, advanced age, congestive heart failure, volume depletion, and coincident use of NSAIDs also increase the risk for CIN.

Unlike the first-generation contrast media, high-osmolar contrast media (HOCM; osmolality of 1500–1800 mOsm/kg), low-osmolar contrast media (LOCM; osmolality of 600–850 mOsm/kg), and isosmolar contrast media (IOCM; osmolality of 270–320 mOsm/kg) are associated with a lower incidence of contrast-induced AKI.^{49–51} However, whether the incidence of contrast-induced AKI is lower in IOCM than LOCM is still debated.^{49–52} One factor that may mitigate against the theoretic value of IOCM versus LOCM is that IOCM is a dimer and has a higher viscosity than LOCM. Viscosities for HOCM, LOCM, and IOCM are 0.00275, 0.00525, and 0.0114 pascal-second (Pa·s) at 37° C, respectively.²⁸ According to Poiseuille's law, an increase in viscosity negatively affects flow. Following intravenous injection, contrast medium becomes diluted in the bloodstream, the viscosity and osmolality are reduced, and therefore nonkidney organs are exposed to low concentrations of contrast medium. In the kidney, however, because of the increase in medullary osmolality, the concentration of contrast medium increases in the peritubular capillary and, following filtration, the concentration of contrast rises in the tubule lumen. The consequence is that (1) the distal tubules are exposed to increasing concentration and viscosity of contrast medium, and (2) the tubule flow rate decreases, leading to prolonged exposure to contrast, potentially enhancing direct tubule nephrotoxicity.^{28,53}

Furthermore, when infused in animals, medullary PO_2 is lower in IOCM when compared with LOCM.³⁴

Aminoglycoside Nephrotoxicity. The nephrotoxicity of aminoglycosides has best been characterized for gentamicin, a polar drug excreted by glomerular filtration. It is thought that cationic amino groups (NH_3^+) on the drug bind to anionic phospholipid residues on the brush border of proximal tubular cells, and the drug is then internalized by endocytosis. Although the precise cellular mechanisms responsible for renal accumulation of aminoglycosides have not been fully elucidated, binding to the endocytic complex formed by megalin and cubilin at the apical surface of proximal tubular cells appears important.^{54,55} The complete elimination of aminoglycoside uptake in mice deficient in megalin has suggested that this is the major pathway responsible for renal aminoglycoside accumulation.⁵⁵ Chloride transporters, including cystic fibrosis transmembrane conductance regulator (CFTR) and the $ClC-5$ have been implicated because mice lacking functional CFTR (*Cftr*^{ΔF/ΔF}) or deficient in the Cl^-/H^+ exchanger (*Cln5*^{Y/-}) have decreased kidney accumulation of gentamicin.⁵⁶ A three-dimensional model has described the complex between megalin and gentamicin. Gentamicin binds to megalin with low affinity and exploits the common ligand-binding motif using the indole side chain of amino acids Trp-1126 and the negatively charged residues of Asp-1129, Asp-1131, and Asp-1133.⁵⁷ Once endocytosed, aminoglycosides inhibit endosomal fusion. They are also directly trafficked to the Golgi apparatus and, through retrograde movement, to the endoplasmic reticulum (ER). From the ER, gentamicin moves into the cytosol in a size- and charge-dependent manner.⁵⁸ Once in the cytosol, either from the ER⁵⁸ or via lysosomal rupture, aminoglycosides distribute to various intracellular organelles and mediate organelle-specific toxicity, such as mitochondrial dysfunction.^{58,59} Gentamicin acts on the mitochondria, activating the intrinsic pathway of apoptosis, disrupting ATP production,^{60,61} and producing hydroxyl radicals and superoxide anions.^{62,63} Also, delivery to the ER via retrograde transport from the Golgi apparatus allows for the binding of aminoglycosides to the 16S rRNA subunit,⁶⁴ resulting in a reduction of protein synthesis^{65,66} and altering postranslational protein folding.⁶⁷ The number of cationic groups on the molecules determines the facility with which these drugs are transported across the cell membrane and is an important determinant of toxicity.^{68,69} Neomycin is associated with the most nephrotoxicity, gentamicin, tobramycin, and amikacin are intermediate, and streptomycin is the least nephrotoxic. Furthermore, blocking megalin with cilastatin is known to reduce drug-induced nephrotoxicity.⁷⁰ Risk factors for aminoglycoside nephrotoxicity include the use of high or repeated doses or prolonged therapy, CKD, volume depletion, diabetes, advanced age, and the coexistence of renal ischemia or use of other nephrotoxins.^{71–73}

Vancomycin Nephrotoxicity. Vancomycin was discovered, developed, and approved by the US Food and Drug Administration (FDA) in 1958^{74,75} and was found to be active against most gram-positive organisms, including penicillin-resistant staphylococci.⁷⁵ Initial formulations of the drug were dubbed “Mississippi mud” due to the brown color presumably due to impurities, which were thought to be responsible for

nephrotoxicity and ototoxicity.^{75–77} Reformulation resulted in a drug that had improved purity and was called *vancomycin* (from the word vanquish).

The incidence of vancomycin nephrotoxicity is variable, ranging from as low as 0% to over 40%, and those with vancomycin-associated nephrotoxicity were more likely to have higher trough levels and prolonged duration of treatment.⁷⁸ Durations of 7 to more than 15 days of treatment are associated with nephrotoxicity.^{79–81} With modern formulations of vancomycin, the variability is due to concomitant drug use, severity of illness, and variability in the definition of AKI. Daily total dose of vancomycin of more than 4 g has been shown to be a risk factor for nephrotoxicity.⁸² Vancomycin-associated nephrotoxicity is thought to be more common when combined with antipseudomonal beta-lactams. In a retrospective matched cohort study of patients receiving vancomycin and cefepime (588 patients) or vancomycin and piperacillin-tazobactam (3605 patients), the unadjusted incidence of AKI was 12.6% versus 21.4%, respectively ($P < .0001$). Vancomycin and piperacillin-tazobactam was associated with a more than twofold increase in the risk of AKI, after matching for severity of illness.⁸⁰

Luque and associates⁸³ have described a patient who received vancomycin without coadministration of an aminoglycoside. The biopsy showed obstructive tubular casts composed of noncrystal nanospheric vancomycin aggregates associated with uromodulin. In eight additional patients with AKI associated with high vancomycin levels, vancomycin-associated casts were found. These findings, which were reproduced in mice given vancomycin, demonstrate a link between vancomycin and AKI.

Cisplatin Nephrotoxicity. Treatment with cisplatin (cisplatinum), a platinum-based chemotherapeutic agent, is commonly associated with nephrotoxicity. The pathophysiologic mechanism of cisplatin-induced tubular damage is complex and involves a number of interconnected factors, such as accumulation of cisplatin mediated by membrane transport, conversion into nephrotoxins, DNA damage, mitochondrial dysfunction, oxidative stress, inflammatory response, activation of signal transducers and intracellular messengers, and activation of apoptotic pathways. Movement of cisplatin through the renal tubular cells occurs in a basolateral to apical direction. Two primary transporters are involved in transporting cisplatin into the tubular cells, the copper transport protein 1 (Ctr1), expressed on proximal and distal tubules, and the organic cation transporter 2 (OCT2), expressed on the basolateral side of the proximal convoluted tubule.⁸⁴ Multidrug and toxin extrusion 1 (MATE1/SLC47A1; MATE1), expressed on the apical membrane of the proximal tubule, is responsible for the efflux of cisplatin.^{84,85} When cisplatin was administered to *Mate1*^{-/-} mice, blood urea nitrogen (BUN) and creatinine levels were higher than in *Mate1*^{+/+} mice. Furthermore cisplatin levels were higher in plasma and in kidneys of compared with *Mate1*^{+/+} mice, suggesting that MATE1 mediates the efflux of cisplatin and could contribute to cisplatin nephrotoxicity.

The S3 segment of the proximal tubule in the corticomedullary region is the most common site of cisplatin nephrotoxicity in rats. More distal sites may be affected in humans, but glomeruli remain unaffected. A recent study in mice has examined the effects of low but frequent doses of cisplatin

given once a week for 4 weeks. Mice who received multiple doses of cisplatin had increased levels of fibrotic markers in kidney tissue, including fibronectin, transforming growth factor- β , and α -smooth muscle actin, as well as interstitial fibrosis.⁸⁶ These studies in mice support observations in humans showing that adult patients with cancer who received multiple doses of experience small but permanent declines in the estimated GFR (eGFR).^{87,88}

Acute Phosphate Nephropathy. AKI has been described as a complication following the administration of oral sodium phosphate solution as a bowel cathartic in preparation for colonoscopy and bowel surgery.^{89–91} Although the mechanism linking oral sodium phosphate administration with AKI remains incompletely understood, the pathogenesis likely relates to a transient and significant rise in the serum phosphate concentration that occurs simultaneously with intravascular volume depletion.

When the urine is oversaturated and buffering factors such as pH, citrate, and pyrophosphate are overwhelmed, renal phosphorus excretion becomes compromised. This may lead to the intratubular precipitation of calcium phosphate salts when the solubility coefficient is exceeded and to obstruction of the tubular lumen, leading to direct tubular damage. ROS generated by the binding of calcium phosphate crystals further promotes tubular damage. Risk factors for acute phosphate nephropathy include preexisting volume depletion, the use of ACE inhibitors and angiotensin receptor blockers (ARBs), NSAIDs, CKD, older age, female gender, and higher dosage of oral sodium phosphate.^{90,91} Patients who develop acute phosphate nephropathy typically present with elevated serum creatinine concentrations days to months following the administration of oral sodium phosphate solution and can experience progression to CKD and end-stage kidney disease. As in other conditions associated with hypercalcemia, hyperphosphatemia, or hyperphosphaturia, calcium phosphate precipitation in renal tubules is seen on renal biopsy as bluish-purple crystals that are nonpolarizable.⁹²

Tubular Disease—Endogenous Nephrotoxins

Myoglobin and Hemoglobin. Myoglobin and hemoglobin are the endogenous toxins most commonly associated with ATN. Myoglobin, a 17.8-kDa heme protein released during muscle injury, is freely filtered and causes red-brown urine, with a dipstick result positive for heme in the absence of RBCs in the urine. Intravascular hemolysis results in circulating free hemoglobin, which, when excessive, is filtered, resulting in hemoglobinuria, hemoglobin cast formation, and heme uptake by proximal tubular cells. The uptake in the proximal tubule may be mediated via endocytosis by megalin and cubilin. Megalin knockout mice have reduced accumulation of injected myoglobin in tubule cells and reduced nephrotoxicity.⁹³ The heme center of myoglobin may directly induce lipid peroxidation and, in addition, the liberation of free ferrous iron, depending on the redox potential, can promote hydroxyl radical formation by the Haber-Weiss (Fenton) reaction, resulting in the oxidation of lipids, proteins, and nucleic acids.^{94,95} Iron is an intermediate accelerator in the generation of free radicals. Studies have suggested that there is increased formation of H₂O₂ in rat kidney models of myohemoglobinuria.⁹⁶ The subsequent hydroxyl

(OH⁻) radical plays a vital role in oxidative stress-induced AKI through mechanisms discussed in detail later in this chapter. In response, heme protein induces heme-degradative enzyme, heme oxygenase, and increased synthesis of ferritin. Ferritin, a major factor in sequestering free iron,⁹⁷ is made up of two types of 24 subunits, heavy chain and light chain. It is the ferritin heavy chain (FtH) that has ferroxidase activity necessary for iron incorporation and to limit toxicity. Zarjou and coworkers have demonstrated that proximal tubule-specific, *FtH*-knockout mice (*FtH*^{PT-/-} mice) have significant mortality in myoglobin-induced AKI, indicating the protective role of proximal tubule FtH in AKI.⁹⁸ Various iron chelators such as deferoxamine and other scavengers of ROS such as glutathione provide protection against myohemoglobinuric AKI.⁹⁹ Similarly, endothelin antagonists also prevent hypofiltration and proteinuria in rats that have undergone glycerol-induced rhabdomyolysis.¹⁰⁰ In addition, NO supplementation may be beneficial by preventing the heme-induced renal vasoconstriction because heme proteins scavenge NO.^{101,102} Finally, precipitation of myoglobin with Tamm-Horsfall protein and shed proximal tubular cells lead to cast formation and tubular obstruction, which is enhanced in acidic urine.¹⁰³ In human studies, volume expansion and perhaps alkalization of urine to limit cast formation are the preventive measures generally used because none of the experimental agents used in animal studies has been convincingly beneficial. This emphasizes the multifactorial nature of these conditions. It is unlikely that a single agent will be beneficial in this setting.¹⁰⁴

Immunoglobulin Light Chains. *Direct tubule toxicity.* Excessive immunoglobulin light chains, produced in diseases such as multiple myeloma, are filtered, absorbed, and then catabolized in proximal tubule cells and can induce proximal tubulopathy.¹⁰⁵ The concentration of light chains leaving the proximal portion of the nephron depends on both the concentration of light chains in the glomerular filtrate and the capacity of the proximal tubule to reabsorb and catabolize them. Certain light chains can be directly toxic to the proximal tubules themselves.¹⁰⁶ In the proximal tubules, free light chains (FLCs) are reabsorbed by binding to the proximal tubule heterodimeric complex consisting of megalin and cubilin.^{107–109} Accumulation of light chains in the endosomes and lysosomes of the proximal tubule leads to cellular desquamation and fragmentation, vacuolization, and focal loss of brush border.¹¹⁰ Mechanisms for tubule toxicity may include blocking of transport of glucose, amino acids, or phosphate.¹⁰⁵ FLCs generate hydrogen peroxide,¹¹¹ which leads to the production of chemokines and cytokines,^{111–114} with nuclear translocation of nuclear factor-kappa B (NF-κB), suggesting that light chain endocytosis leads to the production of inflammatory cytokines through activation of NF-κB.¹¹⁵ Monoclonal FLC also promotes apoptosis through apoptosis signal-regulating kinase (ASK1), also called mitogen-activated protein kinase kinase kinase 5 (MAP3K5).¹¹⁴ Subsequent inflammation leads to tubulointerstitial fibrosis.¹¹⁶

Cast nephropathy. Once the capacity for proximal tubule uptake is overwhelmed, a light chain load is presented to the distal tubule, where, on reaching a critical concentration, the light chains aggregate and coprecipitate with Tamm-Horsfall protein (THP) and form characteristic light chain casts.¹¹⁷ FLCs bind to specific sites on Tamm-Horsfall glycoproteins through

the CDR3 domain of FLC, leading to their coprecipitation in the lumen of the distal nephron and tubule flow.¹¹⁸ There are critical determinants of the binding site between CDR3 and FLCs that lead to the development of a cyclized competitive peptide. This peptide inhibited binding of FLCs to THP and was effective in inhibiting intraluminal cast formation and AKI.¹¹⁸ Some studies have shown that light chains, in the amount seen in plasma cell dyscrasia patients, are capable of catalyzing the formation of hydrogen peroxide in cultured HK-2 cells. Hydrogen peroxide stimulates the production of monocyte chemoattractant protein-1 (MCP-1), a key chemokine involved in monocyte or macrophage recruitment to proximal tubular cells.¹¹¹

Any process that reduces the GFR, such as volume depletion, hypercalcemia, or NSAIDs, will accelerate and aggravate light chain cast formation. It has been proposed that acutely reducing the presented light chain load by plasmapheresis or dialysis using high-cutoff membranes might be beneficial in limiting cast formation and reducing the extent of AKI in certain select patients, allowing for the initiation of chemotherapy to decrease bone marrow-dependent light chain formation.^{119,120}

Uric Acid. Tumor cell necrosis following chemotherapy can release large amounts of intracellular contents such as uric acid, phosphate, and xanthine into the circulation, potentially leading to AKI. Acute uric acid nephropathy with intratubular crystallization leading to obstruction and interstitial nephritis is not seen as commonly as it was in the past, mainly due to the prophylactic use of allopurinol or rasburicase before chemotherapy to lower the serum uric acid concentration acutely.

POSTRENAL ACUTE KIDNEY INJURY

Postrenal azotemia occurs from obstruction of the ureters, bladder outlet, or urethra. AKI from ureteric obstruction requires that the blockage occur bilaterally at any level of the ureters or unilaterally in a patient with a solitary functioning kidney or CKD. Ureteric obstruction can be intraluminal or external. Bilateral ureteric calculi, blood clots, and sloughed renal papillae can obstruct the lumen, whereas external compression from a tumor or hemorrhage can block the ureters as well. Fibrosis of the ureters intrinsically or from the retroperitoneum can narrow the lumen to the point of complete luminal obstruction. The most common cause for postrenal azotemia is structural or functional obstruction of the bladder neck. Prostatic conditions, therapy with anticholinergic agents, and a neurogenic bladder can all cause postrenal AKI. Relief of the obstruction usually causes prompt return of the GFR if the duration of obstruction has not been excessive. The rate and magnitude of functional recovery are dependent on the extent and duration of the obstruction.¹²¹

AKI resulting from obstruction usually accounts for fewer than 5% of cases, although in certain settings (e.g., transplantation), it can be as high as 6% to 10%. Clinically, patients can present with pain and oliguria, although these are neither specific nor sensitive. Because of the availability of retroperitoneal imaging using ultrasonography or computed tomography (CT), the diagnosis is usually straightforward, although, on occasion, a volume-depleted patient or a patient

with severe reduction in the GFR may not show hydronephrosis on radiologic assessment. Because the GFR is typically not affected early in the course of obstructive AKI, volume repletion can increase the sensitivity of diagnosis by increasing the GFR and urine production into the ureter, leading to dilation of the ureter proximal to the obstruction, enhancing ultrasonographic visualization. Early diagnosis and prompt relief of obstruction remain key goals in preventing long-term parenchymal damage because the shorter the period of obstruction, the better the chances for recovery and favorable long-term outcomes.

PATHOPHYSIOLOGY OF ACUTE KIDNEY INJURY

OVERVIEW OF THE PATHOPHYSIOLOGY OF ACUTE KIDNEY INJURY

AKI is a summation of temporally activated systems that together result in inflammation, activation of cell death pathways, tubular obstruction, backleak, altered glomerular hemodynamics and loss of the GFR. Within the kidney, mechanisms pertaining to microvascular compartments, innate immunity, and ATN result in temporary, partial, or permanent loss of the GFR. Furthermore, it is becoming widely accepted that AKI is a systemic process that affects a number of organs, leading to the high morbidity and mortality seen in patients with AKI. Systemic responses to AKI may influence the extent of injury. Hemodynamic alterations (e.g., decrease in cardiac output, low blood pressure, vasoconstriction) may initiate AKI or exacerbate intrinsic microenvironmental mechanisms of AKI. Systemic immunologic mechanisms of proinflammatory or antiinflammatory conditions may affect AKI, and neural mechanisms may attenuate AKI. Thus, the complexity of the pathogenesis of AKI requires careful understanding of its molecular mechanisms through defining important targets in humans and testing in relevant models of AKI (Fig. 28.1).^{122–127}

EXPERIMENTAL MODELS

The goal of preclinical AKI research is to translate basic scientific knowledge of AKI to clinical practice. Despite extensive research in AKI focusing on standardized definitions of AKI (from *Risk, Injury, Failure, Loss of kidney function*, and *End-stage kidney disease* (RIFLE), *Acute Kidney Injury Network* (AKIN), and *Kidney Disease: Improving Global Outcomes* [KDIGO]), biomarkers of AKI, novel drug targets, and improved clinical trial design, our knowledge remains incomplete, and effective therapies are lacking.^{122,123} To advance the field, it is important that we identify relevant targets through the analysis of human tissue biopsy and necropsy specimens, develop relevant disease models of AKI, include proper pharmacokinetic, pharmacodynamic, and dose response studies and, finally, improve preclinical and human clinical trial design.^{123,125,126,128} The National Institutes of Health (NIH) and National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) have initiated a bold project, the *Kidney Precision Medicine Project* (KPMP), whose objectives are to obtain human kidney biopsies ethically from participants with AKI and CKD with a goal to identify critical cells, pathways, and

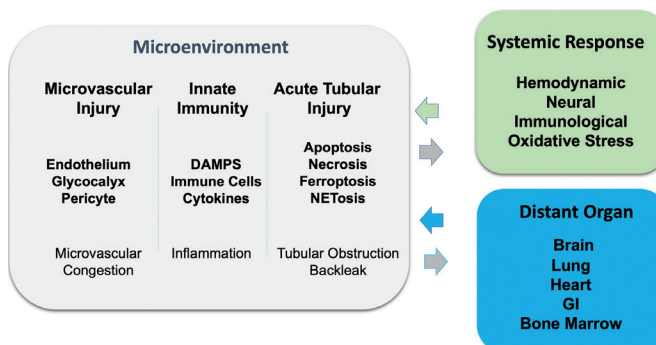


Fig. 28.1 Overview of pathophysiology of acute kidney injury (AKI). AKI is a summation of temporally activated systems that together result in loss of the glomerular filtration rate (GFR). In the kidney, mechanisms pertaining to microvascular compartment, innate immunity, and acute tubular necrosis result in temporary, partial, or permanent loss of the GFR. Systemic responses may influence the extent of injury. Hemodynamic alterations (decrease in cardiac output, low blood pressure, and vasoconstriction) may initiate AKI or exacerbate intrinsic microenvironmental mechanisms of AKI. Systemic immunologic mechanisms of proinflammatory or antiinflammatory conditions may affect AKI, and neural mechanisms may attenuate AKI (see text for details).

targets for therapy. The following section will focus on current animal models.

IN VIVO MODELS OF ACUTE KIDNEY INJURY

Current investigations into the pathophysiology of AKI include a combination of animal and cell culture models of AKI designed to understand the pathophysiology of AKI better and investigate novel therapeutic agents. However, there remains a need to develop in vivo experimental models of AKI that more closely resemble clinical AKI for the development of effective therapies.^{123,126,129,130} Some of the important principles in studying the pathophysiology of AKI in various models include outcome measures at multiple time points and the ability to control physiologic functions known to affect kidney function (e.g., temperature, blood pressure, anesthesia, fluid status). Current models of AKI (e.g., warm ischemia-reperfusion using a pedicle clamp) test fundamental proof of principle concepts that identify potential molecular targets. A simplified model limits confounding variables but can clearly identify potential therapeutic targets. However, additional models of AKI are necessary that should reflect human disease (e.g., aged animals, impaired kidney function, multiorgan failure, preexisting vascular changes, multiple renal insults) that often coexist in human AKI. We will briefly describe the pros and cons of using presently characterized experimental models (Table 28.1).^{131–146}

The warm ischemia-reperfusion renal pedicle clamp model is one of the most widely used experimental models in rats and mice because of its simplicity and reproducibility. However, the inflammatory response differs greatly between mice and rats. It is important to realize that there is considerable variability between mice and rats, mouse strains (C57BL/6 vs. BALB/c), and the same strain from different vendors.¹⁴⁷ Although mice are the primary species used experimentally for examining the immune response to AKI, there are significant differences when compared with the human immune

Table 28.1. Acute Kidney Injury–Induced Distant Organ Injury

Organ	Mechanism	Species	Reference
Lung	CXCL1	Mouse	131
	TNF- α	Mouse	132
	IL-10	Mouse	133
	Neutrophil elastase	Mouse	134
	Lung permeability	Mouse	135
	Pulmonary edema, inflammatory cytokines	Rat	136
	Oxidant stress	Rat	137
Heart	Systemic cytokines and HMGB1	Human (ex vivo)	138
	TNF- α	Rat	139
	Mitochondrial fission protein	Mouse	140
Brain	RAAS	Mouse	141
Liver	Administration of glutathione	Rat	142
	Hepatic oxidant stress	Mouse	143, 144
Intestine	Hepatic oxidant stress	Mouse	145

AKI, Acute kidney injury; CXCL1, chemokine (C-X-C motif) ligand 1; HMGB1, high-mobility group box protein B1; IL, interleukin; RAAS, renin-angiotensin-aldosterone system; TNF- α , tumor necrosis factor α .

Modified from Lee SA, Cozzi M, Bush EL, Rabb H. Distant organ dysfunction in acute kidney injury: a review. *Am J Kidney Dis*. 2018;72(6):846–856.

system.¹⁴⁸ Additionally, there are structural differences between rodents and humans, although pig kidneys are most similar to humans.¹³⁰

These differences need to be taken into consideration when using mice in preclinical models to mimic human AKI. Furthermore, tubular injury and repair and medullary congestion are difficult to compare with human ischemic ATN. In human AKI, pure ischemia alone is seen in the minority of cases, and there is rarely complete cessation of blood flow to the kidneys. The parenteral delivery of prophylactic therapeutic agents is impossible in complete occlusion models. Because oxygen and metabolic substrates are unable to reach the kidney, the generation of ROS and peroxynitrite species, considered to be important mediators of injury, might have a different or delayed role as compared with low-oxygen states in hypoperfusion models. Total blood flow cessation also prevents the degradative products of the ischemic kidney from being washed out. Other factors playing a role in the pathophysiology of AKI, such as inflammatory mediators released from the spleen, ischemic gut, endothelium, and vascular smooth muscle cells, need to be taken into consideration in any experimental model. Bowel proteins released into the circulation can act as inflammatory mediators and increase the susceptibility to AKI.¹⁴⁹ Others have shown that short-chain fatty acids derived from gut bacteria prevent AKI¹⁵⁰ or germ-free conditions render mice more susceptible to AKI.^{151,152} The S3 segment of the proximal tubule is characterized by severe necrosis in clamp models, a finding seen rarely in human AKI. Human biopsies, however, rarely sample the outer medulla, where most of the injury is thought to occur. In contrast to animal models, human AKI histologic biopsy data are lacking at early time points from the onset of insult.¹²⁹ Thus, the NIH/NIDDK KPMP will be vital in this regard.

The cold ischemia–warm reperfusion model resembles human kidney transplantation but is inadequately studied and experimentally difficult to perform. In the isolated

perfused kidney model, the kidney is perfused ex vivo using perfusates with and without erythrocytes, and the model uses ischemic (stopping perfusate) or hypoxic (reduced oxygen tension of erythrocytes) to induce functional impairment. The morphologic patterns are different in erythrocyte-free and erythrocyte-rich perfusates. The latter system is more comparable with what is observed histologically in animal models. Additionally, limitations include exclusion of various inflammatory mediators, neuroendocrine hemodynamic regulation, and systemic cytokine and growth factor interactions known to be present and that play a pathophysiologic role in animal models and likely in human ischemia.

Cardiac arrest commonly leads to AKI. Burne-Taney and colleagues have described a whole-body ischemia-reperfusion injury (IRI) model induced by 10 minutes of cardiac arrest, followed by cardiac compression resuscitation, ventilation, epinephrine, and fluids that lead to a significant rise in serum creatinine level and renal tubular injury at 24 hours.¹⁵³ One of the unique advantages of this model is the crosstalk among vital organs such as the brain, heart, and lung and renal hemodynamics.¹³⁹ A hypoperfusion model of AKI using partial aortic clamping, first described by Zager,¹⁵⁴ may be more representative of human AKI, reflecting a state of reduced blood flow to the kidney, with systolic blood pressure of approximately 20 mm Hg, resulting in reproducible AKI.¹⁵⁴

Toxic models of kidney failure use various known toxins, such as radiocontrast media, gentamicin, cisplatin, glycerol, and pigments, including myoglobin, folic acid, and hemoglobin. Septic models to study AKI include cecal ligation and puncture (CLP), endotoxin infusion, and bacterial infusion into the peritoneal cavity. The endotoxin model, which is simple, inexpensive, and suitable for studying new pharmacologic agents, has certain drawbacks as well. There is variability among sources of lipopolysaccharide (LPS) endotoxin, the rates and methods of administration vary, and it is usually of short duration due to the high mortality associated with the doses required to induce AKI. It also

tends to be a vasoconstrictive model and does not recapitulate the early hemodynamics or inflammation of human sepsis.¹⁵⁵ Wichterman and colleagues were the first to describe a sepsis model in the early 1980s using the CLP laboratory model.¹⁵⁶ In the CLP model, there is considerable similarity to sepsis in humans, with acute lung injury, metabolic derangement, and systemic vasodilation, accompanied initially by increased cardiac output. However, there is some variability depending on the mode and size of cecal perforation. Doi and coworkers have developed a sepsis model that considers the following: (1) animals should receive the same supportive therapy that is standard for intensive care unit (ICU) patients (fluid resuscitation and antibiotics); and (2) age, chronic comorbid conditions, and genetic heterogeneity vary.¹⁵⁷ Another example is a hyperdynamic form of sepsis established in Merino sheep. These studies have provided important new information that cannot be derived from rodent models of sepsis.¹⁵⁸ Complex animal models of human sepsis that introduce these disease-modifying factors are likely more relevant and may be more pharmacologically relevant than simple animal models.¹⁵⁷

This description is intended to remind the reader of the potential pitfalls in each model when evaluating experimental studies or therapeutic interventions using these models. The lack of ability to demonstrate the effectiveness of an agent in humans shown to be efficacious in animal models does not necessarily reflect a flaw with the model or the agent in question. Most often, the agent is administered late in the course of the human disease; patient heterogeneity and the difficulty in stratifying patients by severity of injury makes it even more difficult to establish efficacy.^{123,159} Further studies have led to the development of zebrafish, three-dimensional (3D) microfluidic, and human kidney organoids models that are improving our understanding of AKI or facilitating pharmacologic studies for the treatment of AKI.

Zebrafish

Given the capacity of zebrafish adult nephrons to undergo robust epithelial regeneration and to form nephrons *de novo*, investigations using zebrafish models of AKI have allowed a better understanding of the cellular mechanisms associated with kidney regeneration after AKI. Using chemical genetics, investigators have discovered that histone deacetylase inhibitors (HDACi) are capable of ameliorating gentamicin-induced AKI in the zebrafish embryo with an expansion of renal progenitors that express the genes *Lhx1a*, *Pax2a*, and *Pax8*.¹⁶⁰ Future studies using zebrafish transgenic lines in which injury can be induced in tubular cells via the nitroreductase-metronidazole system, are likely to help characterize how individual populations of cells in the nephron respond to kidney damage.¹⁶¹

Three-Dimensional Microfluidic Models and Organoids

Side effects, and especially drug-induced nephrotoxicity, can often be an important limiting factor in the development of new pharmacotherapy. In addition to each of the current animal models of AKI having limitations in fully recapitulating human AKI, the lack of ability to predict drug-induced AKI has led to failed drug trials. Early nephrotoxicity screening studies included two-dimensional standard well plates with semipermeable filter cups. There were a number of limitations to two-dimensional standard well plates, including the

following: (1) use of cell lines of nonhuman origin (MDCK; LLC-PK 1); (2) use of cell lines with limited proximal tubule characteristics (human kidney 2 [HK 2]); (3) cell lines that have features of epithelial to mesenchymal transition (HK 2); and (4) static conditions. By contrast, proximal tubule cells are subject to continuous tubule fluid flow and fluid shear stress that modulate cellular signal transduction through mechanosensing receptors, organization of the cytoskeleton, actin filaments, and adherens junctions.^{162–166} More physiologically relevant models such as three-dimensional microfluidic models have been developed to bridge the step from standard two-dimensional systems to animal models or as a total replacement for costly, inefficient animal models. These three-dimensional microfluidic models, such as parallel flow-plate models and early three-dimensional perfusion models, include chip technology with cell monolayers or tubular structures that are sandwiched between two microfluidic channels. Recently, Qu and coworkers developed a multilayer microfluidic device to simulate glomerulus, Bowman's capsule, proximal tubular lumen, and peritubular endothelial cells to investigate the pathophysiology of drug-induced AKI. The authors were able to demonstrate time- and dose-dependent induction of cell death by cisplatin and doxorubicin. The use of this biomimetic device has yielded useful information about drug-induced AKI at the preclinical stage.¹⁶⁷

Human kidney organoids are three-dimensional clusters of cells that are functionally and genetically similar to kidney. Human-induced pluripotent stem cells (hiPSCs) are ideally suited for human kidney organoids because of their unlimited self-renewal and ability to generate cells of all three embryonic germ layers. Major challenges to the generation of functional bioengineered kidneys are the incorporation of adequate vascularization to kidney organoids and the establishment of an effective drainage system for the removal of blood filtrate after passage and processing through the tubule system.¹⁶⁸ Protocols that add combinations of growth factors to mimic *in vivo* conditions and growing cells in three-dimensional cultures have generated nephron progenitor cells and kidney organoids from human embryonic stem cells and hiPSCs.^{169–172}

ACUTE TUBULAR NECROSIS

EPITHELIAL CELL INJURY

Although all segments of the nephron may undergo injury during an ischemic insult, the major and most commonly injured epithelial cell involved in AKI related to ischemia, sepsis, and/or nephrotoxins is the PTC. Of the three segments (S1–S3), the S3 segment of the proximal tubule in the outer stripe of the medulla is the cell most susceptible to ischemic injury for several reasons.¹⁷³ First, it has limited capacity to undergo anaerobic glycolysis due to its dependence on fatty acid oxidation as the major source of energy. Second, due to its unique primarily venous capillary regional blood flow, there is marked hypoperfusion and congestion in this medullary region after injury that persists, even though cortical blood flow may have returned to near-normal levels after ischemic injury. Endothelial cell injury and dysfunction are primarily responsible for this phenomenon, often referred to as the “extension phase” of AKI.¹⁵⁹ The other major epithelial cells of the nephron involved are those of the medullary

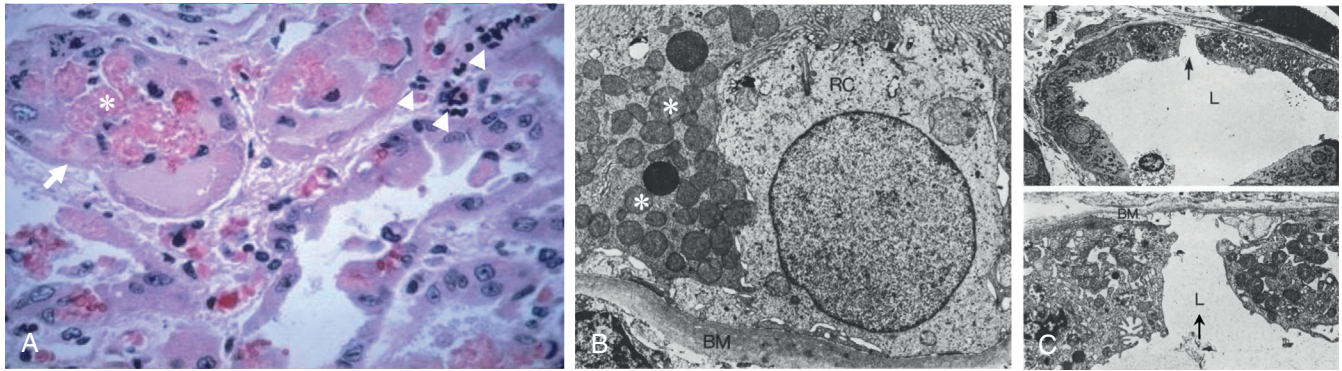


Fig. 28.2 Morphology of human acute tubular necrosis. (A) Human biopsy specimens reveal significant proximal tubular cell damage, with intraluminal accumulation of apical membrane fragments and detached cells (*), thinning of proximal tubular cells to maintain monolayer tubule integrity (arrow), and dividing cells and accumulation of white cells within the microvascular space in the peritubular area (arrowheads). (B) Electron micrograph of a regenerating epithelial cell. Shown are small fragmented mitochondria (*). (C) Electron micrograph of renal epithelial cell showing nonreplacement site (black arrow) that morphologically supports the concept of backleak in the pathophysiology of AKI. (A, Courtesy M. Venkatchalam; B and C from Olsen TS, Olsen HS, Hansen HE. Tubular ultrastructure in acute renal failure in man: epithelial necrosis and regeneration. *Virchows Arch A Pathol Anat Histopathol.* 1985;406(1):75–89.)

thick ascending limb located more distally. Cells of the S1 and S2 segments are usually involved in toxic nephropathy due to their high rates of endocytosis, leading to increased cellular uptake of the toxin. PTCs and thick ascending limb of Henle (TAL) cells have been shown to be involved as sensors, effectors, and injury recipients of AKI stimuli.

Proximal tubular cell injury and dysfunction during ischemia or sepsis lead to a profound drop in the GFR through afferent arteriolar vasoconstriction, mediated by tubular glomerular feedback and proximal tubular obstruction. This phenomenon, along with tubular backleak, leads to a fall in the effective GFR^{174,175} (Figs. 28.1 and 28.2).

Morphologic Changes

The classic histologic hallmark of ATN was described in a landmark study by Oliver and coworkers, in which individual nephrons from autopsy specimens of patients dying of acute renal failure (ARF) were microdissected. Portions of glomerular ultrafiltrate had become sequestered in tubules that were obstructed (necrotic PT cells), which suggested that filtrate leaked back through damaged tubular walls and entered the interstitium, which caused it to become edematous.¹⁷⁶ Early on, there is the loss of the apical brush border of the PTCs. Microvilli disruption and detachment from the apical cell surface forming membrane-bound blebs occurs early, with release into the tubular lumen. Patchy detachment and subsequent loss of tubular cells exposing areas of denuded tubular basement and focal areas of proximal tubular dilation, along with the presence of distal tubular casts, are also major pathologic findings in ATN.¹⁷⁷ The sloughed tubular cells, brush border vesicle remnants, and cellular debris in combination with Tamm-Horsfall glycoprotein form the classic muddy brown granular casts. These distal casts have the potential to obstruct the tubular lumen. Frank necrosis itself is inconspicuous and is restricted to the highly susceptible outer medullary regions. Alternatively, features of apoptosis are more commonly seen in proximal and distal tubular cells. Glomerular epithelial cell injury in ischemic, septic, or nephrotoxic injury is not typically seen, although some studies have shown thickening and coarsening of foot

processes; Wagner and associates have shown podocyte-specific molecular and cellular changes.¹⁸⁹ The future morphologic course of the tubular cell alterations varies according to the type and extent of injury, as discussed in the next section (see Fig. 28.2).

Cytoskeletal and Intracellular Structural Changes

The cytoskeleton in eukaryotic cells consists of intermediate filaments, microtubules, and actin filaments.¹⁷⁸ The microtubule cytoskeleton is composed of α -tubulin and β -tubulin heterodimers that serve to regulate the shape, motility, and division of tubular epithelial cells. A recent study has demonstrated that IRI to the kidney causes α -tubular deacetylation in microtubules, and inhibition of microtubule dynamics induced by changes in tubulin acetylation during the recovery phase retards tubular epithelial cell regeneration.¹⁷⁹ Epithelial cellular structure and function are mediated in part by the actin cytoskeleton, which plays an integral role in surface membrane structure and function, cell polarity, endocytosis, signal transduction, cell motility, movement of organelles, exocytosis, cell division, cell migration, barrier function of the junctional complexes, cell-matrix adhesion, and signal transduction.¹⁸⁰ Based on its role in this multitude of processes, any disruption of the actin cytoskeleton results in changes and/or disruption of the functions mentioned earlier. This is especially important for PTCs, where amplification of the apical membrane by microvilli is essential for normal cell function.

Ischemic insult results in cellular ATP depletion, which in turn leads to a rapid disruption of the apical actin and disruption and redistribution of the cytoskeleton F-actin core, resulting in the formation of membrane-bound extracellular vesicles or blebs.¹⁸¹ These can be exfoliated into the tubular lumen or internalized with the capability of being recycled. The core mechanism of disruption is the depolymerization mediated by actin-binding protein known as actin depolymerizing factor (ADF) or cofilin.¹⁸² This protein family is normally maintained in the inactive phosphorylated form, which cannot bind to actin. Ischemia results in ATP depletion, which has been shown to cause Rho GTPase inactivation.¹⁸³ This can

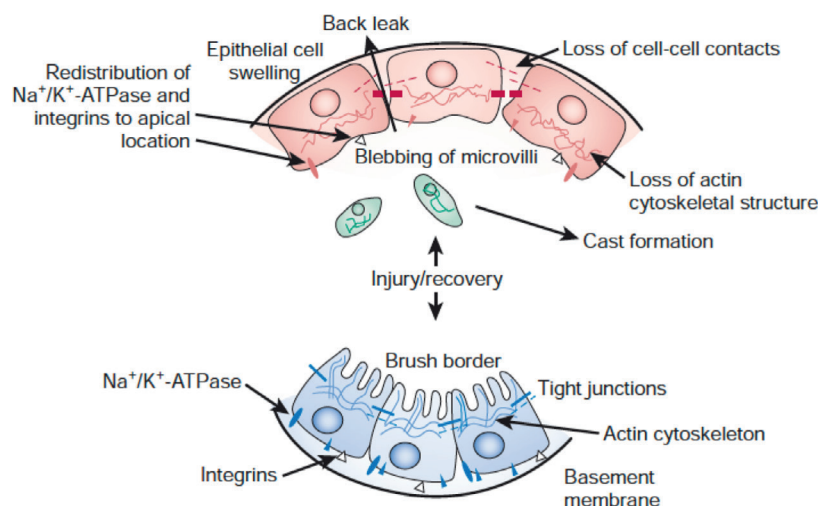


Fig. 28.3 Overview of sublethally injured tubular cells. Sodium-potassium adenosine triphosphatase (Na⁺-K⁺-ATPase) pumps are normally located at the basolateral membrane. In sublethal ischemia, the pumps redistribute to the apical membrane of the proximal tubule. On reperfusion, the pumps reverse back to their basolateral location. (From Sharfuddin A, Molitoris B. Epithelial cell injury. In Vincent JL, Hall JB, eds. *Encyclopedia of Intensive Care Medicine*. New York: Springer; 2012.)

lead to activation and relocation of ADF (cofilin) to the apical membranes, where it can mediate different effects, including depolymerization, severing, capping, and nucleation of F-actin. This destroys the actin filament core structure of microvilli and results in surface membrane instability and blebbing^{184,185} (Fig. 28.3).

A recent study has used human primary tubular epithelial cells in culture to examine the role of hypoxic injury on epithelial cell cytoskeletal organization. Given previous information that stabilization of hypoxia-inducible factor (HIF) simulates hypoxic injury during AKI, the authors found that HIF stabilization in human tubular epithelial cells reduces tubular cell migration and induces the rearrangement of actin filaments and cell adhesion molecules, including paxillin and focal adhesion kinase. These data support the role of HIF stabilization during epithelial migration, which underlies a potential mechanism of renal regeneration in response to AKI.¹⁸⁶ Similarly, tropomyosins physiologically bind to and stabilize the F-actin microfilament core in the terminal web by preventing access to ADF. After ischemia, there is dissociation of tropomyosins from the microfilament core, resulting in access of the microfilaments in the terminal web to the binding, severing, and depolymerizing actions of ADF/cofilin.^{187,188}

Another important consequence of disruption of the actin cytoskeleton is the loss of tight junctions and adherens junctions. These junctional complexes actively participate in numerous functions, including paracellular transport, cell polarity, and cellular shape. The tight junctions, also known as zonula occludens (ZO), are composed of proteins such as occludin, claudin, ZO-1, and protein kinase C with numerous barrier functions, such as adhesion, permeability, and transport. The actin present in the terminal web is linked to ZO, and hence any disruption of the terminal web results in disruption of the tight junctions. Early ischemic injury results in opening of these tight junctions, leading to increased paracellular permeability and causing further backleak of the glomerular filtrate into the interstitium.¹⁸⁰ In the glomerulus, ischemia also induces a rapid loss of

interaction between slit diaphragm junctional proteins NEPH1 and ZO-1,¹⁸⁹ leading to podocyte damage, effacement, and proteinuria.

The molecular mechanisms underlying these changes have been studied as well and show that ATP depletion that results in actin polymerization is followed by a reduction in cellular adhesion ability. Pretreatment with an actin stabilizer prevented ATP depletion-induced actin polymerization and reduction of cell adhesion, indicating that the cytoskeleton reorganization decreased the cellular adhesion ability. Furthermore, the ATP depletion markedly increased the levels of p38 mitogen-activated protein (MAP) kinase and heat shock protein 27 (hsp27) phosphorylation, with enhanced translocation of phosphorylated hsp27 from cytoskeleton to cytoplasm. The inhibition of p38 MAP kinase by specific inhibitor SB203580 blocked the ATP depletion to induce hsp27 phosphorylation and actin polymerization. These findings suggest that ischemia remodels F-actin, leading to desquamation of proximal tubular epithelial cells through p38 MAP kinase–hsp27 signaling.¹⁹⁰

Actin cytoskeleton alterations and dysfunction during ischemia result in changes in cell polarity and function. Normally, sodium-potassium adenosine triphosphatase (Na⁺-K⁺-ATPase) pumps reside in the basolateral membrane of the tubular epithelial cell, but under conditions of ischemia, they redistribute to the apical membrane as early as within 10 minutes.¹⁹¹ This process occurs due to the disruption of the pumps' attachment to the membrane via the spectrin/actin cytoskeleton. Postulated mediated mechanisms include hyperphosphorylation of the protein ankyrin, with consequent loss of the binding protein spectrin, and cleavage of spectrin by the activation of proteases such as calpain. This redistribution of the Na⁺-K⁺-ATPase pump results in bidirectional transport of sodium and water across the epithelial cell apical membrane, as well as the basolateral membrane. This results in transport of cellular Na back into the tubular lumen, one of the major mechanisms of the high FE_{Na} seen in patients with ischemic ATN,¹⁹² and the inefficient use of cellular ATP. ATP is consumed but effective, vectorial Na transport is lost.

Cell Death Pathways

Necrosis and Regulated Cell Death Pathways. Epithelial cell necrosis is a passive nonenergy dependent process that develops secondary to severe ATP depletion from toxic or ischemic insult. It is not dependent on caspase activation but results from a rise in intracellular calcium and the activation of membrane phospholipases.^{193,194} Hence, morphologically, necrotic cells do not exhibit the nuclear fragmentation or chromatin condensation seen in apoptosis, and they do not form apoptotic bodies. Functionally, severe ATP depletion results first in mitochondrial injury, with subsequent arrest of oxidative phosphorylation, causing further depletion of energy stores, and robust formation of ROS, which in turn mediate further cellular injury.

In some cases, defined regulated molecular pathways may lead to necrotic cell death, a process referred to as *necroptosis*.¹⁹⁵ Differentiation between necrosis and necroptosis requires that necroptosis-dependent cell death involves the receptor-interacting protein kinase 3 (RIPK3).¹⁹⁶ Regulated cell death pathways can be grouped into a caspase-controlled cell death system (apoptosis, necroptosis, and pyroptosis) or lipid peroxidation–autooxidation-controlled necrosis system (ferroptosis; Fig. 28.4).¹⁹⁷

Apoptosis. The fate of an epithelial cell after an injury ultimately depends on the extent of the injury. Cells undergoing

sublethal or less severe injury have the capacity for functional and structural recovery if the insult is interrupted. Cells that suffer a more severe (or lethal) injury undergo apoptosis or necrosis. Apoptosis is an energy-dependent, programmed cell death after injury that results in the condensation of nuclear and cytoplasmic material, forming apoptotic bodies. These apoptotic bodies, which are plasma-membrane bound, are rapidly phagocytosed by macrophages and neighboring viable epithelial cells. In necrosis, there is cellular and organelle swelling, with loss of plasma membrane integrity and release of cytoplasmic and nuclear material into the lumen or interstitium¹⁹⁸ (Fig. 28.5).

The caspase family of proteases is an important initiator as well as an effector of apoptosis.^{199,200} Both the intrinsic (mitochondrial) and extrinsic (death receptor) apoptotic pathways are activated in human AKI. Specifically, activation of procaspase-9 primarily depends on intrinsic mitochondrial pathways regulated by the Bcl-2 family of proteins, whereas that of procaspase-8 results from extrinsic signaling via cell surface death receptors, such as Fas and their ligand FADD (Fas-associated protein with death domain). There is also considerable crosstalk between the intrinsic and extrinsic pathways. The other group of caspases—3, 6, and 7—are effector caspases, which are more abundant and catalytically robust, cleaving many cellular proteins and resulting in the classic apoptotic phenotype. Caspase activation in

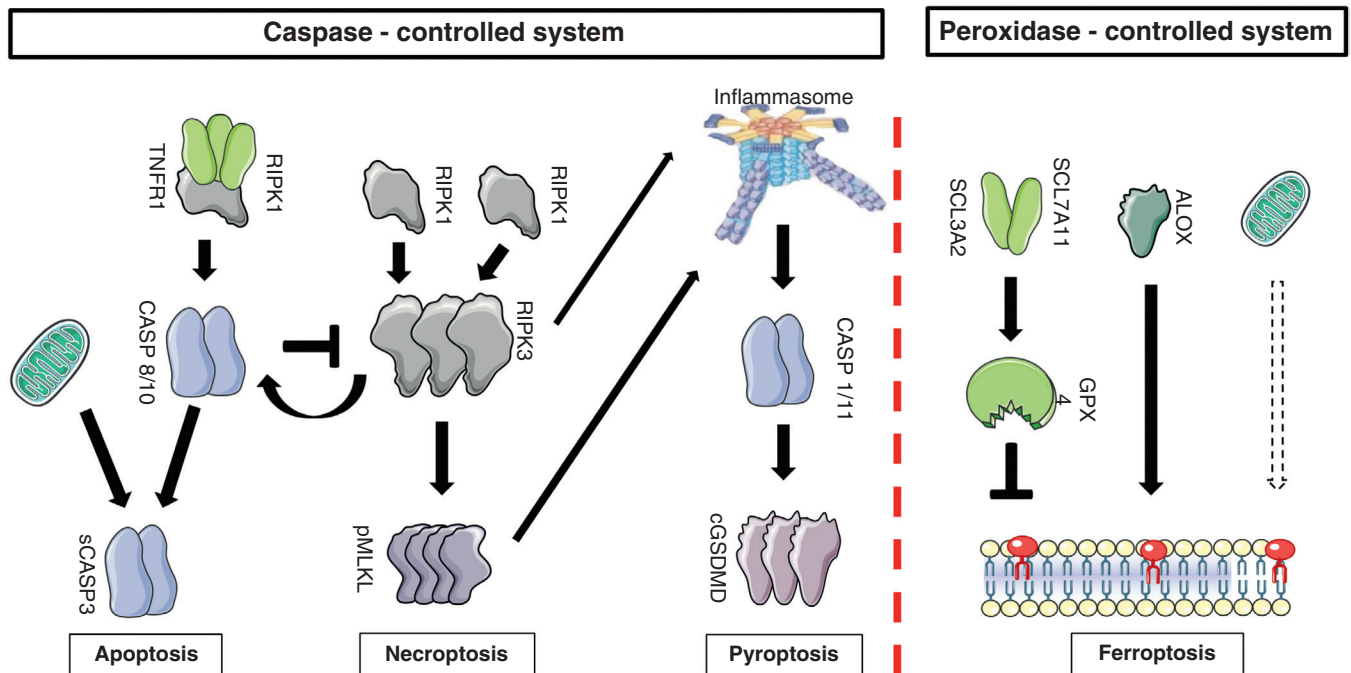


Fig. 28.4 Overview of the pathways of regulated cell death. In general, two systems may be best differentiated when regulated cell death is considered. The caspase-controlled system includes apoptosis, necroptosis, and pyroptosis and has been studied in great detail over the last few decades. In contrast, the peroxidation-controlled system of ferroptotic cell death functions entirely independently of the caspase-controlled network. Importantly, both systems contribute to human disease. Targeting clinically relevant cell death, therefore, should at least require a combination of therapies aimed at these two systems, which may exhibit some redundant functions. Similarly, in the caspase-controlled system, inhibition of either pathway may result in alternative pathways. With the idea of necroinflammation in mind, it may be helpful to shift a deadly signal from a highly immunogenic pathway (e.g., pyroptosis) toward a pathway with less immunogenic potency (e.g., apoptosis). (From Tonnus W, Gembardt F, Latk M, et al. The clinical relevance of necroinflammation—highlighting the importance of acute kidney injury and the adrenal glands. *Cell Death Differ.* 2019;26(1):68–82.)

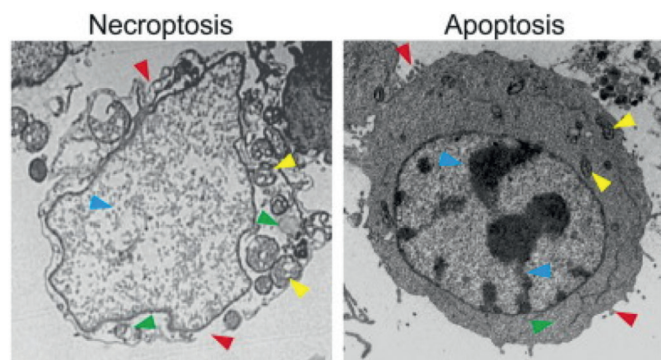


Fig. 28.5 Morphologic features of necroptosis and apoptosis. HT29 colon cancer cells treated with an anticancer drug for 48 hours were analyzed by transmission electron microscopy. The cell undergoing necroptosis shows plasma membrane rupture and permeabilization, compared with the intact plasma membrane, with blebbing in the apoptotic cell (red arrowheads). The necroptotic cell exhibits cytoplasm swelling and vacuolization, which are absent in the apoptotic cell (green arrowheads). The necroptotic cell has swelled mitochondria, in contrast to those in the apoptotic cells (yellow arrowheads). The necroptotic cell also lacks the condensed and fragmented nuclei seen in the apoptotic cell (blue arrowheads). (From Chen D, Yu J, Zhang L. Necroptosis: an alternative cell death program defending against cancer. *Biochim Biophys Acta*. 2016;1865(2):228–236.)

epithelial cells occurs due to ischemic and other cytotoxic insults, whereas inhibition of caspase activity is protective against such injury in cultured and in vivo renal epithelial tubular AKI.^{201,202}

Several pathways, including the intrinsic (Bcl-2 family, cytochrome *c*, caspase-9), extrinsic (Fas, FADD, caspase-8), and regulatory (p53 and NF- κ B) pathways, appear to be activated during ischemic renal tubular cell injury. It has also been shown that the balance between cell survival and death depends on the relative concentrations of the proapoptotic (Bax, Bcl-2-associated death promoter [Bad], and Bid) and antiapoptotic (Bcl-2 and Bcl-xL) members of the Bcl-2 family of proteins. Overexpression of proapoptotic or relative deficiency of antiapoptotic proteins may lead to the formation of mitochondrial pores. Conversely, the inhibition of such pore formation may occur with the opposite imbalance.^{203–205}

Other proteins that have been shown to play a significant role in the apoptotic pathways include NF- κ B and p53.^{206,207} The central proapoptotic transcription factor p53 can be activated by hypoxia, via HIF-1 α , as well as by other noxious stimuli, such as certain drugs (e.g., cisplatin). The kinase-mediated pathways such as ERKs and c-Jun N-terminal kinases (JNKs) are responsible for mediating cellular responses involved in apoptosis, survival, and repair through their interaction with other signals from growth factors, such as hepatocyte growth factor, insulin-like growth factor-1, epidermal growth factor, and vascular endothelial growth factor (VEGF).^{208,209} These independent mechanisms can inhibit proapoptotic proteins such as Bad and activate the antiapoptotic transcription of CREB (cyclic adenosine monophosphate response element binding) factors. More recent studies have indicated that there is rapid delivery of small interfering RNA (siRNA) to proximal tubular cells in AKI; targeting siRNA to minimize p53 production leads to a dose-dependent

attenuation of apoptotic signaling and kidney function, suggesting potential therapeutic benefit for ischemic and nephrotoxic kidney injury.²¹⁰ In vivo, microRNA-24 (miR-24) also regulates the HO-1 and H2A histone family, member X. Overall, these results indicate that miR-24 promotes renal ischemic injury by stimulating apoptosis in endothelial and tubular epithelial cells.²¹¹

Considering the various pathways available for blockade or modulation, the therapeutic implications of targeting apoptosis in preventing epithelial cell injury are significant. However, it is likely that the “window” to avert lethal injury and prevent cells from progressing to necrosis is in the early initiating apoptotic phases.

Numerous studies have shown that ATP depletion leads to a rise in intracellular calcium through impairment of calcium ATPases, whereas inhibition of Na⁺-K⁺-ATPase activity potentiates calcium entry into the cell via the sodium-calcium exchanger. Increased cytosolic calcium causes further mitochondrial injury and cytoskeletal alterations.²¹² This chain of events results in the downstream activation of proteases such as calpain and phospholipases. Phospholipases such as phospholipase A2 cause direct hydrolytic damage to membranes and also release toxic free fatty acids. They also cause release of eicosanoids that have vasoactive and hemokinetic activities, resulting in an intense surrounding inflammatory response. Calpain mediates plasma membrane permeability and hydrolysis of the cytoskeletal proteins.^{213,214} Finally, there is release of lysosomal enzymes and proteases that degrade histones, resulting in accessibility of the endonucleases to the entire segment, typically seen as the smear pattern on gel electrophoresis, in contrast to the typical ladder pattern seen in apoptosis.²¹⁵

Necroptosis. Necroptosis and ferroptosis are two forms of regulated, nonapoptotic cell death. Necrosis is distinguished from apoptosis by the presence of a breakdown of the integrity of the plasma membrane (see Fig. 28.5). As such, necrotic cell death is accompanied by the release of unprocessed intracellular contents, including cellular organelles, highly immunogenic proteins such as ATP, HMGB1, double-stranded DNA, and RNA components, also referred to as damage-associated molecular patterns (DAMPs), a concept known as *neuroinflammation*.²¹⁶ Although tubular necrosis was thought to be accidental, work done over the last 2 decades has revealed several pathways of genetically determined and regulated necrosis in which receptor interacting serine-threonine protein kinase 1 (RIPK1),^{217,218} RIPK3,²¹⁹ and its substrate, mixed-lineage kinase domain-like protein (MLKL), have been directly implicated in the regulation of this novel cell death pathway known as necroptosis.²²⁰ The signaling pathway that triggers necroptosis includes the engagement of death receptors in the presence of caspase inhibition, stimulation of Toll-like receptors, signaling through interferons, with the activation of kinase RIPK3, and phosphorylation of pseudokinase MLKL. Phosphorylation of MLKL by RIPK3 leads to a molecular switch mechanism that induces plasma membrane rupture.²²¹ Demonstration of a protective effect on kidney function with the use of necrostatin-1, an inhibitor of RIPK1, suggests that necroptosis occurs in ischemic AKI.²²² Necroptosis also occurs in the model of cisplatin-induced AKI, as demonstrated by the significant protection of kidney function when using RIPK3 and MLKL-deficient mice.²²³

Ferroptosis. Ferroptosis is a previously unrecognized, non-apoptotic form of regulated cell death that is iron-dependent and characterized by increased lipid peroxidation resulting from lack of activity of the lipid repair enzyme glutathione peroxidase 4 (GPX4).^{224–226} This leads to the accumulation of lipid-based ROS, including lipid hydroperoxides. This form of iron-dependent cell death is distinct from other forms of cell death such as apoptosis, unregulated necrosis, and necroptosis because it mediates cell death in a noncell autonomous and synchronized manner, providing a potential explanation for nephron loss during AKI.²²⁷ Erastin and RSL3 are ferroptosis-inducing compounds that induce cell death in the absence of apoptotic features.^{224,228,229} Erastin and RSL3 induce cell death in the absence of major components of apoptotic machinery, including caspases, BAX, and BAK.²³⁰ In a recent study, ferroptosis was shown to play a key role in folic acid-induced AKI,²³¹ resulting in increased inflammation. Ferroptosis is also importantly regulated by heme oxygenase-1 (HO-1).²³² Immortalized proximal tubule cells from HO-1^{+/+} mice were much more susceptible to erastin or RSL3 than cells from HO-1^{-/-} mice. Iron supplementation decreased cell viability further in HO-1^{-/-} compared with HO-1^{+/+} cells. Finally, ferrostatin (a ferroptosis inhibitor), deferoxamine (an iron chelator), or N-acetyl-L-cysteine (a glutathione replenisher) attenuate erastin-induced ferroptosis²³² (see Fig. 28.4).

Pyroptosis. This highly inflammatory form of regulated cell death requires caspase 1,4/5 (caspase 11 in mice) for activation^{197,233} (see Fig. 28.4). Intracellular damage (sterile

inflammatory, such as AKI) and release of DAMPs (pathogen-associated molecular patterns [PAMPs]) activate NLRP3 inflammasome and, in turn, lead to caspase activation and cleavage of pro-interleukin-1 β (IL-1 β) and pro-IL-18 to their mature forms. In addition, a newly identified pyroptosis executioner, gasdermin D (GSDMD), is activated, releasing the N-terminal fragment (GSDMD-NT).^{197,234} The GSDMD-NT fragment is thought to oligomerize and form a membrane pore on the cell membrane,²³⁵ leading to cell swelling, osmotic lysis, and release of IL-1 β and IL-18 and other intracellular contents.^{197,234} Thus, unlike apoptosis, pyroptosis results in plasma membrane rupture and release of DAMP molecules, leading to the activation of innate immunity and recruitment of immune cells.²³⁶

Understanding cell death pathways can lead to specific interventions that could involve strategies to preserve apoptosis for avoiding inflammation while blocking the more inflammatory pathways such as pyroptosis, considered the most proinflammatory of the regulated pathways.¹⁹⁷

Autophagy. Autophagy is an essential mechanism for normal homeostasis, disease pathogenesis, and aging in kidneys.^{237–240} Fig. 28.6 reviews the three forms of autophagy—macroautophagy, microautophagy, and chaperone-mediated autophagy:

- Macroautophagy starts with the de novo formation of a cup-shaped isolation double membrane that engulfs a portion of cytoplasm.

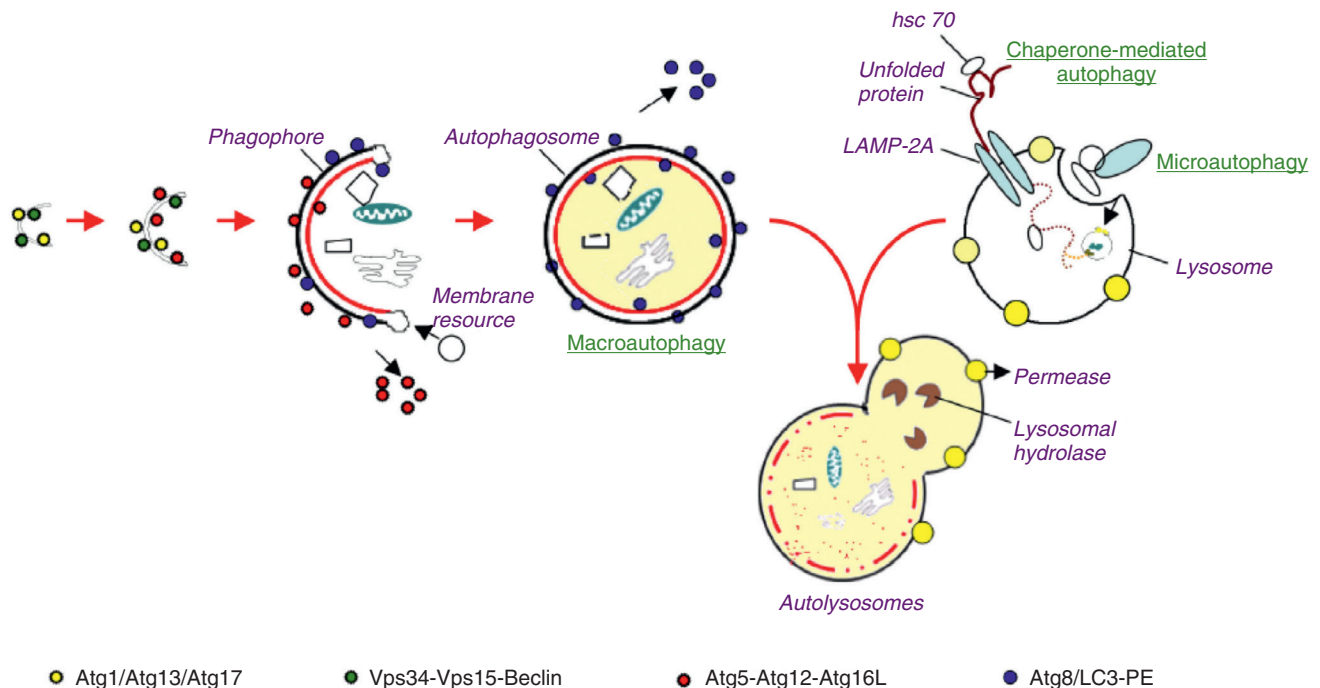


Fig. 28.6 Autophagy. The forms of autophagy are macroautophagy, microautophagy, and chaperone-mediated autophagy. Macroautophagy starts with the de novo formation of a cup-shaped isolation double membrane that engulfs a portion of cytoplasm. Microautophagy involves the engulfment of cytoplasm instantly at the lysosomal membrane by invagination, protrusion, and separation. Chaperone-mediated autophagy is a process of direct transport of unfolded proteins via the lysosomal chaperonin hsc70 and LAMP-2A. All forms of autophagy subsequently lead to the degradation of intraautophagosomal components by lysosomal hydrolases. PE, Phosphatidylethanolamine. (From Periyasamy-Thandavan S, Jiang M, Schoenlein P, Dong Z. Autophagy: molecular machinery, regulation, and implications for renal pathophysiology. *Am J Physiol Renal Physiol*. 2009;297(2):F244–F256.)

- Microautophagy involves the engulfment of cytoplasm instantly at the lysosomal membrane by invagination, protrusion, and separation.
- Chaperone-mediated autophagy is a process of direct transport of unfolded proteins via the lysosomal chaperonin hsc70 and the receptor for chaperone-mediated autophagy, LAMP-2A.

Autophagy is highly conserved and is a lysosomal degradation pathway for the salvation and reuse of degraded proteins, lipids, and organelles to generate basic structural components and energy. It also removes obsolete and dysfunctional organelles and cytotoxic protein aggregates for cellular homeostasis. Under pathogenic conditions, autophagy plays a key role in ischemic, toxic, immunologic, and oxidative insults that lead to the induction of autophagy in renal epithelial cells, which plays a key role in altering the course of disease.²³⁸ Autophagy-mediated leukocyte clearance is an important mechanism for resolving inflammation; it is a function performed by macrophages.²³⁹ Autophagy is characterized by the formation of autophagosomes, a double-membrane vesicles wrapping cytosolic components and organelles that are destined for degradation. Autophagosomes fuse with lysosomes, the contents are degraded, and the products are salvaged and serve as building blocks for cellular function.²³⁷ There are a number of genes and proteins associated in autophagy (autophagy-related; Atg).²³⁷

Autophagy serves a protective function in renal tubular cells during ischemia-reperfusion injury; it precedes the appearance of apoptotic cells, and suppression of autophagy exacerbates renal injury, in part by inhibiting apoptosis.^{241,242} The importance of autophagy has been demonstrated through the generation of proximal tubule-specific ATG5 or ATG7 knockout mice.^{243–245} The absence of ATG5 in proximal tubules led to the accumulation of deformed mitochondria and cytoplasmic inclusions or protein aggregates.²⁴³ The lack of ATG5 in distal tubules did not cause significant alterations in kidney function.²⁴⁴ Similar results were observed in proximal tubule-specific ATG7 knockout mice.²⁴⁶ Importantly, renal ischemia-reperfusion injury was exaggerated in proximal tubule-specific ATG5 or ATG7 knockout mice.^{243,244,246} Autophagy activation was also detected in cisplatin-induced AKI in mice.²⁴⁷ ATG7 knockout mice demonstrated worse kidney function and injury with cisplatin treatment.²⁴⁶ Chloroquine, an inhibitor of autophagy, tested in a cisplatin-induced model of AKI, blocked the lysosomal degradation of LC3 in autophagosomes and worsened kidney function.²⁴⁶ Similarly, autophagy was induced in the septic AKI model of LPS treatment of mice, and inhibition by chloroquine or tubular ATG7 ablation worsened LPS-induced AKI.²⁴⁸ Thus, data from pharmacologic studies and genetically deficient mice have provided strong evidence for a renoprotective role of autophagy. In addition, dysregulated autophagy can result in increased renal degeneration, leading to progressive kidney disease, as characterized by interstitial fibrosis.^{249,250}

NETosis. NETosis is a specific form of regulated neutrophil death in which neutrophils release neutrophil extracellular traps (NETs), which are extracellular structures composed of chromatin and histones that enable immobilization and killing of bacteria.²⁵¹ On their activation by IL-8, LPS, bacteria or activated platelets, neutrophils initiate a program that leads

to their death and the formation of NETs.²⁵² The activation pathways are known to involve Toll-like receptors (TLRs), cytokines, and Fc receptors.^{251,253} This form of cell death is different from apoptosis and necrosis. Cell death is independent of caspases and is not accompanied by fragmentation.^{252,254}

Infiltrating neutrophils undergoing NETosis contribute further to organ damage in ischemic AKI. Using the IRI model, Nakazawa and coworkers have demonstrated that infiltrating neutrophils undergo NETosis, leading to the release of cytotoxic DAMPs, such as histones, which exacerbate tubular epithelial cell injury and interstitial inflammation.²⁵⁵ Using an inhibitor of peptidyl arginine deiminase (PAD), a key enzyme in NET formation, or depleting neutrophils with anti-Ly6G mab, reduced NET formation, tissue necrosis, and biomarkers of AKI and improved kidney function. Furthermore, the authors also demonstrated the presence of NETs in kidney biopsies of patients with ATN.

Innate Immunity and Inflammation

Kidney Microenvironment. The interstitial microenvironment, between the basement membranes of epithelial cells and peritubular capillaries, is a reactive compartment containing mononuclear phagocytes, interstitial fibroblasts, and pericytes, as well as various soluble mediators.²⁵⁶ The mononuclear phagocytic system consists of bone marrow-derived macrophages and dendritic cells,²⁵⁷ which overlap in functional characteristics and surface biomarkers.²⁵⁸ Macrophages are resident tissue phagocytic cells that function generally to clear dying cells and produce cytokines and growth factors.^{259,260} Although this system is well known to be activated in response to foreign pathogens,²⁶¹ the microenvironment responds to endogenous molecules released from dying cells following tissue injury or to changes produced by conditions such as hypoxia, ischemia, or other forms of sterile inflammation. Matzinger has proposed the danger model to explain these exceptions to the classical role of immune responsiveness to foreign antigen.²⁶² Dendritic cells are activated by DAMPs or PAMPs²⁶³ and are the key initiators of the innate immune system, following ischemia-reperfusion (Fig. 28.7).

Mononuclear Phagocytes in Acute Kidney Injury. Dendritic cells, a resident population of bone marrow-derived cells, and macrophages form a network between the basement membranes of tubular epithelia and peritubular endothelial cells.^{256,264} Although dendritic cells and macrophages are often considered as distinct cell types with characteristic functions, recent studies have shown considerable overlap in cell surface markers and function between dendritic cells and macrophages²⁵⁷ (Fig. 28.8). Located in the interstitial space, dendritic cells have access to endogenous and exogenous DAMPs and PAMPs released by epithelial cells, invading organisms, and infiltrating cells, and thus are key initiators, potentiators, and effectors of the innate immune system.^{263,265}

Dendritic cells have enormous plasticity and can be antiinflammatory or proinflammatory.^{266,267} Dendritic cells contribute early in the course of IRI-induced activation of NKT cells and the IL-17/IL-23 signaling pathway.^{268,269} The importance of dendritic cells in activating the innate immune response in AKI has been determined through depletion of dendritic cells. This was accomplished using transgenic mice expressing the human diphtheria toxin receptor ([DTR]; human heparin-binding epidermal growth factor-like growth

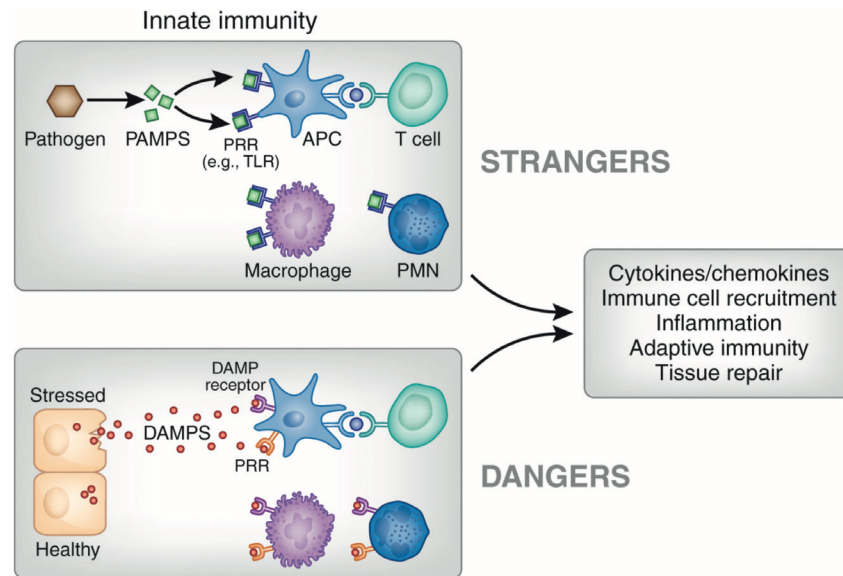


Fig. 28.7 Danger and stranger models. Infections of pathogenic bacteria or viruses cause the release of pathogen-associated molecular patterns (PAMPs) that bind to pattern recognition receptors (PRRs), such as Toll-like receptors (TLRs), on immune cells and stimulate an innate immune response that is accompanied by inflammation and activation of adaptive immunity, and eventually processes to resolve the infection and allow for tissue repair. The danger model recognizes that similar events occur when cells are stressed or injured and that necrotic cells release molecules normally hidden within the cell. In the extracellular space, these damage-associated molecular patterns (DAMPs) can bind to PRRs or to specialized DAMP receptors to elicit an immune response by promoting the release of proinflammatory mediators and recruiting immune cells to infiltrate the tissue. The immune cells that participate in these processes include, for example, antigen-presenting cells (APCs), such as dendritic cells and macrophages, as well as T cells and neutrophils (polymorphonuclear leukocytes [PMNs]). DAMPs may also stimulate adaptive immunity and participate in autoimmune responses and tissue repair. A wide variety of intracellular and extracellular molecules function as DAMPs when released from cells (see Table 28.1). The functions of such a diverse group of molecules may not yet be fully elucidated; it is unknown whether different DAMPs have specific roles, whether specific functions are elicited in different cell types or conditions, or even whether immune responses to DAMPs can be distinguished from those of PAMPs. (From Rosin DL, Okusa MD. Dangers within: DAMP responses to damage and cell death in kidney disease. *J Am Soc Nephrol.* 2011;22(3):416-425.)

factor in CD11c⁺ cells (CD11c-DTR mouse).²⁷⁰ Using diphtheria toxin (DT) and depleting kidney CD11c⁺ dendritic cells prior to IRI, there was significantly less injury in DT-treated CD11c-DTR mice compared with CD11c DTR mice treated with a catalytically inactive mutant DT (mDT),²⁷¹ strongly supporting the concept that dendritic cells contribute to the early innate response in IRI. Depletion of dendritic cells reduced IRI, and deletion of dendritic cell sphingosine 1-phosphate receptor 3 (S1P₃R) or inhibition of S1P₂R resulted in protection from IRI.²⁷¹⁻²⁷³ They are known to be the earliest producers of IL-6, tumor necrosis factor- α (TNF- α), MCP-1, and RANTES (regulated on activation, normal T cell-expressed and secreted).²⁷⁴ In addition to initiating the recruitment of inflammatory cells, dendritic cells also participate in recovery via IL-10 production.²⁷⁵ Dendritic cells can also induce tolerance. Tolerogenic dendritic cells are functionally immature, express inadequate positive or enhanced negative costimulatory signals and reduced proinflammatory cytokines, and can generate immune tolerance by inducing T cell anergy or deletion or by induction or expansion of regulatory T (Treg) cells.^{276,277} Adenosine 2A receptor (A₂AR)-induced tolerized dendritic cells, suppressed NKT cell activation in vivo and attenuated kidney IRI.²⁷⁸ However, mature dendritic cells also can promote tolerance.^{276,279} Both immature and mature dendritic cells can prime Treg cells that prevent autoimmunity. In contrast to IRI, dendritic cell ablation in DTR mice increased injury in a cisplatin-induced model of

AKI, which demonstrates the importance of dendritic cells for tissue protection in a different model of kidney injury.²⁸⁰ Thus, these studies have revealed disparate roles of tissue resident dendritic cells in AKI and suggest that the differing interstitial microenvironment created by different pathogenic circumstances, such as cisplatin toxicity, IRI, or endogenous molecules (e.g., PAMPs, DAMPs, cytokines, autacoids such as adenosine) may regulate dendritic function.²⁶⁶ Finally, dendritic cells migrate away from the kidney via the lymphatic system to present antigen and regulate lymphocytic responses.²⁸¹ Thus, dendritic cells serve at the crossroads of communication between the epithelium and endothelium, regulating both innate and adaptive immunity, self-tolerance, and tissue injury and repair.

Macrophages are phagocytic innate immune cells that contribute to host defense and, based on surface markers, five distinct populations have been identified.^{282,283} Tissue-resident macrophages are derived from a heterogeneous population of bone marrow derived-monocytes.^{260,284-286} They are characterized by low surface expression of chemokine receptor 2 (CCR2), GR-1, and Ly6C and high surface expression of the fractalkine receptor CX3CR1.^{260,287} These monocytes migrate into normal tissue and differentiate into resident dendritic cells and macrophages.²⁸⁶ In contrast, macrophages that infiltrate into inflamed tissue have a phenotype characterized by high surface expression of CCR2, Ly6C, and GR-1 and low surface expression of CX3CR1. It is likely that the

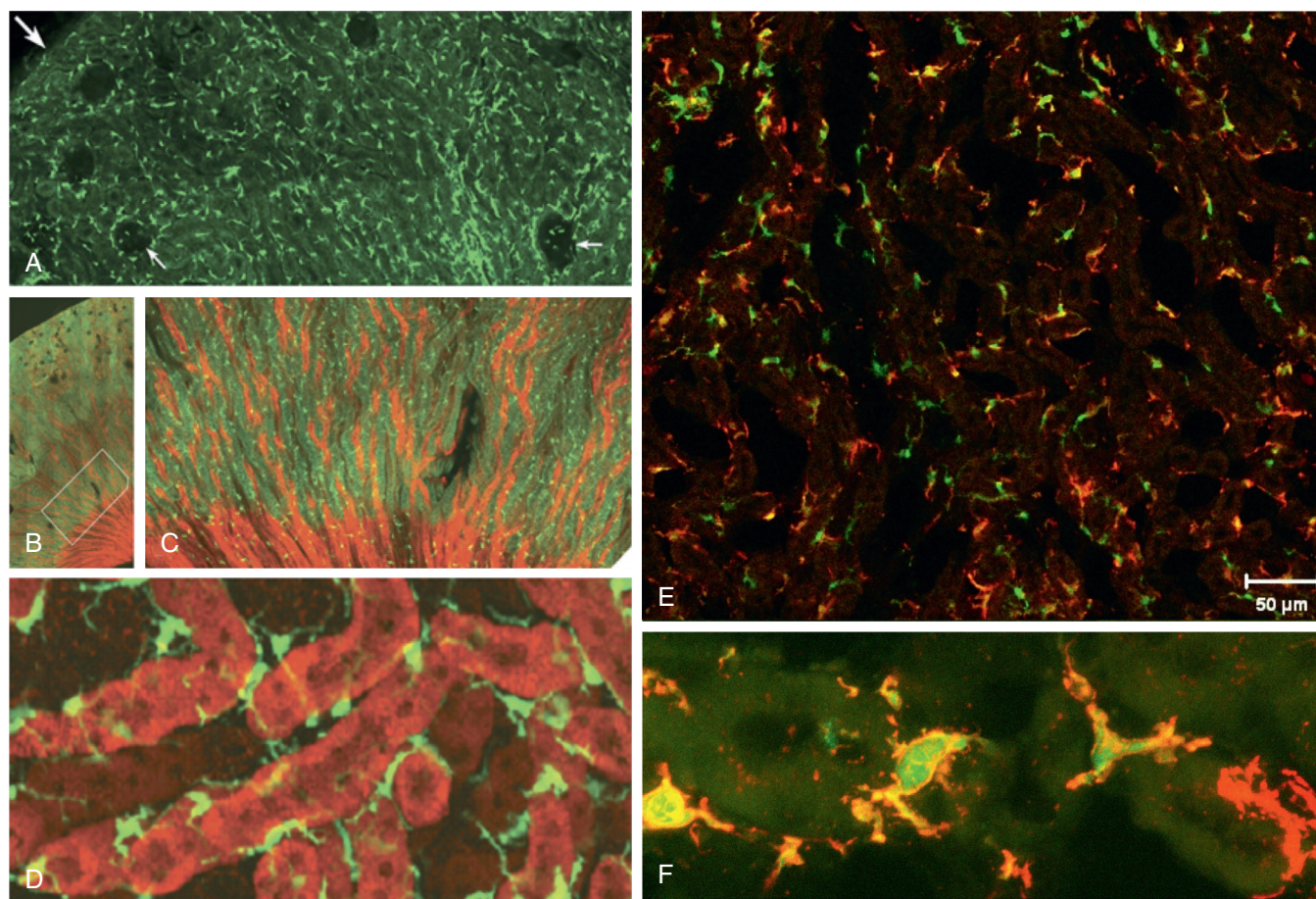


Fig. 28.8 Heterogeneity of mononuclear phagocytes within the microenvironment. Mononuclear phagocytes within the microenvironment, identified as CX3CR1⁺/GFP⁺ cells (labelled green), in the kidney are shown in panels A to D, and the heterogeneity of mononuclear phagocytes is shown in panel E. (A) A heterogeneous population of CX3CR1⁺/GFP⁺ mononuclear phagocytes in superficial cortex includes dendritic cells and macrophages and reside from the cortex and medulla. CX3CR1⁺ GFP⁺ cells extend to the edge of the cortex (large arrow) and populate the entire interstitial space abundantly. Bowman's capsule is encased by CX3CR1⁺/GFP⁺ cells (small arrows). Also notable are CX3CR1⁺ GFP⁺ cells that lie within each glomeruli. (B) Rhodamine-conjugated peanut agglutinin (red fluorescence) highlights distal tubules and collecting ducts, capsule (upper left) to the papilla (lower right). (C) Magnification of the boxed area in panel B showing CX3CR1⁺ GFP⁺ cells in the interstitium of the medulla, including transition into pyramidal tracks. Note the same spatial regularity as in the cortex. (D) Three-dimensional rendering of stellate-shaped CX3CR1⁺ GFP⁺ cells within the interstitium demarcated by tubular segments in the medulla (red fluorescence). (E) Kidney sections from the medulla of CX3CR1⁺/GFP⁺ mice. GFP is expressed mainly on monocyte-macrophages and dendritic cells, and many CX3CR1⁺GFP⁺ green fluorescing cells are seen in the cortex; most IA⁺ cells (red label) are seen in the medulla. Apparent in this image is the heterogeneity of CX3CR1⁺/GFP⁺-only cells, IA⁺-only cells, and dual labeling with CX3CR1⁺/GFP⁺ and IA⁺, which in the latter case represents dendritic cells in the medulla. (F) Higher magnification, Z-stack projection image of five optical slides at 0.69-mm intervals. These results demonstrate the heterogeneity of mononuclear phagocytes within the kidney microenvironment. (A–D from Soos TJ, Sims TN, Barisoni L, et al. CX3CR1⁺ interstitial dendritic cells form a contiguous network throughout the entire kidney. *Kidney Int.* 2006;70(3):591–596; E, F from Li L, Huang L, Sung SS, et al. The chemokine receptors CCR2 and CX3CR1 mediate monocyte/macrophage trafficking in kidney ischemia-reperfusion injury. *Kidney Int.* 2008;74(12):1526–1537.)

microenvironment in the tissue determines their phenotype. TNF- α , IL-4, and IL-15 skew monocyte differentiation toward a dendritic cell phenotype,^{288–290} whereas interferon-gamma (IFN- γ) and IL-6 direct monocyte differentiation toward a macrophage phenotype.^{291,292} Following kidney IRI, through monocyte chemoattractant protein 1 (MCP-1/CCL2) signaling via monocyte CCR2 receptors,²⁹³ monocytes migrate into inflamed kidneys within 24 hours.^{269,271}

Although controversial, and challenged by recent studies, the original classification of M1 and M2 macrophages does provide an important functional framework.²⁹⁴ M1 macrophages, also referred to as “classic macrophages,” are activated by IFN- γ and LPS and express high levels of

inducible nitric oxide synthase (iNOS). M1 macrophages have high microbicidal activity through the production of proinflammatory cytokines such as TNF- α , IL-6, and IL-12 via signal transducer and activator of transcription 1 (STAT1), NF- κ B, and ROS.²⁸³ M2 macrophages, referred to as “alternative macrophages,” are activated by IL-4 or IL-13 and participate in tissue repair and resolution of inflammation through insulin-like growth factor 1 (IGF-1), mannose receptor 1 (Mrc1/CD206), and arginase-1 (Arg1) following STAT6 activation.²⁹⁵

Evidence for these disparate roles played by macrophages can be assessed in vivo through the administration of liposomal clodronate. Liposomal clodronate is engulfed by macrophages

and clodronate is released from the liposome through action of lysosomal phospholipases in the macrophage, becomes toxic, and kills cells by apoptosis.²⁹⁶ The functional role of macrophages was determined by liposomal clodronate depletion studies that focused on the time course for macrophage depletion (early vs. late). Depletion of kidney and spleen macrophages using liposomal clodronate before renal IRI prevented AKI, and adoptive transfer of macrophages reconstituted AKI.²⁹⁷ However, depletion of macrophages 3 to 5 days after IRI slowed tubular cell proliferation and repair,²⁸³ suggesting that M1 macrophages exhibiting a proinflammatory phenotype are important for injury, whereas M2 macrophages exhibiting an antiinflammatory phenotype are important for tissue repair.^{283,298,299}

Following IRI, the proinflammatory monocyte entering into the kidney tissue is activated to a proinflammatory macrophage through the infiltration of polymorphonuclear leukocytes, T cells, and NKT cells.^{268,300,301} Resident dendritic cells activate NKT cells to produce INF- γ and other inflammatory mediators such as ROS, semaphorin-3A, DAMPs, and TNF- α , which conditions the microenvironment and favors the proinflammatory macrophage phenotype.²⁸³ These proinflammatory macrophages then produce proinflammatory cytokines, which lead to further tissue injury. Whether polarization of macrophages from M1 to M2 involves the recruitment of circulating cells or reprogramming is unclear.^{283,302} Studies by Lee and colleagues have supported the concept that macrophage polarization occurs in situ.³⁰³ In these studies, the investigators injected labeled M1 macrophages at the time of injury and examined labeled mac-

rophages that infiltrated the injured kidneys. Most of the labeled cells maintained an M1 phenotype; however, when labeled M1 macrophages were injected 3 days after injury, most of the macrophages expressed an M2 phenotype. These studies support the concept that macrophages may be reprogrammed in situ,^{283,302,304} and that the kidney microenvironment provides important conditioning cues.²⁶⁶

Early inflammation is classically characterized by the margination of leukocytes to the activated vascular endothelium via interactions between selectins and ligands that allows firm adhesion, followed by transmigration.^{305,306} A number of potent mediators are generated by the injured proximal tubular epithelial cell, including proinflammatory cytokines such as TNF- α , IL-6, IL-1 β , MCP-1, IL-8, transforming growth factor- β (TGF- β), and RANTES.^{278,307} TLR2 is an important mediator of endothelial ischemic injury, and TLR4 has been shown to play a similar role in animal models of both ischemic and septic injury,³⁰⁸ especially in PTC.³⁰⁹

Polymorphonuclear Leukocytes in Acute Kidney Injury. Neutrophils accumulate early in ischemic injury in animal models.²⁷¹ Both IL-17A and IFN- γ were shown to be produced by neutrophils and may positively regulate neutrophil transmigration to the injured kidney following kidney IRI²⁶⁸ (Fig. 28.9). Although neutrophils are seen early in rodent models of AKI, whether they play a pathogenic role remains controversial because inhibition or depletion studies of neutrophils leads to protection^{305,310,311} or lack of protection.³¹² Blockade of neutrophil function or neutrophil depletion has been shown to provide only partial protection against injury.

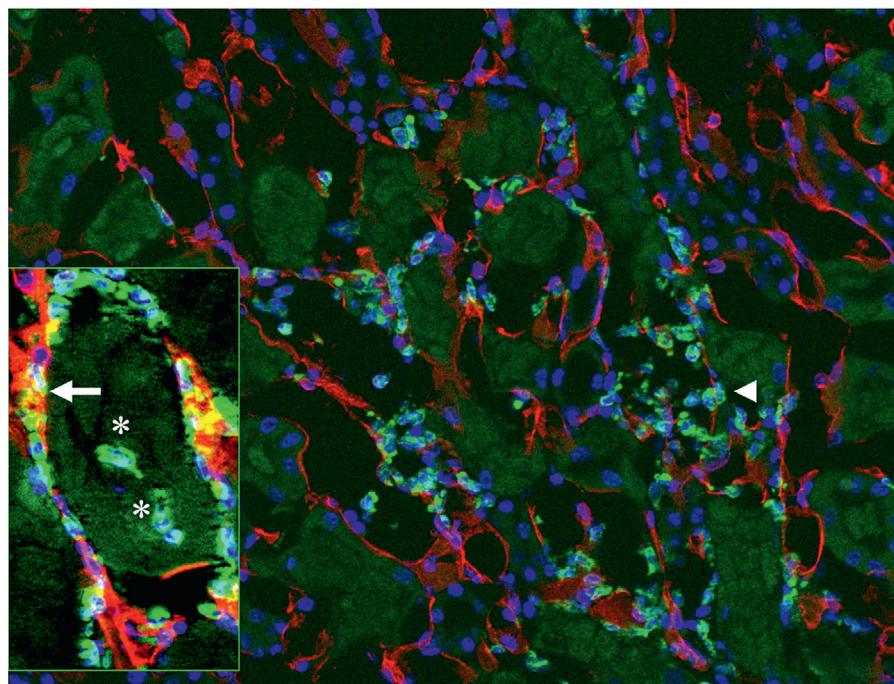


Fig. 28.9 Localization of neutrophils to interstitial and marginated compartments in the kidney. Immunofluorescence staining of kidney outer medulla using antibodies to 7/4 (green, neutrophils) and CD31 (red, vascular endothelium). Nuclei are depicted by DAPI labeling (blue). Neutrophils in both sides of the vascular endothelial wall. *Inset*, Z-stack image (7.0 μ m) of 12 optical slices of the kidney at 0.6- μ m intervals. Shown are kidney neutrophils in the interstitium and peritubular capillaries (arrows); neutrophils that have transmigrated into the lumen of the tubule (*) are shown in the inset. (From Awad, AS, Rouse, M, Huang, L, et al. Compartmentalization of neutrophils in the kidney and lung following acute ischemic kidney injury. *Kidney Int.* 2009;75:689–698.)

Vascular adhesion protein-1 (VAP-1) is an adhesion molecule^{313,314} associated with inflammatory conditions. Tanaka and colleagues have found that VAP-1 is expressed primarily in pericytes, and a specific VAP-1 inhibitor, RTU-1096, attenuated renal IRI and decreased neutrophil infiltration. The protective effect of VAP-1 inhibition was absent in neutrophil-depleted rats, suggesting an important role of neutrophil infiltration.³¹⁵

Other leukocytes, including but not limited to macrophages, NKT cells, B lymphocytes, and T lymphocytes contribute to kidney IRI.^{297,301,316–319} Selective depletion, knockout mice models, and specific blockade have shown that all these cells do mediate tubular injury at various phases and that there are synergistic interactions between different cellular types.^{297,316–319}

Whether released from the endothelium or the epithelial cell, numerous cytokines exert a concerted biochemical effort to augment the inflammatory response seen as a result of ischemic or septic injury.³¹⁷ Furthermore, mouse tubular cells when stimulated with LPS in culture, upregulate TLR2, TLR3, and TLR4 and secrete CC-chemokines such as CC motif chemokine ligand 2 (CCL2)/MCP-1 and CCL5/RANTES. These data suggest that tubular TLR expression might be involved in mediating interstitial leukocyte infiltration and tubular injury during bacterial sepsis.³²⁰

TLR2 and TLR4 are constitutively expressed on renal epithelium, and their expression is enhanced following renal IRI. El-Achkar and coworkers have shown that in a CLP rat model of sepsis, TLR4 expression increases markedly in all tubules (proximal and distal), glomeruli, and the renal vasculature.³²¹ Genetic deletion of TLR2 or TLR4 protects from renal IRI^{309,322} and from cisplatin-induced AKI. In this later study, Zhang et al.,³²³ using bone marrow chimeras, found that renal parenchymal TLR4, rather than hematopoietic-derived TLR4, mediated cisplatin-induced nephrotoxicity. These studies are consistent with studies by Wu and coworkers³⁰⁹ and clearly demonstrate the important role TLR plays in AKI.

T Cells in Acute Kidney Injury. Early work by Burne-Taney and coworkers has demonstrated that T lymphocytes contribute importantly to renal IRI.^{312,324,325} However, conventional CD4⁺ T cells are thought to play an obligatory role in antigen-specific, cognate immunity that requires 2 to 4 days for T cell processing, a time course that cannot explain the rapid, innate immune response following IRI. Natural killer T (NKT) cells are a T cell sublineage³²⁶ and, in mice, NKT cells express an invariant T cell receptor TCR α chain, V α 14-J18. In contrast to conventional T cells, the NKT cell TCR does not interact with peptide antigen presented by classic major histocompatibility complex (MHC) class I or II; rather, it recognizes glycolipids presented by the class I-like molecule, CD1d. On NKT cell activation, vigorous cytokine secretion occurs within 1 to 2 hours, including Th1-type (IFN- γ , TNF) and Th2-type (IL-4, IL-13) cytokines at the same time.^{326–332} The rapid response by NKT cells following activation can amplify and regulate the function of dendritic cells, Treg cells, NK and B cells, and conventional T cells and thus link innate and adaptive immunity.^{333–338} In AKI, NKT cells participate in early innate immune response to kidney IRI.³⁰¹ NKT-produced IFN- γ was found as early as 3 hours following IRI,³⁰¹ supporting previous work demonstrating that CD4⁺ T cell IFN- γ production is responsible for kidney injury.^{339,340} These data demonstrate the central role

of IFN- γ from CD4⁺ T and/or NKT cells in the pathogenesis of renal IRI.

Treg cells^{318,319} have also recently been shown to play a role in ischemic AKI.³⁴¹ Liu and colleagues have shown that in a murine model of ischemic AKI, there is significant trafficking of Treg cells into the kidneys after 3 and 10 days.³⁴² Postischemic kidneys had increased numbers of TCR- β ⁺–CD4⁺ and TCR- β ⁺–CD8⁺ T cells, with enhanced proinflammatory cytokine production. These investigators also noted that Treg depletion starting 1 day after ischemic injury using anti-CD25 antibodies increases renal tubular damage, reduces tubular proliferation at both time points, and enhances infiltrating T lymphocyte cytokine production at 3 days and TNF- α generation by TCR- β ⁺–CD4⁺ T cells at 10 days. In separate mouse studies, infusion of CD4⁺–CD25⁺ Treg cells 1 day after initial injury reduced IFN- γ production by TCR- β ⁺–CD4⁺ T cells at 3 days, improved repair, and reduced cytokine generation at 10 days. These studies demonstrate that Treg cells infiltrate post-IRI kidneys during the healing process and promote repair, likely through modulation of proinflammatory cytokine production of other T cell subsets.³⁴²

The role of Treg cells has been further extended by Kinsey and coworkers, who have shown that partial depletion of Treg cells with an anti-CD25 monoclonal antibody potentiates kidney damage induced by IRI and that reducing the number of Treg cells results in more neutrophils, macrophages, and innate cytokine transcription in the kidney after IRI.³⁴³ Furthermore, FoxP3 (forkhead box P3)⁺ Treg cell-deficient mice accumulated a higher number of inflammatory leukocytes after renal IRI than mice containing Treg cells, and that cotransfer of isolated Treg cells and Scurfy lymph node cells significantly attenuated IRI-induced renal injury and leukocyte accumulation.³⁴³ Studies of adoptively transferred Treg cells have shown that IL-10 production, adenosine production through CD73, expression of the adenosine 2A receptor, and programmed cell death 1 (PD-1) on the cell surface are required by Treg cells to protect recipient mice from IRI.³¹⁹ Studies have demonstrated that PD-1, a negative costimulatory molecule expressed on T lymphocytes, monocytes, dendritic cells, and B cells,^{344,345} is indispensable for Treg function. Administration of monoclonal antibodies to PD-1 or a genetic deficiency of PD-1 on Tregs exacerbated impaired kidney function and ATN after subthreshold ischemia.³⁴⁶ Stremiska and coworkers have synthesized a hybrid cytokine, IL233, which combines IL-2 and IL-33.³⁴⁷ Tregs require IL-2 for homeostasis and upregulate IL-33, which promotes the recruitment and activation of innate lymphoid cells (ILC2s).³⁴⁸ Administration of the novel hybrid cytokine, IL233, increased endogenous Tregs in blood and spleen and prevented IRI more efficiently than a mixture of IL-2 and IL-33 administered separately. IL233 also increased the proportion of ILC2s in blood and kidneys, and adoptive transfer of ILC2s protected mice from IRI.³⁴⁷ Thus, the many barriers to cell-based therapy may be overcome through this hybrid cytokine, which increases endogenous Tregs. Rapid translation to human studies would be a major advancement in the field.

B Lymphocytes. Studies to evaluate the role of B cells and the B1 subset on ischemia-induced AKI have remained controversial, with some studies showing that B cells may be deleterious. Burne-Taney and associates have shown that μ MT

mice, which lack B cells and all immunoglobulins, are protected from AKI despite similar levels of infiltrating granulocytes and macrophages in the postischemic kidneys of wild type (WT) mice.³¹⁶ These investigators demonstrated that μ MT mice were rendered sensitive to ischemia-induced AKI after replenishing these mice with WT serum, but not B cells, thus indicating that a serum factor enhanced the cytotoxic effect of infiltrating granulocytes and macrophages on the ischemic tubular cells. However, Renner and colleagues, using the same μ MT mice, could not show that these mice were protected from ischemia-induced AKI.³⁴⁹ In fact, these mice had more severe injury when compared with control WT mice. Additionally, Lobo and associates failed to show protection from ischemia-induced AKI when WT mice were acutely depleted of B cells (with anti-CD20) and not immunoglobulins.³⁵⁰ More studies are needed to study the role of B cells and immunoglobulins in AKI; it may be preferable to deplete B cells or subsets of B cells acutely to study their role. μ MT mice have other immune deficiencies, including low Tregs and lack of TcR diversification, which may have contributed to more severe injury in the experiments by Thurman and colleagues.^{351,352}

The role of the B1 subset of B cells in murine models of ischemia has also been controversial. Ray, Zhang, and associates have clearly shown that natural IgM produced by B1 cells is deleterious in ischemia-induced injury of murine skeletal muscle, cardiac muscle, and bowel.^{352–354} They demonstrated that ischemia exposed nonmuscle myosin, heavy-chain type IIA and C (NMM) neoantigen in these organs and, by immunohistochemistry, they clearly demonstrated binding of natural IgM and complement to NMM in these ischemic organs. Furthermore, they showed that natural IgM was pathogenic because these same organs were resistant to ischemic injury in immunoglobulin-deficient $\text{Rag1}^{-/-}$ mice, but become susceptible to ischemia if these mice are replenished with purified IgM. On the other hand, Lobo and colleagues have shown that natural IgM protects WT kidneys from ischemic AKI by inhibiting the innate inflammation that occurs after ischemic injury.³⁵⁵ Both they and others have clearly shown that $\text{Rag1}^{-/-}$ mice are also susceptible to ischemic AKI in the absence of IgM.³⁵⁰ Kidneys from IgM knockout mice, deficient in only IgM but not B cells and other immunoglobulins or Tregs, are very sensitive to minimal ischemia that is insufficient to cause AKI in WT mice.³⁵⁵ Additionally, Renner and associates could not show binding of IgM to ischemic tubules or peritubular capillaries in WT mice. It is therefore possible that NMM, which is also present in the kidney, is not exposed after ischemic injury.³⁴⁹ Such observations also demonstrate that the mechanisms that cause ischemic injury in the kidney may be different from those of other organs.

Inflammation

Altered endothelial cell function also mediates inflammation, a hallmark of ischemic injury that has been the subject of numerous studies. Ischemia induces the increased expression of a number of leukocyte adhesion molecules, such as P-selectin, E-selectin, and intercellular adhesion molecule (ICAM) and B7-1. Consequently, it has been shown that strategies to block pharmacologically or genetically ablate the expression of these molecules are protective against ischemic or septic AKI.³¹¹ Investigators have also shown that

T cells play a major role in vascular permeability during ischemic injury. Gene microarray analysis has shown that the production of TNF- α and IFN- γ protein is increased in CD3 and CD4 T cells from the blood and kidney after ischemia. Furthermore, it has also been demonstrated that in CD3, CD4, and CD8 T cell-deficient mice, there is a significantly attenuated rise in renal vascular permeability after ischemic injury. Hence, T cells directly contribute to the increased vascular permeability, potentially through T cell cytokine production.^{342,356} Another feature noted during inflammation and endothelial cell injury is the phenomenon of erythrocyte trapping with rouleaux formation, prolonging the reduction in renal blood flow and exacerbating tubular injury.³⁵⁷ Studies have also shown a role of the sphingosine 1-phosphate receptor (S1PR) in maintaining structural integrity after AKI. Thurman and coworkers have shown that S1PRs in the proximal tubule are necessary for stress-induced cell survival, and S1PR agonists are renoprotective via direct effects on tubular cells.^{358,359}

DNA microarray analysis of ischemic kidneys from TLR4-sufficient and TLR4-deficient mice has shown that pentraxin 3 (PTX3), an endothelial induced protein, is upregulated only in TLR4-sufficient mice. Transgenic knockout of PTX3 ameliorated AKI. PTX3 was shown to be expressed predominantly on the peritubular endothelia of the outer medulla of the kidney in control mice and AKI increased PTX3 protein in the kidney and the plasma. Stimulation studies performed in primary renal endothelial cells have suggested that endothelial PTX3 is induced by pathways involving TLR4 and ROS and demonstrated that these effects could be inhibited by conditional endothelial knockout of MyD88. Compared with WT mice, PTX3 knockout mice had decreased endothelial expression of cell adhesion molecules at 4 hours of reperfusion, possibly contributing to a decreased early maladaptive inflammation in the kidneys of knockout mice, whereas later, at 24 hours of reperfusion, PTX3 knockout increased the expression of endothelial adhesion molecules when regulatory and reparative leukocytes enter the kidney. Thus, endothelial PTX3 plays a pivotal role in the pathogenesis of ischemic AKI.³⁶⁰

The role of glomerular endothelial injury in AKI is unclear. Studies in a mouse model of LPS-induced sepsis have shown decreased abundance of endothelial surface layer heparan sulfate proteoglycans and sialic acid, leading to albuminuria, likely reflecting altered glomerular filtration permselectivity and decreased expression of VEGF. LPS treatment also decreased the GFR, caused ultrastructural alterations in the glomerular endothelium, and lowered the density of glomerular endothelial cell fenestrae. These LPS-induced effects were diminished in TNF receptor 1 (TNFR1) knockout mice, suggesting the role of TNF- α activation of TNFR1, and intravenous administration of TNF also led to a decreased GFR and loss of glomerular endothelial cell fenestrae, increased fenestrae diameter, and damage to the glomerular endothelial surface layer. Thus, glomerular endothelial injury, mediated by higher TNF- α and lower VEGF levels, extends the development and progression of AKI and albuminuria in the LPS model of sepsis in the mouse.³⁶¹

In addition to immune cells responding and sensing DAMPs and PAMPs (see later), proximal tubule cells (PTCs) function as a sensor of both self and nonself; DAMPs and PAMPs serve as recognition signals for pattern recognition receptors (PRRs)

such as TLR4.³⁶² Proximal tubule TLR4 is upregulated and migrates to the apical domain in response to LPS in S1 PTCs, which are the earliest segments of epithelial cell postglomerular filtration.³²¹ Interestingly, the S1 cell internalizes and processes LPS via TLR4 receptors, which is inducible with preexposure to LPS but is protected from injury by upregulated defense mechanisms, including heme oxygenase-1 (HO-1) and sirtuin 1 (SIRT1), two cytoprotective proteins. However, S2 to S3 PTCs undergo oxidative injury with minimal uptake of LPS, implying communication, crosstalk, and coregulation between the segments following LPS exposure.³⁶³ This injury is dependent on CD14, likely due to peroxisomal disruption, perhaps mediated by TNF- α , and the PTC injury was found to be independent of systemic cytokines.

Another role for epithelial TLR4 and MyD88 in mediating ischemic injury has been shown by Wu and colleagues.³⁰⁹ In addition, in these studies, the relative contribution of epithelial versus hematopoietic TLR4 to kidney damage following IRI was assessed using bone marrow chimeras in which TLR4^{-/-} mice were engrafted with WT hematopoietic cells (and vice versa). Both hematopoietic and parenchymal TLR4 contributed to kidney injury, although the effect was more pronounced when TLR4 was expressed only on parenchymal cells. These results suggest that TLR4 signaling on hematopoietic and intrinsic kidney cells contributes to mediating kidney damage but a more significant role is played by intrinsic kidney cells. Similar results with TLR2 knockout mice and chimeric mice have suggested that epithelial TLR2 plays a prominent role during ischemic injury.³⁶⁴ Cytokine and chemokine production was reduced and white blood cell (WBC) infiltration was minimized in chimeric mice using antisense therapy. These studies suggest that renal-associated TLR2 is an important initiator of inflammatory responses that lead to renal injury.

Cytokines and chemokines released by PTCs in response to cell injury have direct effects on endothelial function. Using two-photon microscopy, investigators recorded cellular and physiologic responses to fluorescent cytopathologic *Escherichia coli* microinjected into early proximal tubule segments.^{365–367} Attachment to the apical membrane, but without penetration into or through the PTC monolayer barrier, resulted in rapid and selective termination of blood flow to the adjacent area, and thus resulted in vascular isolation of the infected area with localized hypoxia, leukocyte migration, and necrosis. The same *E. coli* strain, missing only one virulence factor, required a far longer time to initiate this protective process. Tissue concentrations of cytokines were markedly elevated in the affected area compared with none in the injected areas.³⁶⁷ Finally, prevention of this microvascular response resulted in widespread organ dissemination of the injected *E. coli* and death of the rat within 24 hours, something not seen with the intact system.³⁶⁷ Therefore, communication between PTC and endothelial cells may lead to localization of the infecting agent and prevention of systemic spread.

Tamm-Horsfall protein (also known as *uromodulin*), a heavily glycosylated protein uniquely produced in the kidney by TALs,^{368,369} modulates kidney innate immunity and inflammation during kidney injury.^{362,368,370} Uromodulin knockout mice compared with WT controls subjected to kidney IRI showed increased S3 injury and necrosis,^{370,371} neutrophil infiltration in the outer medulla, and expression of TLR4

and CXCL2 by S3 segments.^{370,371} Neutralization of CXCL2 was protective, suggesting that a TLR4-CXCL2 proinflammatory pathway may be important in the pathophysiology and supporting uromodulin-dependent TAL-S3 crosstalk. Indeed, after IRI, a shift of trafficking of uromodulin was demonstrated toward the interstitium and basolateral aspects of S3 segments,³⁷² where a putative receptor for uromodulin is expressed.³⁷⁰ This translocation of uromodulin was not the result of altered polarity of TAL. Uromodulin knockout (THP^{-/-}) mice, in addition to worse injury compared with WT mice, had an impaired transition of renal macrophages toward an M2 healing phenotype, suggesting that interstitial THP may not only regulate mononuclear phagocyte number, but plasticity and phagocytic activity as well.³⁷³ A significant increase in uromodulin expression was shown in the kidney at the onset of recovery, which was concomitant with the suppression of tubular-derived cytokines and chemokines such as MCP-1, supporting the concept that the protective crosstalk mediated by uromodulin may be important in modulating recovery from AKI.³⁷²

COMPLEMENT

The complement system is part of the host defense machinery that protects against microbial invasion after injury.³⁷⁴ The reactivity and specificity of the complement system is accomplished via a series of circulating pattern recognition proteins (PRPs) that sense PAMPs and initiate the complement cascade. The immunomodulatory functions of complement are mediated through three canonical pathways of activation—classic pathway, alternative pathway, and lectin pathway. The classic pathway is initiated by binding of PRPs to immune complexes, whereas the lectin pathway is initiated by binding PRPs to carbohydrate structures exposed in injured cells. The alternative pathway amplifies the initial response and maintains a low level of activity via a tick-over mechanism. On recognition of foreign surfaces, PRP-associated serine proteases cleave soluble components deposited on the activating surface and form C3 convertase complexes. Subsequent convertase-mediated cleavage of C3 leads to the formation of C3 fragments that serve as ligands for a variety of complement receptors that mediate increased phagocytosis and stimulation of the adaptive immune response. In addition to C3 convertase formation, opsonization also leads to the formation of C5 convertase, which activates C5 and initiates the formation of the pore-forming membrane attack complex (MAC) that lyses susceptible microorganisms or damages cells.³⁷⁵

Previous studies have suggested that complement is only located in the intravascular space, with most components synthesized in the liver. Many reports have now documented the presence of a functionally intact intracellular complement system within lymphocytes and epithelial, endothelial, and other cell types, and several studies have identified C3- and C5-mediated activation and signaling events in the intracellular space.³⁷⁶

Complement activation causes kidney injury through direct effects on renal cells and through interactions with cells of the innate and adaptive immune systems. Small soluble peptides, named “anaphylatoxins” (C3a and C5a) are generated during complement activation. These fragments trigger a systemic inflammatory response through their receptors, including vascular changes and chemotaxis of immune cells.³⁷⁷ C3a and C5a receptors are expressed on leukocytes, endothelial cells,

mesangial cells, and tubular epithelial cells.³⁷⁸ During ischemic injury, CR5a expression is markedly upregulated on proximal tubule epithelial cells, as well as interstitial macrophages. C5a is a powerful chemoattractant that recruits inflammatory cells. Complement cascades are activated during sepsis, and C5a, a potent complement component with procoagulant properties, is elevated in rodent models of sepsis. Blocking C5a or its receptor has shown some promise in improving survival with sepsis.³⁷⁹

To protect host cells from uncontrolled complement activation, several complement regulatory proteins are expressed in the surface of kidney cells and can directly inactivate complement convertase. Previous studies have documented complement activation in the pathogenesis of IRI-induced AKI.³⁸⁰ These investigators used mice with a global deficiency to factor B or antibodies to factor B to demonstrate that the inhibition of C3 formation in kidney tissue protects mice from IRI-induced AKI. In addition to AKI, recent studies have demonstrated increased expression of complement C3 fragments and anaphylatoxin receptors C3aR and C5aR1 in PDGFR β -positive pericytes and immune cells that were isolated from fibrotic kidneys. Inhibition of complement activation using mice with a global deletion of C3 has resulted in reduced tubulointerstitial fibrosis.³⁸¹ Given the increased awareness of the role of complement activation, not only with AKI but also in models of CKD, it is likely that emerging therapies aimed at inhibiting complement activation could be used in future studies to reduce the evolution of AKI and/or the progression of CKD.

INTRACELLULAR MECHANISMS

Reactive Oxygen Species

Oxidative stress plays an important role in the pathogenesis of AKI and progressive kidney disease. Low-level ROS can function as signaling molecules for cellular proliferation and vascular homeostasis under healthy conditions; however, increased ROS generation during IRI by mitochondria, nicotinamide adenine dinucleotide phosphate (NADPH) oxidases, or inflammatory cells can aggravate tissue injury. Therefore, reducing oxidative stress is considered an important therapeutic strategy to ameliorate loss of kidney function.³⁸²

ROS such as OH $^{\cdot}$, peroxynitrite (ONOO $^{-}$), and hypochlorous acid (HOCl) are generated in epithelial cells during ischemic injury by catalytic conversion. These ROS can damage cells in a variety of ways (e.g., via peroxidation of lipids in plasma and intracellular membranes). They can also destabilize the cytoskeletal proteins and integrins required to maintain cell-cell adhesion, as well as extracellular matrix. These ROS can also have vasoconstrictive effects by their capacity to scavenge NO.³⁸³

Heat Shock Proteins

Much of the previous discussion has been on proteins or mechanisms that promote injury. However, there are protective mechanisms that allow cells to have a defense against numerous stresses. The complex heat shock protein (HSP) system is induced to exceptionally high levels during stress conditions. The HSPs are believed to facilitate the restoration of normal function by assisting in the refolding of denatured proteins, along with aiding the appropriate folding of newly synthesized

proteins. They also help in the degradation of irreparable proteins and toxins to limit their accumulation. Overexpression of HSPs before injury has protective effects.^{384–386} The proteins HSP90, HSP72, and HSP25 in particular have been extensively studied (e.g., overexpression of HSP25 is protective against actin cytoskeleton disruption).³⁸⁷ After renal ischemia, cytosolic HSP90 is rapidly induced in PTCs, particularly in late stages, leading to the conclusion that HSP90 may be crucial for the disposition of damaged proteins and the assembly of newly formed peptides. Intrarenal transfection with HSP90 protects against IRI with the restoration of endothelial nitric oxide synthase (eNOS)–HSP90 coupling, eNOS activating phosphorylation, and Rho kinase levels, suggesting that HSPs can regulate the NO–eNOS pathway and intrarenal vascular tone.³⁸⁸ In nephrotoxic models, HSP72 has been shown to limit apoptosis through an increased Bcl-2/Bax ratio, implicating HSP72 in cell death as well.³⁸⁷

Nayak's group has described the role of myoinositol oxygenase (MIOX), a proximal tubule enzyme that participates in oxidant injury by increasing the generation of ROS in kidney tissue.³⁸⁹ In a more recent study,³⁹⁰ the investigators used the cisplatin model of AKI in tubule-specific MIOX-overexpressing transgenic (MIOX-TG) mice and MIOX knockout (MIOX $^{-/-}$) mice. Compared with cisplatin-treated WT mice, cisplatin-treated MIOX-TG mice had more pronounced kidney injury, evidenced by higher serum concentrations of urea, creatinine, and kidney injury molecule 1 (KIM-1), along with more prominent apoptosis, but these effects were attenuated in cisplatin-treated MIOX $^{-/-}$ mice. MIOX $^{-/-}$ mice also had reduced NADPH oxidase-4 expression and ROS generation after cisplatin treatment. Increased inflammatory cells and upregulation of cytokines were found in kidneys of cisplatin-treated MIOX-TG mice. These findings suggest that MIOX overexpression leads to worsening of AKI in cisplatin-induced injury via multiple mechanisms and that MIOX gene disruption ameliorates toxic PTC injury.

To protect cells from harmful oxidative stress, the Keap1-Nrf2 system allows the cell to sense and respond to oxidative stress. The Keap1-Nrf2 pathway is a master regulator of cytoprotective genes and a promising target for therapeutic intervention. Nrf2 (nuclear factor–like 2) is a transcription factor that can bind antioxidant response elements (AREs) in target gene regulatory regions. In kidney IRI, Nrf2 signaling is upregulated in response to injury, and Nrf2 null mice developed worse disease.³⁹¹ It was also found that bardoxolone methyl ameliorates IRI through Nrf2.³⁹²

Given the known adverse effects of using pharmacologic enhancers of the Keap1/Nrf2 pathway,³⁹³ a recent study used genetically altered mice that express low levels of Keap1 protein.³⁹⁴ Reduction in levels of this inhibitor of Nrf2 signaling leads to enhanced Nrf2 target transcription. The investigators demonstrated a protective effect of Nrf2 enhancement in the model of ischemia-mediated AKI, as well as in the model of unilateral ureteral obstruction. These results underscore the importance of developing drugs that selectively target increased Nrf2 transcription to ameliorate AKI.

Iron, Ferritin, and Heme Oxygenase

Iron plays a central role in fundamental biologic functions, such as in mitochondrial respiration and DNA repair, yet at the same time plays a detrimental role in the pathophysiology of various diseases.³⁹⁵ Circulating iron is mostly transferrin-bound.

A small amount bound to low-molecular-weight chelates, referred to as *labile* or *catalytic iron* is available in biologic systems and is thought to be responsible for kidney injury.³⁹⁶ Although iron must undergo cyclic oxidation and reduction to perform its normal functions, this redox activity generates free radicals, leading to lipid peroxidation or generation of superoxide radicals through reaction with hydrogen peroxide (Haber-Weiss reaction).³⁹⁷ The toxicity of catalytic iron to macromolecular components of the cell and its causal role in mediating disease have been demonstrated through the protection achieved by iron chelators.^{62,396,398–402} Heme iron is derived primarily from erythrocyte turnover and is the major contributor to iron cycling in the body.³⁹⁵

Kidney tissue iron content increases after AKI,^{403–406} suggesting that regulators of iron metabolism such as hemojuvelin and hepcidin could be used as therapeutic targets for the treatment of AKI. In a recent study, Scindia and coworkers reported that IRI caused an increase in serum iron and kidney nonheme iron levels.⁴⁰⁴ They further demonstrated a significant reduction in IRI-induced apoptosis, oxidative stress, and inflammatory cell infiltration when hepcidin was administered 24 to 48 hours before IRI. Similar protection by the use of hepcidin was reported in the model of hemoglobin-mediated AKI.⁴⁰⁷ Ferritin is the major regulator of intracellular iron and is composed of heavy-chain ferritin (H-ferritin) and light-chain ferritin (L-ferritin).⁹⁸ H-ferritin has ferroxidase activity converting Fe^{2+} to Fe^{3+} , which permits the incorporation of reactive iron into ferritin to avoid iron-induced ROS generation. Mice with proximal tubules deficient of H-ferritin had higher mortality, worse tissue injury and renal function, and increased apoptosis in both cisplatin and rhabdomyolysis models of AKI. Notably, there was disrupted iron trafficking, with altered distribution of ferroportin (iron export transporter) in the proximal tubule of H-ferritin-deficient mice. These results support an important role of proximal tubule H-ferritin and iron trafficking during AKI.⁹⁸

Heme Oxygenase-1

The enzyme HO-1 has also emerged as a prominent player in renal tubular epithelial cell injury.^{408–411} The biologic actions of HO-1 include antiinflammatory, vasodilatory, cytoprotective, and antiapoptotic effects and regulation of cellular proliferation in the setting of AKI. HO-1 is arguably one of the most readily inducible genes, responding to numerous stressors, including, but not limited to, hypoxia, hyperthermia, oxidative stress, and exposure to LPS. Induction of HO-1 has been described in various forms of AKI, including ischemic, endotoxic, and nephrotoxic models. Following ischemia-reperfusion, aged mice exhibit reduced induction of renal medullary HO-1 and worse renal injury than younger mice. A number of studies have indicated a protective effect of induction of HO-1 in AKI.^{408–411} Prior induction of HO-1 by hemoglobin can reduce endotoxemia-induced renal dysfunction and mortality. Inhibition of HO-1 activity in the intact, disease-free kidney reduces medullary blood flow without exerting any effect on cortical blood flow. Overexpression of HO-1 by hemin results in a significant reduction in cisplatin-induced cytotoxicity,^{412,413} and $\text{TNF-}\alpha$ -induced apoptosis in endothelial cells is also attenuated by the induction of HO-1.

These findings have been supported by studies of HO-1-deficient mice where glycerol-induced AKI resulted in marked exacerbation of renal insufficiency and mortality.⁴¹⁴ Mechanisms of HO-1-mediated protection have been extensively

studied by Inguaggiato and coworkers, who showed that overexpression of HO-1 in cultured renal epithelial cells induces upregulation of the cell cycle inhibitory protein p21 and confers resistance to apoptosis.⁴¹⁵ Recently, HO-1 has been shown to attenuate ferroptosis in proximal tubular cells.⁴¹⁶ Induction of ferroptosis (erastin and RSL3) was less profound in HO-1^{+/+} compared with HO-1^{-/-} PTCs. Treatment with ferrostatin-1, a ferroptosis inhibitor, significantly improved viability. Macrophages in which HO-1 is upregulated by adenoviral strategies also protect against ischemic AKI. Fibroblasts from organs of transgenic pigs expressing HO-1 are resistant to proapoptotic stressors and exhibit a blunted proinflammatory response to LPS or $\text{TNF-}\alpha$. Pretreatment with hemin augments glomerular HO-1 expression and renal expression of thrombomodulin and endothelial cell protein C receptor (EPCR) while reducing LPS-induced renal dysfunction, glomerular thrombotic microangiopathy, and the procoagulant state. Hemin also increases plasma levels of activated protein C in this model, suggesting its important role in the endothelial-epithelial axis in AKI.

Perhaps more importantly, HO-1 might also contribute to the repair and regeneration of tubular cells.^{101,417} Following an acute insult, HO-1 is rapidly induced, but its expression subsides before renal recovery fully occurs; such abatement in HO-1 expression may allow the continued expression of proinflammatory and fibrogenic genes. In this regard, HO-1 deficiency promotes epithelial-mesenchymal transition, a process that may underlie the transition of AKI to CKD.⁴⁰⁹ HO-1 gene expression is regulated differently in mice and humans; however, a recent study with a novel humanized transgenic mouse has confirmed the rescue of the pathologic phenotypes observed in HO^{-/-} mice by the human HO-1 gene.⁴⁰⁸ These mice offer an important tool to study the mechanisms of regulation of human HO-1 gene. In addition, the protective effects of HO-1 promoter polymorphisms in AKI could allow us to understand its significance better in clinical contexts and enable the identification of potential novel therapeutic targets.

REPAIR AND REGENERATION

The human kidney subjected to mild injury has the ability to recover renal function, as assessed by serum creatinine levels. Experimentally, the renal tubule has the capacity to undergo regeneration within a few days after AKI.⁴¹⁸ Early studies that focused on progenitor-stem cells in tubular epithelial cell injury found that different types of stem cells may reside in the renal architecture. In the human kidney, CD133⁺ progenitor-stem cells with regenerative potential have been identified.^{419–423} Chimeric studies using enhanced GFP to label donor bone marrow have revealed the lack of significant contribution of hematopoietic cells in the repair of epithelium following injury.⁴²⁴

Currently, there are likely two major hypotheses to explain the recovery of renal function by tubule regeneration⁴²⁵ (Fig. 28.10). The first hypothesis suggests that tubules regenerate from any surviving tubular epithelial cells without the contribution of a preexistent intratubular stem cell or progenitor population. Genetic fate mapping studies have demonstrated that regeneration following injury is achieved primarily by surviving epithelial cells.^{426,427} Lineage-tracing studies have demonstrated that fully differentiated epithelial cells that

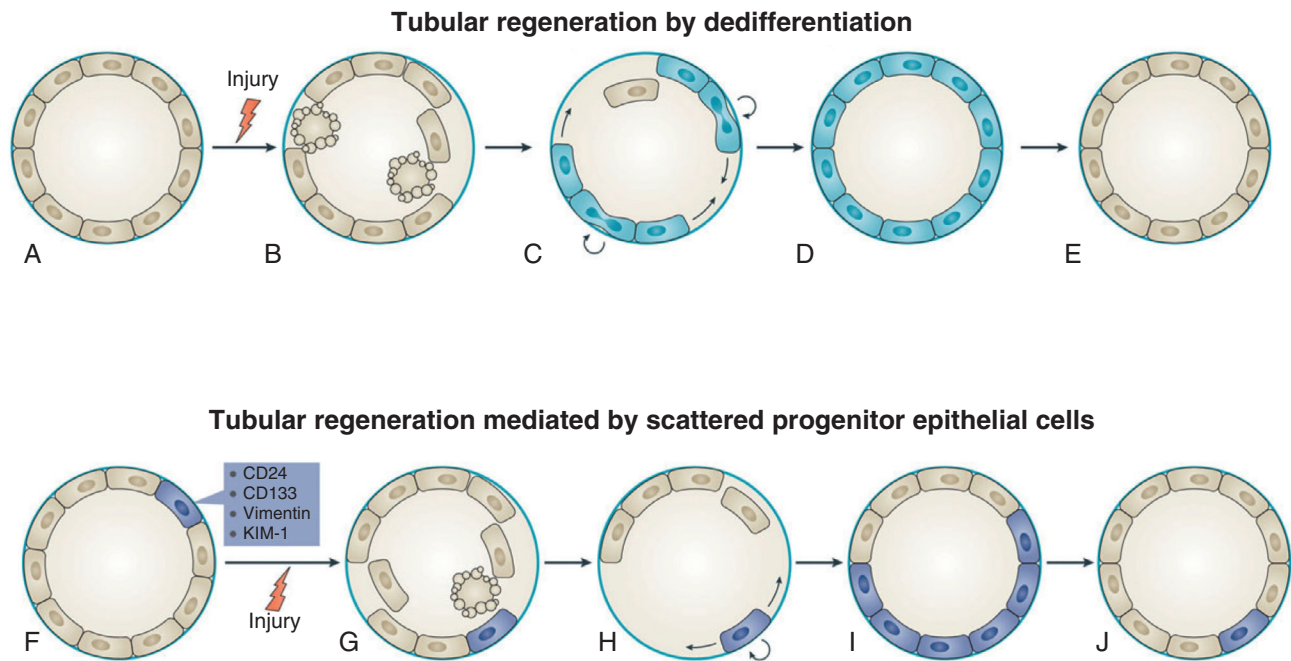


Fig. 28.10 Tubular regeneration following acute kidney injury. *Top panel*, Tubular regeneration by dedifferentiation. (A) Healthy tubules consist of nonproliferative mature epithelial cells that express markers of differentiation. (B) After injury, the epithelium is lost through apoptosis and necrosis. (C) Surviving epithelial cells dedifferentiate, either in response to sublethal injury signals or owing to signals from other injured cells, and acquire a proliferative phenotype. (D) The surviving dedifferentiated cells reconstitute the nephron epithelium. (E) Ultimately, most dedifferentiated cells redifferentiate and downregulate the expression of dedifferentiation genes. *Lower panel*, Tubular regeneration mediated by scattered progenitor epithelial cells. (F) Scattered tubular epithelial cells express signal transducer CD24, prominin-1 (CD133), and other genes that are characteristic of proximal tubule dedifferentiation, such as vimentin and KIM-1. (G) After injury, mature tubular cells, but not scattered cells, undergo apoptosis. (H, I) Scattered tubular cells expand in response to injury and their progeny reconstitute the tubule. (J) The small subpopulation of scattered tubule cells is preserved after regeneration. (From Chang-Panesso M, Humphreys BD. Cellular plasticity in kidney injury and repair. *Nat Rev Nephrol.* 2017;13(1):39–46.)

survive an acute kidney injury episode undergo a process of reversible dedifferentiation and proliferation during repair.⁴²⁸ Another independent group used the same methodology of genetic cell mapping and doxycycline-inducible parietal epithelial cell mapping (PEC)–specific transgenic mouse to label proximal tubules in the model of IRI. The authors found a significant increase in genetic labeling of proximal tubules, with increased expression of scattered tubular cell biomarkers such as CD24 and CD133, during ischemia and subsequent recovery, suggesting that scattered tubular cells arise from any surviving tubular cell.⁴²⁹

The second hypothesis suggests that tubules regenerate from a specific tubular cell subpopulation with high regenerative potential or so-called *scattered tubular cells*^{428,430–432} (see Fig. 28.10). A recent study supports this hypothesis and challenges the current paradigm that functional recovery after AKI relates to a regenerative capacity of all tubular epithelial cells. These investigators applied a lineage tracing approach using conditional Pax8-Confetti mice and the models of ischemia-reperfusion and glycerol-induced AKI to demonstrate the following: (1) AKI involves a permanent loss of tubular epithelial cells (TECs), even when GFR recovery occurs; (2) Pax2-positive TECs located in the S3 segment of the proximal tubule are endowed with a higher resistance to death and are responsible for the spontaneous regeneration of necrotic tubules after AKI; (3) only Pax2-positive cells complete mitosis while other TECs go through a process of endoreplication-mediated hypertrophy that accounts for the

recovery of renal function; and (4) this process of endoreplication is the dominant cell response on AKI in mice and can be detected in renal biopsy tissue of patients who developed CKD after AKI.⁴³³ The therapeutic targeting of tubule progenitors carrying specific biomarkers of scattered tubular cell phenotype could be a promising new strategy to improve long-term outcomes after AKI.

Growth factors and signals from injured cells are crucial at this stage to promote the timely and appropriate regenerative capacity of the viable cells. In animal models, administration of exogenous growth factors accelerates renal recovery from injury. These include epidermal growth factor, IGF-1, α -melanocyte stimulating hormone, erythropoietin, hepatocyte growth factor, and bone morphogenic protein-7 (BMP-7).^{434–437} These effects have not yet been validated in human clinical trials of ATN.^{438,439} They all likely increase the GFR through direct hemodynamic effects and may therefore hasten tubular epithelial cell recovery.

Extracellular vesicles (EVs) derived from bone marrow mesenchymal stromal cells promote the regeneration of kidneys in models of AKI.⁴⁴⁰ EVs represent a heterogeneous group of vesicles derived from nearly all mammalian cells. Smaller vesicles or exosomes range in size from 30 to 100 nm and larger microparticles from 100 to 1000 nm.⁴⁴¹ Microvesicles administered to severe combined immunodeficient (SCID) mice subjected to glycerol-induced AKI induced proliferation of tubule cells and accelerated morphologic and functional recovery.⁴⁴² Tubule proliferation was mediated by

RNA-dependent effects because RNase abolished the effects of microvesicles on this process.⁴⁴² When various populations of EVs were isolated by differential centrifugation, Bruno and coworkers found that transfer of 10 K (isolation at 10,000 g) and 100 K (isolation at 100,000 g) preparations into mice had different effects on AKI. The 100 K EVs contained mRNAs regulating proliferative, antiapoptotic, and growth factors, whereas the 10-K EVs had lower expression of proproliferative molecules and were unable to induce renal regeneration.⁴⁴⁰

Microvascular Function

The microvasculature consists of the endothelium and pericyte. Each of these structures contributes to barrier integrity for normal homeostatic function. During AKI, disruption may occur at each of these sites, leading to increased permeability and cell death. Normally, the microvasculature controls vascular tone, regulation of blood flow to local tissue beds, modulation of coagulation and inflammation, and permeability. Both ischemia and sepsis have profound effects on the endothelium. The renal vasculature and endothelium are particularly sensitive to these insults. When such an insult occurs, the endothelial bed becomes ineffective in performing its function, and the ensuing vascular dysregulation leads to continued ischemic conditions and further injury following the initial insult, which has been termed the “extension phase” of AKI.⁴⁴³ Histopathologically, this is seen as vascular congestion, edema formation, diminished microvascular blood flow, and margination and adherence of inflammatory cells to endothelial cells.

Vascular Tone. Conger and associates were among the first to demonstrate that postischemic rat kidneys manifest vasoconstriction in response to decreased renal perfusion pressure and hence cannot autoregulate blood flow, even when total renal blood flow had returned to baseline values up to 1 week after injury.^{444,445} Single-fiber, laser Doppler flow cytometry has revealed that blood flow is reduced to 60% and 16% of preischemic values in cortex and medulla, respectively and increased 125% in inner medulla following ischemia.⁴⁴⁶ Selective inhibition, depletion, or deletion of iNOS has clearly shown renoprotective effects during ischemia.^{447,448} NO production from the endothelium (eNOS) may be impaired at the level of enzyme activity or modified by ROS to impair normal vasodilatory activity.⁴⁴⁹ In ischemic AKI,⁴⁵⁰ there is an imbalance of eNOS and iNOS, and a relative decrease in eNOS, secondary to endothelial dysfunction and damage, leads to a loss of antithrombotic properties of the endothelium and increased susceptibility to microvascular thrombosis.⁴⁵⁰ Administration of L-arginine, the NO donor molsidomine, or the eNOS cofactor tetrahydrobiopterin can preserve medullary perfusion and attenuate AKI induced by IRI. Conversely the administration of *N*^o-nitro-L-arginine methyl ester, an NO blocker, aggravates the course of AKI following IRI.^{451,452}

In a review of experimental sepsis, the pattern of renal blood flow is inconsistent. Renal blood flow was decreased in 62% of studies and unchanged or increased in 38%.⁴⁵³ Langenberg and colleagues administered *E. coli* to sheep and induced hyperdynamic sepsis associated with increased cardiac output and a decrease in mean arterial pressure and AKI, as evidenced by a rise in creatinine levels.⁴⁵³ Furthermore,

there was marked increase in renal blood flow, decrease in urine output, and decrease in creatinine clearance. When angiotensin was dose-titrated and restored blood pressure to presepsis levels, these effects were reversed; there was a significant decrease in renal blood flow, increase in urine output, and increase in creatinine clearance.⁴⁵⁴ These studies suggest that improvement of function may be more than simply an improvement in mean arterial pressure and that the increase in glomerular filtration pressure may be through selective constriction of efferent arterioles and/or increase in mesangial cell tone.⁴⁵⁴ Based on these principles, in a recent clinical trial, patients with vasodilatory shock who were receiving another vasopressor were randomized to Ang II or placebo. The group that received Ang II had higher mean blood pressure⁴⁵⁵ and, in a subgroup of patients with AKI who required renal replacement therapy, 28-day survival, mean arterial pressure, and rate of discontinuation of renal replacement were higher in the angiotensin group. These results together show the potential benefit of Ang II in hyperdynamic sepsis, which may in part be due to improved glomerular hemodynamics.

The endothelial cell structure consists of a cytoskeletal structure of actin filament bundles that form a supportive ring around the periphery, along with the adhesion complexes that provide the integrity of the endothelial layer. The assembly and disassembly of actin filaments is regulated by a large family of actin-binding proteins, including ADF-cofilin. With ischemic injury, the normal architecture of the actin cytoskeleton is markedly changed, along with endothelial cell swelling, impaired cell-cell and cell-substrate adhesion, and loss of tight junction barrier functions. ATP depletion of cultured endothelial cells has been shown to induce dephosphorylation and activation of ADF-cofilin in a direct and concentration-dependent fashion. This activity results in depolymerized and severed actin filaments, seen as F-actin aggregates at the basolateral aspects of the cell.⁴⁵⁶

Kidney pericytes are extensively branched cells that are scattered on the outer wall of capillaries and embedded in the capillary basement membrane; they serve a homeostatic role in angiogenesis and vessel maturation.⁴⁵⁷ Furthermore, following kidney injury, they detach and migrate into the kidney interstitium⁴⁵⁸ and, in some cases, transform into myofibroblasts, leading to progressive kidney fibrosis. Thus, the endothelial-pericyte crosstalk among the vascular endothelial growth factor receptors and platelet-derived growth factor receptors contribute to normal microvascular health and disease. The importance of kidney pericyte microvascular function has been demonstrated through the use of diphtheria toxin delivery to FoxD1Cre::RsDTR transgenic mice in which the diphtheria toxin receptor (heparin-binding epidermal growth factor receptor) was selectively expressed in FoxD1 cells, which are primarily pericytes.⁴⁵⁹ The administration of diphtheria toxin selectively ablated pericytes only. Within 96 hours, there was an abrupt decline in renal function and an increase in albuminuria. These results demonstrate that pericytes are essential for normal microvascular function.

The glycocalyx is located on the blood side of the endothelium and has a depth of 1 to 3 μ m. The glycocalyx is composed of cell-bound proteoglycans, glycosaminoglycan side chains, and sialoproteins.⁴⁶⁰ Sandwiched between blood components and the endothelium, the glycocalyx is situated to play a key role in microvascular and endothelial homeostasis

and endothelial physiology.⁴⁵⁹ The specific functions served by the endothelial glycocalyx include shear stress mechanotransduction to endothelial cells, regulation of vascular permeability, inhibition of intravascular thrombosis, and protection of the endothelium from platelet and leukocyte adhesion.^{460,461} The glycocalyx may be damaged by local changes in shear stress, ROS, altered oncotic pressure, altered fluid composition, and inflammatory molecules. Shedding

of the glycocalyx occurs, exposing the endothelial adhesion molecules to circulating immune cells and leading to immune cell activation and inflammation and lipoprotein leakage to the subendothelial space. Coronary ischemia-reperfusion leads to shedding of the glycocalyx and an increase in neutrophil adhesion with subsequent myocardial injury.⁴⁶² Similar events are likely to occur in septic AKI and kidney IRI (Fig. 28.11).

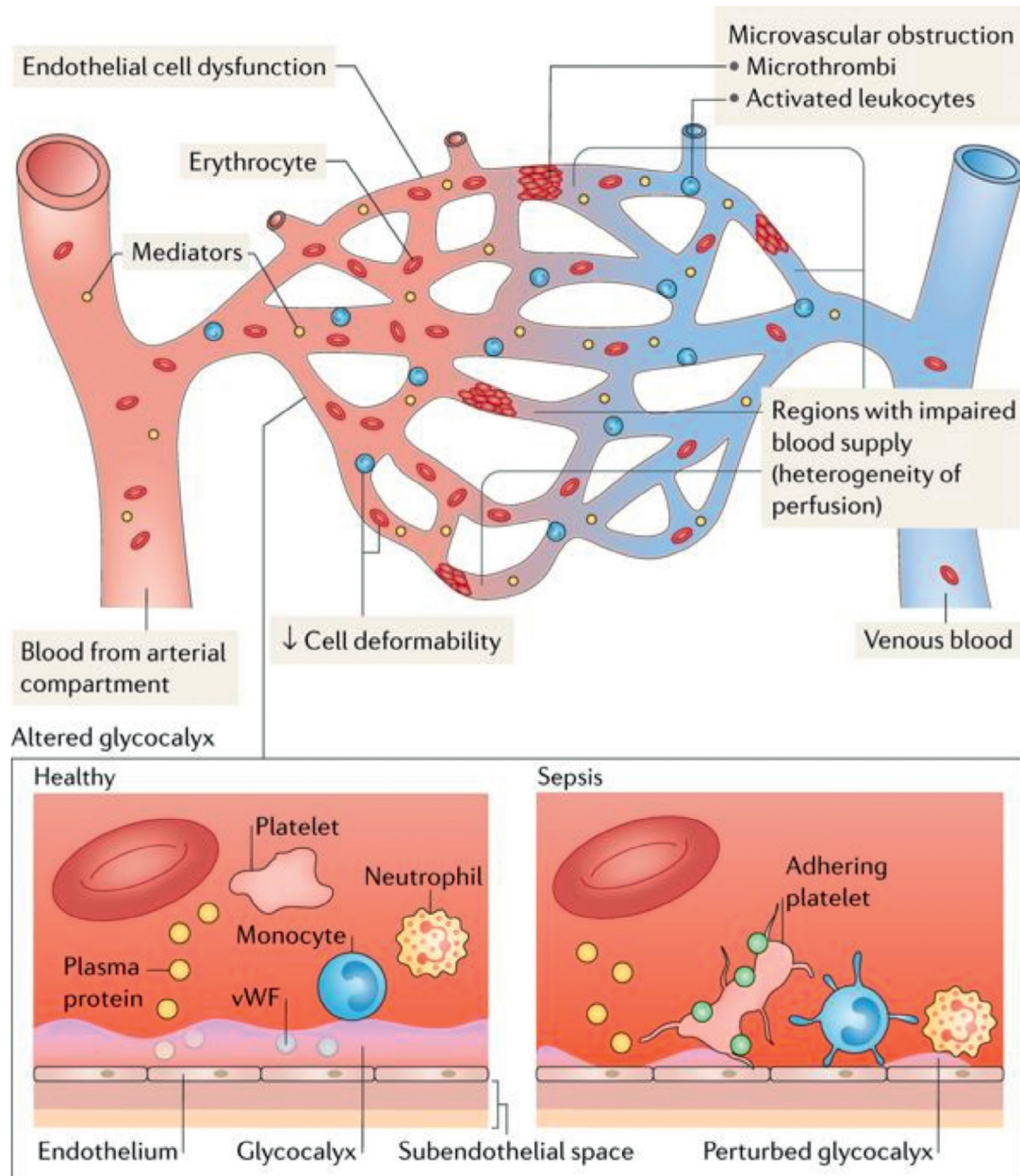


Fig. 28.11 Microvascular dysfunction in septic acute kidney injury. Multiple mechanisms are involved in the development of sepsis-related microvascular dysfunction, among which endothelial dysfunction (related partly to circulating host-derived and pathogen-derived mediators, as well as to reactive oxygen species) and an altered glycocalyx have major roles. The glycocalyx is a thin layer of glycosaminoglycans that covers the endothelial surface, facilitating the flow of red blood cells and limiting the adhesion of leukocytes and platelets to the endothelium. The glycocalyx may be substantially altered during sepsis, so interactions between the vascular endothelium and circulating cells (e.g., leukocytes and platelets) are impaired, and leukocyte rolling and adhesion to the endothelium may occur. Activation of coagulation and the generation of microthrombi might also participate in sepsis-induced microvascular alterations, as well as alterations in erythrocyte deformability and/or their adhesion to the endothelium. All these phenomena might cause heterogeneity in microvascular blood flow, with a decrease in vascular density and nonperfused capillaries, resulting in an increased diffusion distance for oxygen and in alterations in oxygen extraction. vWF, von Willebrand factor. (From Lelubre, C, Vincent, JL: Mechanisms and treatment of organ failure in sepsis. *Nat Rev Nephrol.* 2018;14:417–427.)

Within 24 hours after ischemic injury, there is loss of localization in vascular endothelial cadherin immunostaining, suggesting severe alterations in the integrity of the adherens junctions of the renal microvasculature.⁴⁶³ In vivo two-photon imaging has demonstrated a loss of capillary barrier function within 2 hours of reperfusion, as evidenced by leakiness of high-molecular-weight dextrans (>300,000 Da) into the interstitial space.

Critical constituents of the perivascular matrix, including collagen IV, are known to be substrates of matrix metalloproteinase-2 (MMP-2) and MMP-9, which are collectively known as gelatinases. Breakdown of barrier function may also be due to MMP-2 or MMP-9 activation, and this upregulation is temporally correlated with an increase in microvascular permeability.^{443,463} MMP-9 gene deletion stabilizes microvascular density following ischemic AKI, in part by preserving tissue VEGF levels.⁴⁶⁴ In addition, minocycline, a broad-based MMP inhibitor and the gelatinase-specific inhibitor ABT-518, both ameliorated the increase in microvascular permeability in this model. Taken together, many studies have indicated that the loss of endothelial cells following ischemic injury is not a major contributor to altered microvascular permeability, although renal microvascular endothelial cells are vulnerable to the initiation of apoptotic mechanisms following ischemic injury, which can ultimately reduce microvascular density.⁴⁶⁵

The endothelial cell plays a central role in coagulation via its interaction with protein C through the EPCR and thrombomoduli. The protein C pathway helps maintain normal homeostasis and limits inflammatory responses. Protein C is activated by thrombin-mediated cleavage, and the rate of this reaction is further augmented (by 1000-fold) when thrombin binds to the endothelial cell surface receptor protein thrombomodulin. The activation rate of protein C is further increased by approximately 10-fold when EPCR binds protein C and presents it to the thrombin-thrombomodulin complex. Essentially, activated protein C then has antithrombotic actions and profibrinolytic properties and participates in numerous antiinflammatory and cytoprotective pathways to restore normal homeostasis.⁴⁶⁶ Based on these properties, the endothelial cell plays an absolutely essential and critical role in maintaining a normal and healthy vasculature and endothelial bed.

In AKI, microvascular function is ultimately compromised, resulting in disseminated intravascular coagulation and microvascular thrombosis, decreased tissue perfusion, and hypoxemia, leading to organ dysfunction and failure. It has been shown that both pretreatment and postinjury treatment with soluble thrombomodulin attenuates renal injury, with minimization of vascular permeability defects and improvement in capillary renal blood flow.⁴⁶⁷

The leukocyte and endothelial cell are dynamically involved in the process of adherence of leukocytes to the vascular endothelium. Leukocyte activation and cytokine release require signals through chemokines circulating in the bloodstream or through direct contact with the endothelium. Rolling leukocytes can be activated by chemoattractants such as complement C5a and platelet-activating factor. Once activated, leukocyte integrins bind to endothelial ligands to promote firm adhesion. β_2 -Integrin (CD18) seems to be the most important ligand for neutrophil adherence. These interactions with the endothelium are mediated through

endothelial adhesion molecules that are upregulated during ischemic conditions.⁴⁶⁸

The initial phase starts with slow neutrophil migration mediated by tethering interactions between adhesion molecules and their endothelial cell ligands. Within 2 to 4 hours of IRI or sepsis, endothelial P-selectin and intercellular adhesion molecule 1 (ICAM1) are expressed on endothelial cells associated with leukocyte trapping in peritubular capillaries.³⁰⁶ Singbartl and Ley have found that platelet P-selectin and not endothelial P-selectin is the main determinant in neutrophil-mediated ischemic kidney injury.⁴⁶⁸ There is also significant protection from both ischemic injury and mortality by blockade of the shared ligand to all three selectins (E-, P-, and L-selectin), which seems to be dependent on the presence of a key fucosyl sugar on the selectin ligand.⁴⁶⁹ In a CLP model of septic azotemia, mice deficient for E-selectin, P-selectin, or both were completely protected. Furthermore, selectin-deficient mice demonstrated similar intraperitoneal leukocyte recruitment but altered cytokine levels when compared with WT mice,⁴⁷⁰ in addition to engagement in leukocyte-endothelial cell interactions.⁴⁷⁰ Other adhesion molecules such as VAP-1 appear to play a role in AKI.³¹⁵

Recently, in an attempt to transition to human studies, Dehnadi and coworkers have demonstrated the critical role that leukocyte integrin CD11b-CD18 contribute to both AKI and CKD in nonhuman primates.⁴⁷¹ In their model, using a CD11b-CD18 inhibitor, mAb107, ischemia-reperfusion-induced a no-reflow time; plasma creatinine and ATN were markedly diminished in the mAb107 groups, indicating that the leukocyte integrin CD11b-C18 contributes to AKI. Moreover, subsequent assessment up to 9 months after AKI revealed improvement in microvascular perfusion and histology.

Recent studies have examined the role of microparticles (MPs) in sepsis-related AKI. MPs are cell membrane-derived particles of 0.2 to 2 μ m in diameter that can promote coagulation and inflammation.⁴⁷² Using the cecal ligation model of sepsis, the authors found that calpastatin overexpression improved survival, organ dysfunction (including lung, kidney, and liver damage), and lymphocyte apoptosis. Increased calpastatin expression also decreased the sepsis-induced systemic proinflammatory response and disseminated intravascular coagulation by reducing the number of procoagulant circulating microparticles and therefore delaying thrombin generation.⁴⁷³ The deleterious effect of microparticles in this model was confirmed by transferring microparticles from septic WT to septic transgenic mice, worsening their survival and coagulopathy. The development of therapeutics aimed to prevent the release of MPs from endothelium or other inflammatory cells may be a reasonable target in microvascular dysfunction during sepsis-induced AKI.

ACUTE KIDNEY INJURY TO CHRONIC KIDNEY DISEASE TRANSITION

AKI is one of many initiating events that contribute to chronic, progressive kidney disease.^{474,475} Some patients with AKI fully recover kidney function, whereas in others, the development of CKD is accompanied by a progressive decline in kidney function, leading ultimately to end-stage kidney disease (ESKD).^{476–478} Regardless of the cause of CKD (e.g., nephrotoxic kidney injury, ischemia, infection, genetics; paraneoplastic, immunologic processes), there is a stereotypical response leading to interstitial fibrosis, tubular atrophy, and

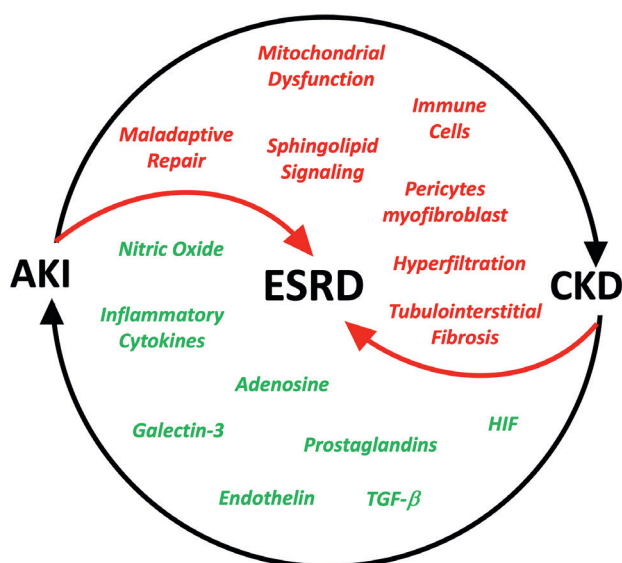


Fig. 28.12 Conceptual framework of acute kidney injury (AKI) to chronic kidney disease (CKD) transition. AKI can lead to CKD and then to end-stage renal disease (ESRD). Conversely, CKD can lead progressively to ESKD or can predispose to AKI, which then enters a vicious circle and predisposes to CKD. A number of factors contribute to AKI to CKD transition, including traditional concepts (green) and emerging (red) concepts.

peritubular rarefaction and inflammation.^{474,479,480} Incomplete repair following AKI contributes to the progression to CKD and ESKD^{418,481} (Fig. 28.12).

Role of the Endothelium

Studies have begun to elucidate the importance of the endothelium in the initiation and progression of fibrosis. Studies on human biopsies have associated microvascular rarefaction with progressive kidney failure.⁴⁸² There is also evidence in experimental AKI that acute injury to endothelial cells may have long-term implications; Basile have shown a significant reduction in blood vessel density following ischemic injury, leading to the phenomenon of vascular dropout.⁴⁸³ Vascular dropout was verified by Horbelt and associates, who found a drop in the vascular density by almost 45% at 4 weeks after an ischemic insult.⁴⁶⁵ Ehling and coworkers, using micro-CT imaging of vascular alterations, found a progressive decline of the renal relative blood volume in three models of progressive kidney disease (IRI, unilateral ureteral obstruction [UUO], and Alport) of up to 61% in late disease stages and that it preceded fibrosis.⁴⁸⁴ Using ex vivo micro-CT imaging and three-dimensional quantification of the renal vasculature, they demonstrated macrovascular to microvascular alterations in progressive renal disease, as evidenced by a reduction in vessel diameter and in vessel branching and increased vessel tortuosity (Fig. 28.13). These findings suggest that unlike the renal epithelial tubular cells, the renal vascular system lacks comparable regenerative potential. Recent studies have suggested that renal endothelial injury can result in capillary rarefaction due to hypoxia and a lack of upregulation of VEGF in response to injury.^{485–487} It is not yet clear whether apoptosis and/or necrosis play a major role in endothelial cell dropout. Ischemia has been shown to inhibit the

angiogenic protein VEGF while inducing ADAMTS1, which is thought to be a VEGF inhibitor.⁴⁸⁸ It was then postulated that the lack of vascular repair could be due to lack of VEGF, as shown by experiments where administration of VEGF-121 preserved the microvascular density.⁴⁸⁸ Reduction of the microvasculature density increases hypoxia-mediated fibrosis and alters usual hemodynamics, which may lead to hypertension. Thus, loss of microvasculature density and its consequential effects may play a critical role in the progression of CKD following initial recovery from ischemia-reperfusion-induced AKI.^{465,483,484} Using a genetic approach in the IRI mouse model of AKI, the temporal effects of endothelial sphingosine 1-phosphate receptor 1 (S1P1) during AKI were examined. Delayed deletion of S1P1 in endothelial cells after IRI, using a tamoxifen-inducible system to delete S1P1 prevented kidney recovery, resulting in chronic inflammation and progressive fibrosis. Specifically, S1P1 directly suppressed the endothelial activation of leukocyte adhesion molecule expression and inflammation. Altogether, the data indicate that activation of endothelial S1P1 is necessary to protect from IRI and permit recovery from AKI.³⁵⁹

Epigenetic Modification

AKI is associated with histone modification⁴⁸⁹ and DNA methylation,⁴⁹⁰ leading to altered transcription of genes thought to contribute to renal injury.^{491,492} There is evidence that epigenetic changes are closely related to renal hypoxia and CKD progression.^{485,486}

Renal Oxidative Stress in Acute Kidney Injury–Chronic Kidney Disease Transition

Renal oxidative stress has been shown to persist with resultant increased Ang II sensitivity in rats following recovery from ischemia-reperfusion-induced AKI after 5 weeks. Post-AKI rats showed significantly enhanced renal vasoconstrictor responses to Ang II, and treatment of AKI rats with apocynin normalized these responses. Expression of mRNA for common reduced NADPH oxidase subunits was not altered in kidneys following recovery from AKI; however, mRNA screening using polymerase chain reaction assays has suggested that post-AKI rats have decreased renal glutathione peroxidase 3 mRNA and an increased expression of other pro-oxidant genes such as lactoperoxidase, myeloperoxidase, and dual oxidase-1. Following infusion with Ang II, renal fibrosis was enhanced in post-AKI rats. The profibrotic response was significantly attenuated in rats treated with apocynin. These data suggest that there is sustained renal oxidant stress following recovery from AKI that alters the hemodynamic and fibrotic responses to Ang II and may contribute to the transition to CKD following AKI.⁴⁹³ This also highlights the importance of managing recovering AKI very carefully to avoid further insults that might enhance ROS or sensitivity to Ang II.

Cell Cycle Arrest

As discussed earlier, in successful repair, mature surviving cells dedifferentiate, proliferate, and repopulate proximal tubules in response to kidney injury.^{426,427} Upregulation of KIM-1 is antiinflammatory through its engulfment and clearance of dead and apoptotic cells⁴⁹⁴ but becomes maladaptive when chronically expressed.⁴⁹⁵ Repetitive injury may result in the upregulation of cyclin-dependent kinase (CDK) inhibitors, p16^{Ink4} and P53-p21^{Cip1/Waf1}, leading to cell cycle arrest.⁴⁹⁶

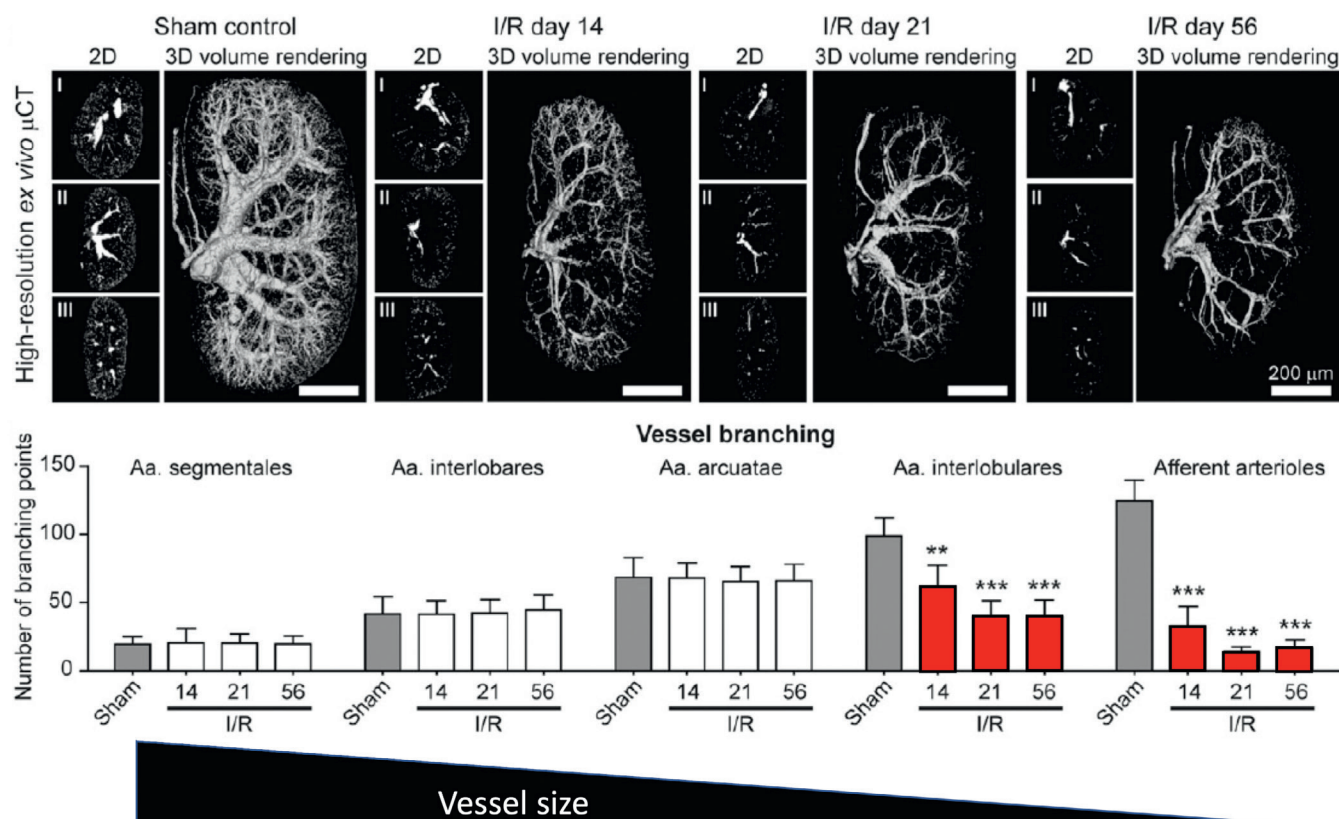


Fig. 28.13 Ex vivo micro-computed tomography (CT) imaging and three-dimensional quantification of the renal vasculature reveal significant alterations of renal arteries in progressive ischemia-reperfusion (IRI) induced renal injury. (Lower panel) Representative high-resolution, ex vivo, micro-CT images of sham control and I/R kidneys from days 14, 21, and 56 after Microfil perfusion (two-dimensional cross-sectional images in transverse [I], coronal [II], and sagittal [III] planes, as well as three-dimensional volume renderings). Note the progressive rarefaction of functional vessels besides the continuous shrinkage of fibrotic kidneys over time. Scale bar, 200 μ m. (Upper panel) Micro-CT-based quantification of vascular branching points per increasing vessel order (from hilus to periphery). A continuous reduction in vessel branching and vessel size was linked to an increased vessel tortuosity during fibrosis progression. **, $P < .01$; ***, $P < .001$. AKI, Acute kidney injury. (From Ehling J, Babickova J, Gremse F, et al. Quantitative Micro-computed tomography imaging of vascular dysfunction in progressive kidney diseases. *J Am Soc Nephrol*. 2016;27(2):520–532.)

An increase in p16^{Ink4} inhibits CDK4 and CDK6 and is associated with tubulointerstitial changes leading to fibrosis; kidneys from *INK4a*^{-/-} mice develop less fibrosis following kidney injury.²⁵⁰ Thus, cell cycle arrest and cellular senescence lead to resistance to apoptosis and persistent metabolic activity, promoting a senescence-type secretory phenotype with the secretion of TGF- β , cytokines, proteases, and other proinflammatory and profibrotic factors.^{418,497} Yang, Bonventre and associates have used four models of kidney injury, including severe bilateral IRI, unilateral IRI, aristolochic acid, and UUO and demonstrated that G2/M arrest contributes to maladaptive repair and fibrosis associated with AKI-CKD transition. Similar findings have been reported and, importantly, pharmacologic inhibitors of G2/M arrest reduce fibrosis.^{498,499}

Inflammation in Acute Kidney Injury–Chronic Kidney Disease Transition

During kidney recovery, renal and extrarenal cells participate in the wound-healing response and can initiate fibrosis. Immune cells of the mononuclear phagocyte system, including macrophages and dendritic cells, not only contribute to injury but have emerged as important cells in the recovery of kidney function during adaptive repair or fibrosis during maladaptive

repair. It is the balance between wound healing and progressive fibrosis that dictates the final outcome. The intrinsic plasticity of monocytes-macrophages and dendritic cells, as well as attempts to relate in vitro studies to in vivo findings, makes the functional definition and phenotype of this myeloid population in kidney pathophysiology complex.^{500–503} In vitro studies have led to two well-defined mononuclear phagocytes. Classically activated macrophages—M1 mononuclear phagocytes consisting of macrophages and dendritic cells—are produced by exposure to LPS or INF- γ and are largely thought to be proinflammatory and contribute to initial kidney injury. Alternatively, activated macrophages (M2 mononuclear phagocytes) are induced by IL-4 and IL-10 and appear later, after AKI has been established, and have a genetic signature associated with wound healing and/or fibrosis.⁵⁰⁴ These mononuclear phagocyte phenotypes depend on the complex local tissue microenvironment, which may induce phenotype switching.³⁰³

A key feature of fibrosis is the activation of extracellular matrix-producing myofibroblasts.^{505,506} Other factors relevant in CKD progression include endothelial cell damage and vascular damage in AKI,⁴⁸³ hypoxia-inducible factor (HIF),⁵⁰⁷ innate and adaptive immunity,⁵⁰⁸ cell cycle arrest,⁵⁰⁹ and epigenetic mechanisms.^{510,511}

Although some injured tubules undergo repair and regeneration, injury may also be accompanied by inflammation, maturation and proliferation of fibroblasts, and extracellular matrix deposition as part of the process of fibrosis. The source of fibroblasts in the injured kidney—fibrocytes, epithelial cells through epithelial mesenchymal transformation (EMT), intrinsic fibroblasts, and pericytes—remains controversial.^{512,513}

Tissue fibrosis is a common component of inflammatory diseases, including various forms of CKD. Collagen deposition, characteristically formed during normal wound healing, is reversible but progressive irreversible fibrosis may occur through repeated injury due to dysregulated inflammation.⁵¹⁴ Some studies have focused on the role of pericytes or resident fibroblasts that contribute to normal homeostatic microvascular function but transform into myofibroblasts on activation.^{513,515,516} Myofibroblasts are the primary collagen-producing cells.⁵¹²

Although the myofibroblast is the cell type responsible for depositing collagen, investigators have been searching for its precursors. Other than interstitial fibroblasts that are known to transition, pericytes, dendritic, endothelial, and epithelial cells have been identified as potential contributors. For example, in vivo and in vitro studies have demonstrated that endothelial cells develop a myofibroblast phenotype.^{517,518} Epithelial cells grown in culture can produce genes expressed in myofibroblasts, suggesting epithelial to mesenchymal transition.^{519–521} Fate-mapping studies by Humphreys and associates have shown that although epithelial cells can obtain mesenchymal markers in vitro, they do not penetrate the basement membrane to enter into the interstitial space and differentiate into myofibroblasts in vivo.⁵¹³ Furthermore, lineage analysis has demonstrated that platelet-derived growth factor receptor (PDGFR)- β -positive pericytes differentiate into myofibroblasts in a UUO model. Following AKI, a potential step in the AKI-CKD transition is the dissociation of pericytes from endothelial cells.⁴⁵⁸ Using bigenic Gli1-CreERT2; R26tdTomato reporter mice, Kramann and coworkers observed dissociation between Gli1⁺ pericytes and endothelial cells after AKI.⁴⁵⁸ Furthermore, following pericyte ablation, endothelial damage was observed, and pericyte loss led to a significantly reduced capillary number, a hallmark of CKD.

Two studies have shed light on this controversy. *Snail1* (encodes snail family zinc finger 1, Snail1) expressed in mouse renal epithelial cells is required for kidney fibrosis.⁵²² Twist (encodes twist family bHLH transcription factor 1, known as Twist) genes are essential regulators of EMT.⁵²³ Inhibition or deletion of these genes or overexpression has led to induction or inhibition of EMT, respectively.^{524,525} Lineage tracing studies have demonstrated that during activation, labeled epithelial cells do not contribute to myofibroblasts or interstitial cells; however, there was downregulation of epithelial markers and loss of epithelial polarity and cell differentiation.⁵²⁵ Thus, partial EMT leads to cell cycle arrest, blocks proliferation and repair, and blocks secretion of growth factors, such as TGF- β that drive proliferation of myofibroblasts.⁵²⁶ (Fig. 28.14).

Signaling pathways Wnt, hedgehog (Hh), and Notch pathways also play important roles during renal fibrosis. In CKD, numerous Wnt ligands are upregulated, resulting in prolonged activation of the Wnt/ β -catenin pathway.^{527,528}

Epithelial β -catenin activation leads to epithelial dedifferentiation and interstitial fibrosis.⁵²⁹ Wnt derived from tubules is necessary for myofibroblast activation and interstitial fibrosis.⁵³⁰ Recently, the role of the sphingolipid pathway in progressive fibrosis. Sphingosine 1-phosphate (S1P), a pleiotropic lysophospholipid that is involved in diverse functions such as cell growth and survival, lymphocyte trafficking, and vascular stability,^{531–536} is the product of sphingosine phosphorylation by two sphingosine kinase isoforms (SphK1 and SphK2). SphK2 has been localized to the nucleus and plays an important role in the production of nuclear S1P, which acts as a histone deacetylase (HDAC) inhibitor,⁵³⁷ thereby allowing gene expression to be induced. Bawja demonstrated that following unilateral IRI or folic acid (FA)-induced kidney fibrosis, protection of *Sphk2*^{-/-} mice from fibrosis at day 14 was thought to be due to IFN- γ . Kidneys of *Sphk2*^{-/-} mice exhibited greater expression of *Ifng* and IFN- γ -responsive genes (Cxcl9 and Cxcl10) than kidneys of WT or *Sphk1*^{-/-} mice. IFN- γ blocking antibody administered to *Sphk2*^{-/-} mice or deletion of *Ifng* using a double-knockout mouse (*Sphk2*^{-/-} *Ifng*^{-/-} mice) blocked the protective effect of *Sphk2* deficiency in fibrosis. Moreover, adoptive transfer of *Sphk2*^{-/-} CD4 T cells into WT mice blocked FA-induced fibrosis. Finally, a selective SphK2 inhibitor blocked FA-induced kidney fibrosis in WT mice. These studies have demonstrated that SphK2 inhibition may serve as a novel therapeutic approach for attenuating kidney fibrosis.^{538,539}

Activation of myofibroblasts is also mediated through paracrine signals from immune cells (lymphocytes and antigen-presenting cells), which include cytokines, chemokines, angiogenic factors, growth factors, peroxisome proliferator-activated receptors (PPARs), and other molecules.⁵⁴⁰ Although Th2 cytokines may induce fibrosis, Th1 cytokines such as IL-12 or IFN- γ have been shown to inhibit pulmonary^{541,542} and experimental renal fibrosis.⁵⁴³

Using an unbiased approach to identify upregulated and downregulated pericyte genes in injury, Schrimpf and colleagues found increased activation and expression of ADAMTS1 (a disintegrin-like metalloprotease with thrombospondin type 1) and downregulation of its inhibitor MMP-3 (TIMP-3).⁵⁴⁴ TIMP-3-stabilized pericytes maintained collagen capillary tube networks, whereas ADAMTS1-treated pericytes led to enhanced destabilization. TIMP-3 has many functions, including regulating VEGF signaling, pericyte migration, and MMP-2 and MMP-9 activity.⁵⁴⁵ Furthermore, TGF- β ₁ has been shown to activate the pericyte-myofibroblast transition, thus adding to the stimuli for pericyte involvement in fibrosis.⁵⁴⁶ Injured epithelial cells reduce production of VEGF, a trophic factor for endothelial maintenance, and increase production of TGF- β and PDGFR, which enhance pericyte dedifferentiation into myofibroblasts. Finally, PDGFR blockade on pericytes or VEGF receptor 2 (VEGFR-2) on endothelial cells led to a reduction in fibrosis and stabilization of the microvasculature in the UUO model.⁵⁴⁷ Thus, numerous factors favor microvascular maladaptation after injury, including TGF- β production and lack of VEGF production by epithelial cells, reductions in PDGF production by endothelial cells, and increased ADAMTS1 expression, with reduced TIMP-3 production by pericytes. These events also suggest several therapeutic targets that could limit microvascular dropout and loss of kidney function secondary to the fibrotic process. Normal pericyte function therefore becomes deranged following transformation to

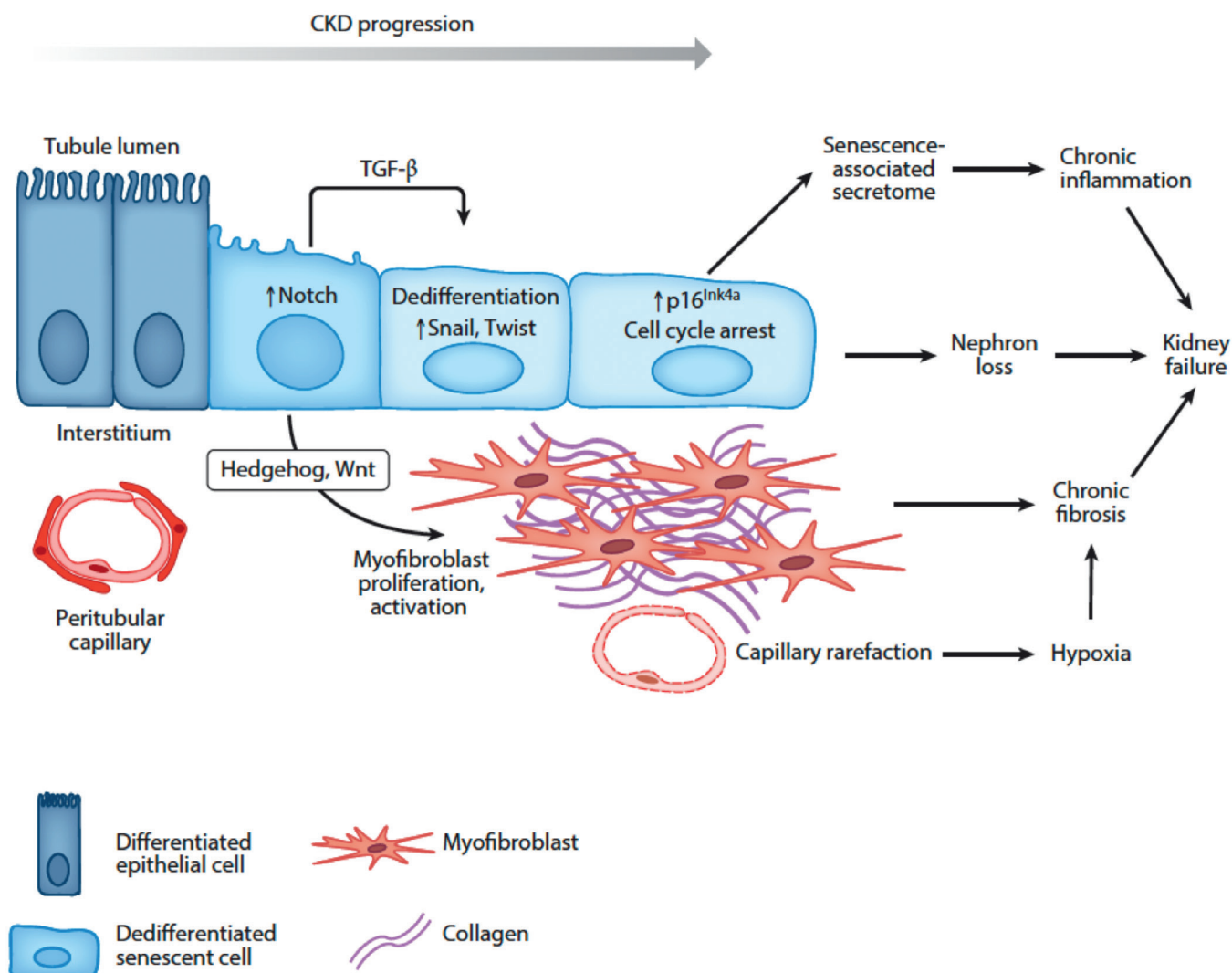


Fig. 28.14 Critical cells and signaling pathways activated in progressive chronic kidney disease (CKD). CKD is initiated by cellular injury, either to the epithelial or endothelial compartment. In the tubulointerstitium, injury leads to epithelial dedifferentiation that may be induced by transforming growth factor- β (TGF- β) and Notch pathway upregulation. Dedifferentiated epithelial cells secrete paracrine signaling factors, such as hedgehog and Wnt ligands, that act on interstitial pericytes and mesenchymal stem cell-like cells to activate myofibroblast differentiation, proliferation, and matrix secretion. This in turn causes peritubular capillary rarefaction and ongoing hypoxia. Chronic tubular injury leads to epithelial cell cycle arrest and senescence, with accompanying secretion of proinflammatory cytokines that amplify inflammation. These interrelated events ultimately drive nephron loss, ongoing interstitial fibrosis, and kidney failure. (From Humphreys BD. Mechanisms of renal fibrosis. *Annu Rev Physiol.* 2018;80:309–326.)

activated myofibroblasts, which leads to loss of microvascular stability, ineffective angiogenesis, increased permeability, and capillary rarefaction, which are all processes leading to fibrosis.^{457,474}

Proximal tubules are highly metabolically active, relying on aerobic metabolism. Mitochondrial fatty acid oxidation is a major source of energy, and mitochondrial dysfunction leads to lipid accumulation, ATP depletion, and fibrosis in proximal tubule cells.⁵⁴⁸ A number of therapeutic approaches are in development, including some agents in clinical trials for diabetic nephropathy and other kidney diseases,^{549,550} but as yet there are no approved treatments to prevent the development of kidney fibrosis or accelerate repair. The development of effective treatments requires a better understanding of the inflammatory, injury, wound healing, matrix

deposition, and cellular repair processes that accompany fibrosis.^{497,514,526,551–553} For example, mitochondria may be an important therapeutic target in AKI to CKD progression.⁵⁵⁴ In one approach, Birk and associates developed a novel mitoprotective drug, SS31.⁵⁵⁵ SS peptides bind selectively to cardiolipin in the mitochondrial inner membrane to protect mitochondrial cristae. SS31 allows better function of the respiratory chain complexes, prevents peroxidase activity, and enhances electron transfer through cytochrome c to improve oxidative phosphorylation and ATP production.^{555–557} SS31 administered 1 month after renal IRI attenuated glomerulosclerosis and senescence and reduced parietal epithelial cell activation and changes in podocyte and endothelial structure.⁵⁵⁸ Perry and associates have examined the role of dynamin-related protein 1 (DRP1), a mediator of cell fission.

Proximal tubule-specific deletion of *Drp1* prevented renal IRI, inflammation, and programmed cell death and promoted epithelial recovery, which was associated with activation of the renoprotective β -hydroxybutyrate signaling pathway.⁵⁵⁹ Importantly, delayed deletion of proximal tubule *Drp1* using a tamoxifen-inducible system after IRI attenuated kidney fibrosis. These results highlight DRP1 and mitochondrial dynamics as an important mediator of AKI and progression to fibrosis and suggest that DRP1 may serve as a therapeutic target for AKI.

Effects of Organ Crosstalk

AKI is associated with significant morbidity and mortality. In the intensive care unit, where patients with AKI sustain multiorgan dysfunction, the mortality rate is around 60%.⁵⁶⁰ Although AKI is associated with consequences such as accumulation of uremic toxins, metabolic acidosis, fluid and electrolyte imbalance, and fluid overload, these factors alone do not explain the mortality associated with AKI.^{146,561} There is accumulating evidence that in many cases, AKI is a systemic condition in which AKI induces dysfunction in distant organs, including lung, heart, brain, liver, and intestines (Fig. 28.15). Moreover, there is evidence that distant organ function can alter kidney function. Recent studies have demonstrated that neural control of inflammation and AKI is thought to be mediated by a pathway referred to as the *cholinergic antiinflammatory pathway*, which requires an intact spleen.^{562–566}

Kelly has demonstrated the effects of renal ischemia on cardiac tissues.¹³⁹ Induction of IL-1, TNF- α , and ICAM-1 mRNA was seen in cardiac tissues as early as 6 hours after renal ischemic injury and remained elevated up to 48 hours after renal ischemic injury. There was also a significant increase in myeloperoxidase activity in the heart and liver, apart from the kidneys. The increase in cardiac myeloperoxidase activity could be prevented by the administration of anti-ICAM-1 antibody at the time of renal ischemia. At 48 hours, cardiac function evaluation by echocardiography also revealed increases in left ventricular end-systolic and diastolic diameter and decreased fractional shortening. As little as 15 minutes of ischemia also resulted in significantly more apoptosis in cardiac tissue.¹³⁹ In transgenic sickle mice, bilateral renal IRI results in marked cardiac vascular congestion and increased serum amyloid P component (the murine equivalent of C-reactive protein). Kramer and colleagues have shown that renal ischemic injury leads to an increase in pulmonary vascular permeability defects that are mediated through macrophages.¹³⁵ Furthermore, they noted that in a rat model of bilateral renal ischemic injury or nephrectomy, there was downregulation of the lung epithelial Na channel, Na⁺-K⁺-ATPase, and aquaporin-5 expression, but not in unilateral ischemic models, suggesting the role of uremic toxins in modulating these effects in the lung.⁵⁶⁷ Liu and colleagues have also shown effects of AKI on functional changes in the brain.⁵⁶⁸ Mice with AKI had increased neuronal pyknosis and microgliosis in the brain; there were increased levels of the proinflammatory chemokines keratinocyte-derived chemoattractant and granulocyte colony-stimulating factor in the cerebral cortex and hippocampus and increased expression of glial fibrillary acidic protein in astrocytes in the cortex and corpus callosum. In addition, extravasation of Evans blue dye into the brain suggested that the blood-brain barrier was disrupted in mice with AKI.

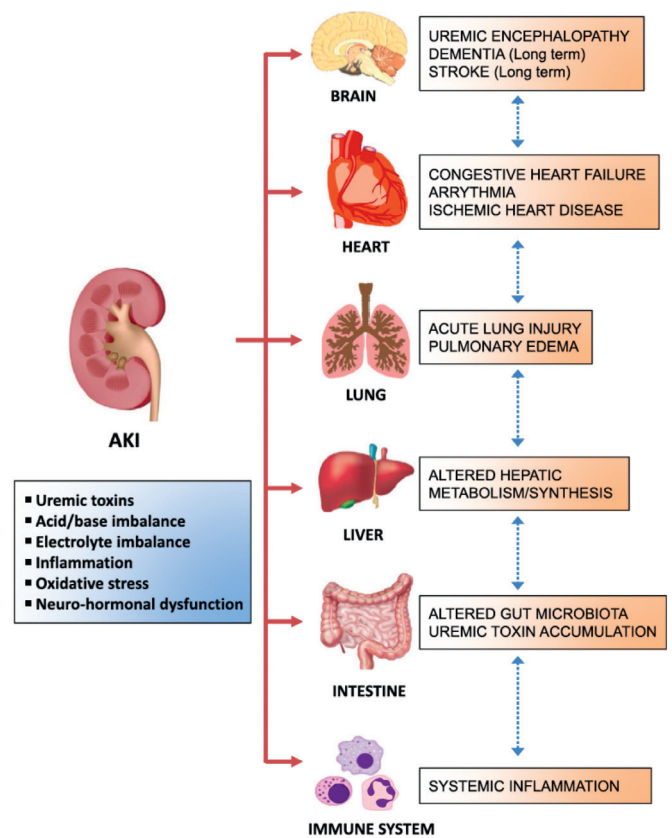


Fig. 28.15 The impact of acute kidney injury (AKI) on distant organs. AKI causes hemodynamic, humoral, and immunologic changes, which lead to dysfunction of distant organs, including lung, heart, brain, liver, and intestine and the immune system. (From Lee SA, Cozzi M, Bush EL, Rabb H. Distant organ dysfunction in acute kidney injury: a review. *Am J Kidney Dis.* 2018;72(6):846–856.)

Many of the same processes involved in kidney-lung, kidney-heart, and kidney-brain interactions have been observed in the liver—increased neutrophil infiltration, vascular congestion, and vascular permeability after AKI. Following experimental AKI, levels of hepatic TNF- α , IL-6, IL-17A, ICM-1, keratinocyte-derived chemoattractant, IL-10, and MCP-1 are increased. The presence of AKI before a second-hit ischemic hepatic injury exacerbates liver injury as well, suggesting ongoing crosstalk between the kidney and liver¹⁴⁶ (Table 28.1).

Other organs also regulate ischemic renal injury. Patients with acute lung injury frequently have AKI, in part due to mechanical ventilation. There are three potential mechanisms whereby mechanical ventilation induces AKI—the effects of changes in arterial blood gases, barotrauma-induced systemic release of inflammatory agents, and the influence on systemic and renal blood flow.⁵⁶⁹ Imai and coworkers have demonstrated the role of lung injury in inducing renal damage.⁵⁷⁰ They found that in rabbits, injurious lung ventilatory strategies (high tidal volume and low peak end-expiratory pressure) alone were sufficient to induce renal epithelial cell apoptosis. This finding was further substantiated by the fact that plasma obtained from rabbits that underwent the injurious ventilation strategy induced more pronounced apoptosis in cultured LLC-RK₁ cells in vitro, suggesting that circulating soluble

factors associated with mechanical ventilation might be involved in this process.

An example of extrarenal organs regulating ischemic AKI is evidenced by the effect of brain death on kidney transplants. Traumatic brain injury elicits a cytokine and inflammatory response. These cytokines result in renal inflammation in kidney transplants from brain-dead donors, distinct from living or cardiac-death donors.⁵⁷¹ Pretransplantation biopsies of brain-dead donor kidneys contain more infiltrating T lymphocytes and macrophages. Reperfusion of kidneys from brain-dead donors is associated with the instantaneous release of inflammatory cytokines, such as granulocyte colony-stimulating factor, IL-6, IL-9, IL-16, and MCP-1. In contrast, kidneys from living and cardiac death donors show a more modest cytokine response, with the release of IL-6 and small amounts of MCP-1.⁵⁷²

An important example of organ crosstalk is the one that occurs between the nervous and immune systems. Rather than two independent systems, these two systems are linked to maintain normal homeostasis and to respond to stress and pathophysiologic disorders. Studies have identified neural pathways that regulate immunity and inflammation via the inflammatory reflex pathway, thus identifying specific molecular targets that can be modulated by stimulating neurons electrically,^{562,573,574} by ultrasound,^{566,583} or optically.⁵⁶⁴ The cholinergic antiinflammatory pathway (CAP) is the efferent limb of the so-called inflammatory reflex pathway mediated through the vagus nerve.^{575–577} The afferent limb is activated by bacterial products (PAMPs), DAMPs, proinflammatory cytokines, immunoglobulins, and ATP.^{578,579} Following detection of these inflammatory molecules at receptors, afferent signals from injured tissue and the immune cells are transmitted to the brain,^{577,580} which then activates the vagus efferent nerve. Then the inflammatory reflex controls peripheral cytokine levels and inflammation through key cellular components such as macrophages and CD4⁺ T cells. Indeed, vagus nerve stimulation shows antiinflammatory and organ-protective effects in various disorders, such as arthritis,⁵⁸¹ colitis,⁵⁸² ileus,⁵⁸² AKI,⁵⁶³ and others. The spleen plays an integral role in linking the nervous and immune systems via the CAP. Prior splenectomy worsens AKI,⁵⁸³ blocks the antiinflammatory effect of the CAP,^{563,564} and worsens lung inflammation following kidney IRI.⁵⁸⁴ Because AKI is associated with high mortality and morbidity, these studies indicate that multiorgan crosstalk that occurs in the setting of AKI is likely to be a major contributor to nonrenal organ dysfunction and may mediate clinically observable events, such as cardiac, pulmonary, and central nervous system events.

Finally, we address the issue of unilateral AKI affecting contralateral kidney function. In a model where the ischemia-reperfusion-injured kidney was removed after 5 weeks to isolate effects on the untouched kidney, challenge with higher dietary sodium yielded significant increase in blood pressure relative to sham-operated controls. Similarly, contralateral kidneys had impaired pressure natriuresis and hemodynamic responses, but reductions in vascular density were observed in the contralateral kidney. However, contralateral kidneys contained interstitial cells, some of which were identified as activated (low CD62L/CD4⁺) T lymphocytes. Taken together, these data suggest that the salt-sensitive features of AKI on hypertension and CKD can be segregated so that effects on

hemodynamics and hypertension may develop independently of direct renal damage.⁵⁸⁵



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

26. Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med*. 2017;376:1770–1781.
58. Sandoval RM, Molitoris BA. Gentamicin traffics retrograde through the secretory pathway and is released in the cytosol via the endoplasmic reticulum. *Am J Physiol Renal Physiol*. 2004;286:F617–F624.
83. Luque Y, Louis K, Jouanneau C, et al. Vancomycin-associated cast nephropathy. *J Am Soc Nephrol*. 2017;28:1723–1728.
97. Nath KA, Balla G, Vercellotti GM, et al. Induction of heme oxygenase is a rapid, protective response in rhabdomyolysis in the rat. *J Clin Invest*. 1992;90:267–270.
98. Zarjou A, Bolisetty S, Joseph R, et al. Proximal tubule H-ferritin mediates iron trafficking in acute kidney injury. *J Clin Invest*. 2013;123:4423–4434.
103. Sanders PW, Booker BB, Bishop JB, et al. Mechanisms of intrarenal proteinaceous cast formation by low molecular weight proteins. *J Clin Invest*. 1990;85:570–576.
116. Zeisberg M, Neilson EG. Mechanisms of tubulointerstitial fibrosis. *J Am Soc Nephrol*. 2010;21:1819–1834.
122. Bonventre JV, Basile D, Liu KD, et al; on behalf of the Kidney Research National, D. AKI: a path forward. *Clin J Am Soc Nephrol*. 2013;8:1606–1608.
123. Okusa MD, Rosner MH, Kellum JA, et al; Acute Dialysis Quality Initiative, XW. Therapeutic targets of human AKI: harmonizing human and animal AKI. *J Am Soc Nephrol*. 2016;27:44–48.
125. Perrin S. Preclinical research: make mouse studies work. *Nature*. 2014;507:423–425.
126. de Caestecker M, Humphreys BD, Liu KD, et al. Bridging translation by improving preclinical study design in AKI. *J Am Soc Nephrol*. 2015.
139. Kelly KJ. Distant effects of experimental renal ischemia/reperfusion injury. *J Am Soc Nephrol*. 2003;14:1549–1558.
157. Doi K, Leelahavanichkul A, Yuen PS, et al. Animal models of sepsis and sepsis-induced kidney injury. *J Clin Invest*. 2009;119:2868–2878.
158. Langenberg C, Wan L, Egi M, et al. Renal blood flow in experimental septic acute renal failure. *Kidney Int*. 2006;69:1996–2002.
162. Wilmer MJ, Ng CP, Lanz HL, et al. Kidney-on-a-chip technology for drug-induced nephrotoxicity screening. *Trends Biotechnol*. 2016;34:156–170.
169. Takasato M, Er PX, Chiu HS, et al. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature*. 2015;526:564–568.
177. Solez K, Morel-Maroger L, Sraer JD. The morphology of ‘acute tubular necrosis’ in man: analysis of 57 renal biopsies and a comparison with the glycerol model. *Medicine (Baltimore)*. 1979;58:362–376.
194. Portilla D. Role of fatty acid beta-oxidation and calcium-independent phospholipase A2 in ischemic acute renal failure. *Curr Opin Nephrol Hypertens*. 1999;8:473–477.
210. Molitoris BA, Dagher PC, Sandoval RM, et al. siRNA targeted to p53 attenuates ischemic and cisplatin-induced acute kidney injury. *J Am Soc Nephrol*. 2009;20:1754–1764.
216. Sarhan M, Land WG, Tonnus W, et al. Origin and consequences of necroinflammation. *Physiol Rev*. 2018;98:727–780.
237. Havasi A, Dong Z. Autophagy and Tubular Cell Death in the Kidney. *Semin Nephrol*. 2016;36:174–188.
240. Ravichandran KS. Beginnings of a good apoptotic meal: the find-me and eat-me signaling pathways. *Immunity*. 2011;35:445–455.
256. Kaissling B, Le Hir M. The renal cortical interstitium: morphological and functional aspects. *Histochem Cell Biol*. 2008;130:247–262.
259. Geissmann F, Manz MG, Jung S, et al. Development of monocytes, macrophages, and dendritic cells. *Science*. 2010;327:656–661.
262. Matzinger P. Tolerance, danger, and the extended family. *Annu Rev Immunol*. 1994;12:991–1045.
269. Li L, Huang L, Sung SS, et al. The chemokine receptors CCR2 and CX3CR1 mediate monocyte/macrophage trafficking in kidney ischemia-reperfusion injury. *Kidney Int*. 2008;74:1509–1511.
284. Geissmann F, Jung S, Littman DR. Blood monocytes consist of two principal subsets with distinct migratory properties. *Immunity*. 2003;19:71–82.

301. Li L, Huang L, Sung SJ, et al. NKT cell activation mediates neutrophil IFN-gamma production and renal ischemia-reperfusion injury. *J Immunol.* 2007;178:5899–5911.
303. Lee S, Huen S, Nishio H, et al. Distinct macrophage phenotypes contribute to kidney injury and repair. *J Am Soc Nephrol.* 2011;22:317–326.
324. Burne MJ, Daniels F, El Ghandour A, et al. Identification of the CD4+ T cell as a major pathogenic factor in ischemic acute renal failure. *J Clin Invest.* 2001;108:1283–1290.
347. Stremiska ME, Jose S, Sabapathy V, et al. IL233, a novel IL-2 and IL-33 hybrid cytokine, ameliorates renal injury. *J Am Soc Nephrol.* 2017;28:2681–2693.
355. Lobo PI, Bajwa A, Schlegel KH, et al. Natural IgM anti-leukocyte autoantibodies attenuate excess inflammation mediated by innate and adaptive immune mechanisms involving Th-17. *J Immunol.* 2012;188:1675–1685.
377. Thurman JM, Nester CM. All things complement. *Clin J Am Soc Nephrol.* 2016;11:1856–1866.
381. Xavier S, Sahu RK, Landes SG, et al. Pericytes and immune cells contribute to complement activation in tubulointerstitial fibrosis. *Am J Physiol Renal Physiol.* 2017;312:F516–F532.
391. Liu M, Grigoryev DN, Crow MT, et al. Transcription factor Nrf2 is protective during ischemic and nephrotoxic acute kidney injury in mice. *Kidney Int.* 2009;76:277–285.
395. Swaminathan S. Iron, hormesis, and protection in acute kidney injury. *Kidney Int.* 2016;90:16–17.
417. Kapturczak MH, Wasserfall C, Brusko T, et al. Heme oxygenase-1 modulates early inflammatory responses: evidence from the heme oxygenase-1-deficient mouse. *Am J Pathol.* 2004;165:1045–1053.
418. Ferenbach DA, Bonventre JV. Mechanisms of maladaptive repair after AKI leading to accelerated kidney ageing and CKD. *Nat Rev Nephrol.* 2015;11:264–276.
426. Humphreys BD, Valerius MT, Kobayashi A, et al. Intrinsic epithelial cells repair the kidney after injury. *Cell Stem Cell.* 2008;2:284–291.
441. Erdbugger U, Le TH. Extracellular vesicles in renal diseases: more than novel biomarkers? *J Am Soc Nephrol.* 2016;27:12–26.
442. Bruno S, Grange C, Deregibus MC, et al. Mesenchymal stem cell-derived microvesicles protect against acute tubular injury. *J Am Soc Nephrol.* 2009;20:1053–1067.
443. Molitoris BA, Sutton TA. Endothelial injury and dysfunction: role in the extension phase of acute renal failure. *Kidney Int.* 2004;66:496–499.
474. Chawla LS, Eggers PW, Star RA, et al. Acute kidney injury and chronic kidney disease as interconnected syndromes. *N Engl J Med.* 2014;371:58–66.
475. Venkatachalam MA, Griffin KA, Lan R, et al. Acute kidney injury: a springboard for progression in chronic kidney disease. *Am J Physiol Renal Physiol.* 2010.
501. Zhang MZ, Yao B, Yang S, et al. CSF-1 signaling mediates recovery from acute kidney injury. *J Clin Invest.* 2012;122:4519–4532.
524. Lovisa S, LeBlau VS, Tampe B, et al. Epithelial-to-mesenchymal transition induces cell cycle arrest and parenchymal damage in renal fibrosis. *Nat Med.* 2015;21:998–1009.
525. Grande MT, Sanchez-Laorden B, Lopez-Blau C, et al. Snail1-induced partial epithelial-to-mesenchymal transition drives renal fibrosis in mice and can be targeted to reverse established disease. *Nat Med.* 2015;21:989–997.
562. Okusa MD, Rosin DL, Tracey KJ. Targeting neural reflex circuits in immunity to treat kidney disease. *Nat Rev Nephrol.* 2017;13:669–680.
566. Gigliotti JC, Huang L, Ye H, et al. Ultrasound prevents renal ischemia-reperfusion injury by stimulating the splenic cholinergic anti-inflammatory pathway. *J Am Soc Nephrol.* 2013;24:1451–1460.
585. Basile DP, Leonard EC, Tonade D, et al. Distinct effects on long-term function of injured and contralateral kidneys following unilateral renal ischemia-reperfusion. *Am J Physiol Renal Physiol.* 2012;302:F625–F635.

REFERENCES

- Nash K, Hafeez A, Hou S. Hospital-acquired renal insufficiency. *Am J Kidney Dis.* 2002;39:930–936.
- Liano F, Pascual J. Epidemiology of acute renal failure: a prospective, multicenter, community-based study. Madrid Acute Renal Failure Study Group. *Kidney Int.* 1996;50:811–818.
- Sesso R, Roque A, Vicioso B, et al. Prognosis of ARF in hospitalized elderly patients. *Am J Kidney Dis.* 2004;44:410–419.
- Kastner PR, Hall JE, Guyton AC. Control of glomerular filtration rate: role of intrarenally formed angiotensin II. *Am J Physiol.* 1984;246:F897–F906.
- Cogan MG. Angiotensin II: a powerful controller of sodium transport in the early proximal tubule. *Hypertension.* 1990;15:451–458.
- Quan A, Baum M. Regulation of proximal tubule transport by endogenously produced angiotensin II. *Nephron.* 2000;84:103–110.
- Liu FY, Cogan MG. Angiotensin II stimulation of hydrogen ion secretion in the rat early proximal tubule. *J Clin Invest.* 1988;82:601–607.
- Liu FY, Cogan MG. Angiotensin II stimulates early proximal bicarbonate absorption in the rat by decreasing cyclic adenosine monophosphate. *J Clin Invest.* 1989;84:83–91.
- Schuster VL, Kokko JP, Jacobson HR. Angiotensin II directly stimulates sodium transport in rabbit proximal convoluted tubules. *J Clin Invest.* 1984;73:507–515.
- Blantz RC. Pathophysiology of pre-renal azotemia. *Kidney Int.* 1998;53:512–523.
- Badr KF, Ichikawa I. Prerenal failure: a deleterious shift from renal compensation to decompensation. *N Engl J Med.* 1988;319:623–629.
- Yared A, Kon V, Ichikawa I. Mechanism of preservation of glomerular perfusion and filtration during acute extracellular fluid volume depletion. Importance of intrarenal vasopressin-prostaglandin interaction for protecting kidneys from constrictor action of vasopressin. *J Clin Invest.* 1985;75:1477–1487.
- Oliver JA, Sciacca RR, Cannon PJ. Renal vasodilation by converting enzyme inhibition. Role of renal prostaglandins. *Hypertension.* 1983;5:166–171.
- Pascual J, Liano F, Ortuno J. The elderly patient with acute renal failure. *J Am Soc Nephrol.* 1995;6:144–153.
- Gines P, Guevara M, Arroyo V, et al. Hepatorenal syndrome. *Lancet.* 2003;362:1819–1827.
- Gines P, Cardenas A, Arroyo V, et al. Management of cirrhosis and ascites. *N Engl J Med.* 2004;350:1646–1654.
- Wadei HM, Mai ML, Ahsan N, et al. Hepatorenal syndrome: pathophysiology and management. *Clin J Am Soc Nephrol.* 2006;1:1066–1079.
- Malbrain ML, Cheatham ML, Kirkpatrick A, et al. Results from the International Conference of Experts on Intra-abdominal Hypertension and Abdominal Compartment Syndrome. I. Definitions. *Intensive Care Med.* 2006;32:1722–1732.
- Kirkpatrick AW, Roberts DJ, De Waele J, et al; Pediatric Guidelines Sub-Committee for the World Society of the Abdominal Compartment, S. Intra-abdominal hypertension and the abdominal compartment syndrome: updated consensus definitions and clinical practice guidelines from the World Society of the Abdominal Compartment Syndrome. *Intensive Care Med.* 2013;39:1190–1206.
- Xu K, Rosenstiel P, Paragas N, et al. Unique transcriptional programs identify subtypes of AKI. *J Am Soc Nephrol.* 2016.
- Thadhani RI, Camargo CA Jr, Xavier RJ, et al. Atheroembolic renal failure after invasive procedures. Natural history based on 52 histologically proven cases. *Medicine (Baltimore).* 1995;74:350–358.
- van der Wal MA, Wisselink W, Rauwerda JA. Traumatic bilateral renal artery thrombosis: case report and review of the literature. *Cardiovasc Surg.* 2003;11:527–529.
- Dinchman KH, Spirnak JP. Traumatic renal artery thrombosis: evaluation and treatment. *Semin Urol.* 1995;13:90–93.
- Piette JC, Cacoub P, Wechsler B. Renal manifestations of the antiphospholipid syndrome. *Semin Arthritis Rheum.* 1994;23:357–366.
- Kodner CM, Kudrimoti A. Diagnosis and management of acute interstitial nephritis. *Am Fam Physician.* 2003;67:2527–2534.
- Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med.* 2017;376:1770–1781.
- Moledina DG, Perazella MA. PPIs and kidney disease: from AIN to CKD. *J Nephrol.* 2016;29:611–616.
- Fahling M, Seeliger E, Patzak A, et al. Understanding and preventing contrast-induced acute kidney injury. *Nat Rev Nephrol.* 2017;13:169–180.
- Wanchoo R, Jhaveri KD, Deray G, et al. Renal effects of BRAF inhibitors: a systematic review by the Cancer and the Kidney International Network. *Clin Kidney J.* 2016;9:245–251.
- Belliere J, Meyer N, Mazieres J, et al. Acute interstitial nephritis related to immune checkpoint inhibitors. *Br J Cancer.* 2016;115:1457–1461.
- Sturmer T, Elseviers MM, De Broe ME. Nonsteroidal anti-inflammatory drugs and the kidney. *Curr Opin Nephrol Hypertens.* 2001;10:161–163.
- Paeksakon P, Fogo AB. Drug-induced nephropathies. *Histopathology.* 2017;70:94–108.
- Michel DM, Kelly CJ. Acute interstitial nephritis. *J Am Soc Nephrol.* 1998;9:506–515.
- Persson PB, Hansell P, Liss P. Pathophysiology of contrast medium-induced nephropathy. *Kidney Int.* 2005;68:14–22.
- Bettmann MA. Contrast medium-induced nephropathy: critical review of the existing clinical evidence. *Nephrol Dial Transplant.* 2005;20(suppl 1):i12–i17.
- McCullough PA. Radiocontrast-induced acute kidney injury. *Nephron Physiol.* 2008;109:p61–p72.
- Liss P, Carlsson PO, Nygren A, et al. Et-A receptor antagonist BQ123 prevents radiocontrast media-induced renal medullary hypoxia. *Acta Radiol.* 2003;44:111–117.
- Heyman SN, Reichman J, Brezis M. Pathophysiology of radiocontrast nephropathy: a role for medullary hypoxia. *Invest Radiol.* 1999;34:685–691.
- Liss P, Nygren A, Hansell P. Hypoperfusion in the renal outer medulla after injection of contrast media in rats. *Acta Radiol.* 1999;40:521–527.
- Nygren A, Ulfendahl HR. Effects of high- and low-osmolar contrast media on renal plasma flow and glomerular filtration rate in euvoletic and dehydrated rats. A comparison between ioxithalamate, iopamidol, iohexol and ioxaglate. *Acta Radiol.* 1989;30:383–389.
- Nygren A, Ulfendahl HR, Hansell P, et al. Effects of intravenous contrast media on cortical and medullary blood flow in the rat kidney. *Invest Radiol.* 1988;23:753–761.
- Sendeski MM, Persson AB, Liu ZZ, et al. Iodinated contrast media cause endothelial damage leading to vasoconstriction of human and rat vasa recta. *Am J Physiol Renal Physiol.* 2012;303:F1592–F1598.
- Sendeski M, Patzak A, Pallone TL, et al. Iodixanol, constriction of medullary descending vasa recta, and risk for contrast medium-induced nephropathy. *Radiology.* 2009;251:697–704.
- Ueda J, Nygren A, Sjoquist M, et al. Iodine concentrations in the rat kidney measured by X-ray microanalysis. Comparison of concentrations and viscosities in the proximal tubules and renal pelvis after intravenous injections of contrast media. *Acta Radiol.* 1998;39:90–95.
- Lancelot E, Idee JM, Couturier V, et al. Influence of the viscosity of iodixanol on medullary and cortical blood flow in the rat kidney: a potential cause of nephrotoxicity. *J Appl Toxicol.* 1999;19:341–346.
- Humes HD, Hunt DA, White MD. Direct toxic effect of the radiocontrast agent diatrizoate on renal proximal tubule cells. *Am J Physiol.* 1987;252:F246–F255.
- Messana JM, Cieslinski DA, Nguyen VD, et al. Comparison of the toxicity of the radiocontrast agents, iopamidol and diatrizoate, to rabbit renal proximal tubule cells in vitro. *J Pharmacol Exp Ther.* 1988;244:1139–1144.
- Andersen KJ, Christensen EI, Vik H. Effects of iodinated x-ray contrast media on renal epithelial cells in culture. *Invest Radiol.* 1994;29:955–962.
- Aspelin P, Aubry P, Fransson SG, et al. Nephrotoxic effects in high-risk patients undergoing angiography. *N Engl J Med.* 2003;348:491–499.
- Solomon R. Contrast media: are there differences in nephrotoxicity among contrast media? *Biomed Res Int.* 2014;934947:2014.
- Heinrich MC, Haberle L, Muller V, et al. Nephrotoxicity of iso-osmolar iodixanol compared with nonionic low-osmolar contrast media: meta-analysis of randomized controlled trials. *Radiology.* 2009;250:68–86.
- Weisbord SD, Palevsky PM. Strategies for the prevention of contrast-induced acute kidney injury. *Curr Opin Nephrol Hypertens.* 2010;19:539–549.
- Jost G, Lengsfeld P, Lenhard DC, et al. Viscosity of iodinated contrast agents during renal excretion. *Eur J Radiol.* 2011;80:373–377.
- Nagai J, Tanaka H, Nakanishi N, et al. Role of megalin in renal handling of aminoglycosides. *Am J Physiol Renal Physiol.* 2001;281:F337–F344.

55. Schmitz C, Hilpert J, Jacobsen C, et al. Megalin deficiency offers protection from renal aminoglycoside accumulation. *J Biol Chem*. 2002;277:618–622.
56. Raggi C, Fujiwara K, Leal T, et al. Decreased renal accumulation of aminoglycoside reflects defective receptor-mediated endocytosis in cystic fibrosis and Dent's disease. *Pflugers Arch*. 2011;462:851–860.
57. Dagil R, O'Shea C, Nykjaer A, et al. Gentamicin binds to the megalin receptor as a competitive inhibitor using the common ligand binding motif of complement type repeats: insight from the nmr structure of the 10th complement type repeat domain alone and in complex with gentamicin. *J Biol Chem*. 2013;288:4424–4435.
58. Sandoval RM, Molitoris BA. Gentamicin traffics retrograde through the secretory pathway and is released in the cytosol via the endoplasmic reticulum. *Am J Physiol Renal Physiol*. 2004;286:F617–F624.
59. Sandoval RM, Dunn KW, Molitoris BA. Gentamicin traffics rapidly and directly to the Golgi complex in LLC-PK(1) cells. *Am J Physiol Renal Physiol*. 2000;279:F884–F890.
60. Simmons CF Jr, Bogusky RT, Humes HD. Inhibitory effects of gentamicin on renal mitochondrial oxidative phosphorylation. *J Pharmacol Exp Ther*. 1980;214:709–715.
61. Morales AI, Dettaille D, Prieto M, et al. Metformin prevents experimental gentamicin-induced nephropathy by a mitochondria-dependent pathway. *Kidney Int*. 2010;77:861–869.
62. Walker PD, Shah SV. Evidence suggesting a role for hydroxyl radical in gentamicin-induced acute renal failure in rats. *J Clin Invest*. 1988;81:334–341.
63. Cuzzocrea S, Mazzon E, Dugo L, et al. A role for superoxide in gentamicin-mediated nephropathy in rats. *Eur J Pharmacol*. 2002;450:67–76.
64. Zingman LV, Park S, Olson TM, et al. Aminoglycoside-induced translational read-through in disease: overcoming nonsense mutations by pharmacogenetic therapy. *Clin Pharmacol Ther*. 2007;81:99–103.
65. Monteil C, Leclerc C, Fillastre JP, et al. Characterization of gentamicin-induced dysfunctions in vitro: the use of optimized primary cultures of rabbit proximal tubule cells. *Ren Fail*. 1993;15:475–483.
66. Bennett WM, Mela-Riker LM, Houghton DC, et al. Microsomal protein synthesis inhibition: an early manifestation of gentamicin nephrotoxicity. *Am J Physiol*. 1988;255:F265–F269.
67. Horibe T, Matsui H, Tanaka M, et al. Gentamicin binds to the lectin site of calreticulin and inhibits its chaperone activity. *Biochem Biophys Res Commun*. 2004;323:281–287.
68. Humes HD. Aminoglycoside nephrotoxicity. *Kidney Int*. 1988;33:900–911.
69. Bennett WM. Mechanisms of aminoglycoside nephrotoxicity. *Clin Exp Pharmacol Physiol*. 1989;16:1–6.
70. Hori Y, Aoki N, Kuwahara S, et al. Megalin blockade with cilastatin suppresses drug-induced nephrotoxicity. *J Am Soc Nephrol*. 2017;28:1783–1791.
71. Rea RS, Capitano B. Optimizing use of aminoglycosides in the critically ill. *Semin Respir Crit Care Med*. 2007;28:596–603.
72. Rougier F, Claude D, Maurin M, et al. Aminoglycoside nephrotoxicity. *Curr Drug Targets Infect Disord*. 2004;4:153–162.
73. Hatala R, Dinh T, Cook DJ. Once-daily aminoglycoside dosing in immunocompetent adults: a meta-analysis. *Ann Intern Med*. 1996;124:717–725.
74. Mergenhagen KA, Borton AR. Vancomycin nephrotoxicity: a review. *J Pharm Pract*. 2014;27:545–553.
75. Levine DP. Vancomycin: a history. *Clin Infect Dis*. 2006;42(suppl 1):S5–S12.
76. Hazlewood KA, Brouse SD, Pitcher WD, et al. Vancomycin-associated nephrotoxicity: grave concern or death by character assassination? *Am J Med*. 2010;123(182):e181–e187.
77. Moellering RC Jr. Vancomycin: a 50-year reassessment. *Clin Infect Dis*. 2006;42(suppl 1):S3–S4.
78. Jeffres MN, Isakow W, Doherty JA, et al. A retrospective analysis of possible renal toxicity associated with vancomycin in patients with health care-associated methicillin-resistant *Staphylococcus aureus* pneumonia. *Clin Ther*. 2007;29:1107–1115.
79. Pritchard L, Baker C, Leggett J, et al. Increasing vancomycin serum trough concentrations and incidence of nephrotoxicity. *Am J Med*. 2010;123:1143–1149.
80. Rutter WC, Cox JN, Martin CA, et al. Nephrotoxicity during Vancomycin Therapy in Combination with Piperacillin-Tazobactam or Cefepime. *Antimicrob Agents Chemother*. 2017;61.
81. Bosso JA, Nappi J, Rudisill C, et al. Relationship between vancomycin trough concentrations and nephrotoxicity: a prospective multicenter trial. *Antimicrob Agents Chemother*. 2011;55:5475–5479.
82. Lodise TP, Lomaestro B, Graves J, et al. Larger vancomycin doses (at least four grams per day) are associated with an increased incidence of nephrotoxicity. *Antimicrob Agents Chemother*. 2008;52:1330–1336.
83. Luque Y, Louis K, Jouanneau C, et al. Vancomycin-associated cast nephropathy. *J Am Soc Nephrol*. 2017;28:1723–1728.
84. Manohar S, Leung N. Cisplatin nephrotoxicity: a review of the literature. *Journal of Nephrology*. 2017.
85. Nakamura T, Yonezawa A, Hashimoto S, et al. Disruption of multidrug and toxin extrusion MATE1 potentiates cisplatin-induced nephrotoxicity. *Biochem Pharmacol*. 2010;80:1762–1767.
86. Sharp CN, Doll MA, Dupre TV, et al. Repeated administration of low-dose cisplatin in mice induces fibrosis. *Am J Physiol Renal Physiol*. 2016;310:F560–F568.
87. Latcha S, Jaimes EA, Patil S, et al. Long-term renal outcomes after cisplatin treatment. *Clin J Am Soc Nephrol*. 2016;11:1173–1179.
88. Arany I, Safirstein RL. Cisplatin nephrotoxicity. *Semin Nephrol*. 2003;23:460–464.
89. Hurst FP, Abbott KC. Acute phosphate nephropathy. *Curr Opin Nephrol Hypertens*. 2009;18:513–518.
90. Markowitz GS, Perazella MA. Acute phosphate nephropathy. *Kidney Int*. 2009;76:1027–1034.
91. Markowitz GS, Stokes MB, Radhakrishnan J, et al. Acute phosphate nephropathy following oral sodium phosphate bowel purgative: an underrecognized cause of chronic renal failure. *J Am Soc Nephrol*. 2005;16:3389–3396.
92. Fogo AB, Lusco MA, Najafian B, et al. AJKD atlas of renal pathology: nephrocalcinosis and acute phosphate nephropathy. *Am J Kidney Dis*. 2017;69:e17–e18.
93. Gburek J, Birn H, Verroust PJ, et al. Renal uptake of myoglobin is mediated by the endocytic receptors megalin and cubilin. *Am J Physiol Renal Physiol*. 2003;285:F451–F458.
94. Zorova LD, Pevzner IB, Chupyrkina AA, et al. The role of myoglobin degradation in nephrotoxicity after rhabdomyolysis. *Chem Biol Interact*. 2016;256:64–70.
95. Plotnikov EY, Chupyrkina AA, Pevzner IB, et al. Myoglobin causes oxidative stress, increase of NO production and dysfunction of kidney's mitochondria. *Biochim Biophys Acta*. 1792;796:803:2009.
96. Holt S, Moore K. Pathogenesis of renal failure in rhabdomyolysis: the role of myoglobin. *Exp Nephrol*. 2000;8:72–76.
97. Nath KA, Balla G, Vercellotti GM, et al. Induction of heme oxygenase is a rapid, protective response in rhabdomyolysis in the rat. *J Clin Invest*. 1992;90:267–270.
98. Zarjou A, Bolisetty S, Joseph R, et al. Proximal tubule H-ferritin mediates iron trafficking in acute kidney injury. *J Clin Invest*. 2013;123:4423–4434.
99. Zager RA, Burkhart KM. Differential effects of glutathione and cysteine on Fe²⁺, Fe³⁺, H₂O₂ and myoglobin-induced proximal tubular cell attack. *Kidney Int*. 1998;53:1661–1672.
100. Karam H, Bruneval P, Clozel JP, et al. Role of endothelin in acute renal failure due to rhabdomyolysis in rats. *J Pharmacol Exp Ther*. 1995;274:481–486.
101. Hill-Kapturczak N, Chang SH, Agarwal A. Heme oxygenase and the kidney. *DNA Cell Biol*. 2002;21:307–321.
102. Ogawa T, Nussler AK, Tuzuner E, et al. Contribution of nitric oxide to the protective effects of ischemic preconditioning in ischemia-reperfused rat kidneys. *J Lab Clin Med*. 2001;138:50–58.
103. Sanders PW, Booker BB, Bishop JB, et al. Mechanisms of intrarenal proteinaceous cast formation by low molecular weight proteins. *J Clin Invest*. 1990;85:570–576.
104. Bosch X, Poch E, Grau JM. Rhabdomyolysis and acute kidney injury. *N Engl J Med*. 2009;361:62–72.
105. Sanders PW. Mechanisms of light chain injury along the tubular nephron. *J Am Soc Nephrol*. 2012;23:1777–1781.
106. Winearls CG. Acute myeloma kidney. *Kidney Int*. 1995;48:1347–1361.
107. Moro S, van Rhee AM, Sanders LH, et al. Flavanoid derivatives as adenosine receptor antagonists: a comparison of the hypothetical receptor binding site based on a comparative molecular field analysis model. *J Med Chem*. 1998;41:46–52.
108. Klassen RB, Allen PL, Batuman V, et al. Light chains are a ligand for megalin. *J Appl Physiol*. 2005;98:257–263.
109. Li M, Balamuthusamy S, Simon EE, et al. Silencing megalin and cubilin genes inhibits myeloma light chain endocytosis and

- ameliorates toxicity in human renal proximal tubule epithelial cells. *Am J Physiol Renal Physiol*. 2008;295:F82–F90.
110. Sanders PW, Herrera GA, Galla JH. Human Bence Jones protein toxicity in rat proximal tubule epithelium in vivo. *Kidney Int*. 1987;32:851–861.
 111. Wang PX, Sanders PW. Immunoglobulin light chains generate hydrogen peroxide. *J Am Soc Nephrol*. 2007;18:1239–1245.
 112. Basnayake K, Ying WZ, Wang PX, et al. Immunoglobulin light chains activate tubular epithelial cells through redox signaling. *J Am Soc Nephrol*. 2010;21:1165–1173.
 113. Ying WZ, Wang PX, Aaron KJ, et al. Immunoglobulin light chains activate nuclear factor-kappaB in renal epithelial cells through a Src-dependent mechanism. *Blood*. 2011;117:1301–1307.
 114. Ying WZ, Wang PX, Sanders PW. Pivotal role of apoptosis signal-regulating kinase 1 in monoclonal free light chain-mediated apoptosis. *Am J Pathol*. 2012;180:41–47.
 115. Sengul S, Zwizinski C, Simon EE, et al. Endocytosis of light chains induces cytokines through activation of NF-kappaB in human proximal tubule cells. *Kidney Int*. 2002;62:1977–1988.
 116. Zeisberg M, Neilson EG. Mechanisms of tubulointerstitial fibrosis. *J Am Soc Nephrol*. 2010;21:1819–1834.
 117. Chauveau D, Choukroun G. Bence Jones proteinuria and myeloma kidney. *Nephrol Dial Transplant*. 1996;11:413–415.
 118. Ying WZ, Allen CE, Curtis LM, et al. Mechanism and prevention of acute kidney injury from cast nephropathy in a rodent model. *J Clin Invest*. 2012;122:1777–1785.
 119. Kaplan AA. Therapeutic apheresis for the renal complications of multiple myeloma and the dysglobulinemias. *Ther Apher*. 2001;5:171–175.
 120. Hutchison CA, Cockwell P, Stringer S, et al. Early reduction of serum-free light chains associates with renal recovery in myeloma kidney. *J Am Soc Nephrol*. 2011;22:1129–1136.
 121. Mustonen S, Ala-Houhala IO, Tammela TL. Long-term renal dysfunction in patients with acute urinary retention. *Scand J Urol Nephrol*. 2001;35:44–48.
 122. Bonventre JV, Basile D, Liu KD, et al; on behalf of the Kidney Research National, D. AKI: a path forward. *Clin J Am Soc Nephrol*. 2013;8:1606–1608.
 123. Okusa MD, Rosner MH, Kellum JA, et al; Acute Dialysis Quality Initiative, XW. Therapeutic targets of human AKI: harmonizing human and animal AKI. *J Am Soc Nephrol*. 2016;27:44–48.
 124. Molitoris BA. Therapeutic translation in acute kidney injury: the epithelial/endothelial axis. *J Clin Invest*. 2014;124:2355–2363.
 125. Perrin S. Preclinical research: make mouse studies work. *Nature*. 2014;507:423–425.
 126. de Caestecker M, Humphreys BD, Liu KD, et al. Bridging translation by improving preclinical study design in AKI. *J Am Soc Nephrol*. 2015.
 127. Zuk A, Palevsky PM, Fried L, et al. Overcoming translational barriers in acute kidney injury: a report from an NIDDK workshop. *Clin J Am Soc Nephrol*. 2018;13:1113–1123.
 128. Leaf DE, Waikar SS. End points for clinical trials in acute kidney injury. *Am J Kidney Dis*. 2017;69:108–116.
 129. Rosen S, Heyman SN. Difficulties in understanding human “acute tubular necrosis”: limited data and flawed animal models. *Kidney Int*. 2001;60:1220–1224.
 130. Lieberthal W, Nigam SK. Acute renal failure. II. Experimental models of acute renal failure: imperfect but indispensable. *Am J Physiol Renal Physiol*. 2000;278:F1–F12.
 131. Ahuja N, Andres-Bernando A, Altmann C, et al. Circulating IL-6 mediates lung injury via CXCL1 production after acute kidney injury in mice. *Am J Physiol Renal Physiol*. 2012;303:F864–F872.
 132. White LE, Santora RJ, Cui Y, et al. TNFR1-dependent pulmonary apoptosis during ischemic acute kidney injury. *Am J Physiol Lung Cell Mol Physiol*. 2012;303:L449–L459.
 133. Hoke TS, Douglas IS, Klein CL, et al. Acute renal failure after bilateral nephrectomy is associated with cytokine-mediated pulmonary injury. *J Am Soc Nephrol*. 2007;18:155–164.
 134. Ishii T, Doi K, Okamoto K, et al. Neutrophil elastase contributes to acute lung injury induced by bilateral nephrectomy. *Am J Pathol*. 2010;177:1665–1673.
 135. Kramer AA, Postler G, Salhab KF, et al. Renal ischemia/reperfusion leads to macrophage-mediated increase in pulmonary vascular permeability. *Kidney Int*. 1999;55:2362–2367.
 136. Mitaka C, Si MK, Tulafu M, et al. Effects of atrial natriuretic peptide on inter-organ crosstalk among the kidney, lung, and heart in a rat model of renal ischemia-reperfusion injury. *Intensive Care Med Exp*. 2014;2:28.
 137. Oztay F, Kara-Kisla B, Orhan N, et al. The protective effects of prostaglandin E1 on lung injury following renal ischemia-reperfusion in rats. *Toxicol Ind Health*. 2016;32:1684–1692.
 138. Yasuda N, Goto K, Yamamoto S, et al. Removal of 17 cytokines, HMGB1, and albumin by continuous hemofiltration using a cellulose triacetate membrane: an ex vivo study. *J Surg Res*. 2012;176:226–231.
 139. Kelly KJ. Distant effects of experimental renal ischemia/reperfusion injury. *J Am Soc Nephrol*. 2003;14:1549–1558.
 140. Sumida M, Doi K, Ogasawara E, et al. Regulation of mitochondrial dynamics by dynamin-related protein-1 in acute cardiorenal syndrome. *J Am Soc Nephrol*. 2015;26:2378–2387.
 141. Cao W, Li A, Li J, et al. Reno-cerebral reflex activates the renin-angiotensin system, promoting oxidative stress and renal damage after ischemia-reperfusion injury. *Antioxid Redox Signal*. 2017;27:415–432.
 142. Golab F, Kaddhodae M, Zahmatkesh M, et al. Ischemic and non-ischemic acute kidney injury cause hepatic damage. *Kidney Int*. 2009;75:783–792.
 143. Khastar H. Protective effects of vitamin E against liver damage caused by renal ischemia reperfusion. *Ren Fail*. 2015;37:494–496.
 144. Awad AS, Kamel R, Sherief MA. Effect of thymoquinone on hepatorenal dysfunction and alteration of CYP3A1 and spermidine/spermine N-1-acetyl-transferase gene expression induced by renal ischaemia-reperfusion in rats. *J Pharm Pharmacol*. 2011;63:1037–1042.
 145. Kim M, Park SW, Kim M, et al. Isoflurane activates intestinal sphingosine kinase to protect against renal ischemia-reperfusion-induced liver and intestine injury. *Anesthesiology*. 2011;114:363–373.
 146. Lee SA, Cozzi M, Bush EL, et al. Distant organ dysfunction in acute kidney injury: a review. *Am J Kidney Dis*. 2018.
 147. Wei Q, Dong Z. Mouse model of ischemic acute kidney injury: technical notes and tricks. *Am J Physiol Renal Physiol*. 2012;303:F1487–F1494.
 148. Mestas J, Hughes CC. Of mice and not men: differences between mouse and human immunology. *J Immunol*. 2004;172:2731–2738.
 149. Adrie C, Adib-Conquy M, Laurent I, et al. Successful cardiopulmonary resuscitation after cardiac arrest as a “sepsis-like” syndrome. *Circulation*. 2002;106:562–568.
 150. Andrade-Oliveira V, Amano MT, Correa-Costa M, et al. Gut bacteria products prevent AKI induced by ischemia-reperfusion. *J Am Soc Nephrol*. 2015;26:1877–1888.
 151. Jang HR, Gandolfo MT, Ko GJ, et al. Early exposure to germs modifies kidney damage and inflammation after experimental ischemia-reperfusion injury. *Am J Physiol Renal Physiol*. 2009;297:F1457–F1465.
 152. Noel S, Martina-Lingua MN, Bandapalle S, et al. Intestinal microbiota-kidney cross talk in acute kidney injury and chronic kidney disease. *Nephron Clin Pract*. 2014;127:139–143.
 153. Burne-Taney MJ, Kofler J, Yokota N, et al. Acute renal failure after whole body ischemia is characterized by inflammation and T cell-mediated injury. *Am J Physiol Renal Physiol*. 2003;285:F87–F94.
 154. Zager RA. Partial aortic ligation: a hypoperfusion model of ischemic acute renal failure and a comparison with renal artery occlusion. *J Lab Clin Med*. 1987;110:396–405.
 155. Kikeri D, Pennell JP, Hwang KH, et al. Endotoxemic acute renal failure in awake rats. *Am J Physiol*. 1986;250:F1098–F1106.
 156. Wichterman KA, Baue AE, Chaudry IH. Sepsis and septic shock—a review of laboratory models and a proposal. *J Surg Res*. 1980;29:189–201.
 157. Doi K, Leelahavanichkul A, Yuen PS, et al. Animal models of sepsis and sepsis-induced kidney injury. *J Clin Invest*. 2009;119:2868–2878.
 158. Langenberg C, Wan L, Egi M, et al. Renal blood flow in experimental septic acute renal failure. *Kidney Int*. 2006;69:1996–2002.
 159. Heyman SN, Lieberthal W, Rogiers P, et al. Animal models of acute tubular necrosis. *Curr Opin Crit Care*. 2002;8:526–534.
 160. de Groh ED, Swanhart LM, Cosentino CC, et al. Inhibition of histone deacetylase expands the renal progenitor cell population. *J Am Soc Nephrol*. 2010;21:794–802.
 161. McKee RA, Wingert RA. Zebrafish renal pathology: emerging models of acute kidney injury. *Curr Pathobiol Rep*. 2015;3:171–181.
 162. Wilmer MJ, Ng CP, Lanz HL, et al. Kidney-on-a-chip technology for drug-induced nephrotoxicity screening. *Trends Biotechnol*. 2016;34:156–170.
 163. Essig M, Terzi F, Burtin M, et al. Mechanical strains induced by tubular flow affect the phenotype of proximal tubular cells. *Am J Physiol Renal Physiol*. 2001;281:F751–F762.

164. Du Z, Yan Q, Duan Y, et al. Axial flow modulates proximal tubule NHE3 and H-ATPase activities by changing microvillus bending moments. *Am J Physiol Renal Physiol*. 2006;290:F289–F296.
165. Bhat VD, Windridge PA, Cherry RS, et al. Fluctuating shear stress effects on stress fiber architecture and energy metabolism of cultured renal cells. *Biotechnol Prog*. 1995;11:596–600.
166. Maggiorani D, Dissard R, Belloy M, et al. Shear stress-induced alteration of epithelial organization in human renal tubular cells. *PLoS ONE*. 2015;10:e0131416.
167. Qu Y, An F, Luo Y, et al. A nephron model for study of drug-induced acute kidney injury and assessment of drug-induced nephrotoxicity. *Biomaterials*. 2018;155:41–53.
168. Morizane R, Miyoshi T, Bonventre JV. Concise review: kidney generation with human pluripotent stem cells. *Stem Cells*. 2017;35:2209–2217.
169. Takasato M, Er PX, Chiu HS, et al. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature*. 2015;526:564–568.
170. Lam AQ, Freedman BS, Morizane R, et al. Rapid and efficient differentiation of human pluripotent stem cells into intermediate mesoderm that forms tubules expressing kidney proximal tubular markers. *J Am Soc Nephrol*. 2014;25:1211–1225.
171. Takasato M, Er PX, Croft M, et al. Directing human embryonic stem cell differentiation towards a renal lineage generates a self-organizing kidney. *Nat Cell Biol*. 2014;16:118–126.
172. Taguchi A, Kaku Y, Ohmori T, et al. Redefining the in vivo origin of metanephric nephron progenitors enables generation of complex kidney structures from pluripotent stem cells. *Cell Stem Cell*. 2014;14:53–67.
173. Venkatachalam MA, Bernard DB, Donohoe JF, et al. Ischemic damage and repair in the rat proximal tubule: differences among the S1, S2, and S3 segments. *Kidney Int*. 1978;14:31–49.
174. Alejandro V, Scandling JD Jr, Sibley RK, et al. Mechanisms of filtration failure during postischemic injury of the human kidney. A study of the reperfused renal allograft. *J Clin Invest*. 1995;95:820–831.
175. Ramaswamy D, Corrigan G, Polhemus C, et al. Maintenance and recovery stages of postischemic acute renal failure in humans. *Am J Physiol Renal Physiol*. 2002;282:F271–F280.
176. Oliver J, Mac DM, Tracy A. The pathogenesis of acute renal failure associated with traumatic and toxic injury; renal ischemia, nephrotoxic damage and the ischemic episode. *J Clin Invest*. 1951;30:1307–1439.
177. Solez K, Morel-Maroger L, Sraer JD. The morphology of 'acute tubular necrosis' in man: analysis of 57 renal biopsies and a comparison with the glycerol model. *Medicine (Baltimore)*. 1979;58:362–376.
178. Kornreich M, Avinery R, Malka-Gibor E, et al. Order and disorder in intermediate filament proteins. *FEBS Lett*. 2015;589:2464–2476.
179. Han SJ, Kim JH, Kim JI, et al. Inhibition of microtubule dynamics impedes repair of kidney ischemia/reperfusion injury and increases fibrosis. *Sci Rep*. 2016;6:27775.
180. Molitoris BA. Actin cytoskeleton in ischemic acute renal failure. *Kidney Int*. 2004;66:871–883.
181. Molitoris BA, Dahl R, Hosford M. Cellular ATP depletion induces disruption of the spectrin cytoskeletal network. *Am J Physiol*. 1996;271:F790–F798.
182. Atkinson SJ, Hosford MA, Molitoris BA. Mechanism of actin polymerization in cellular ATP depletion. *J Biol Chem*. 2004;279:5194–5199.
183. Raman N, Atkinson SJ. Rho controls actin cytoskeletal assembly in renal epithelial cells during ATP depletion and recovery. *Am J Physiol*. 1999;276:C1312–C1324.
184. Ashworth SL, Southgate EL, Sandoval RM, et al. ADF/cofilin mediates actin cytoskeletal alterations in LLC-PK cells during ATP depletion. *Am J Physiol Renal Physiol*. 2003;284:F852–F862.
185. Ashworth SL, Sandoval RM, Hosford M, et al. Ischemic injury induces ADF relocalization to the apical domain of rat proximal tubule cells. *Am J Physiol Renal Physiol*. 2001;280:F886–F894.
186. Muller S, Djudjaj S, Lange J, et al. HIF stabilization inhibits renal epithelial cell migration and is associated with cytoskeletal alterations. *Sci Rep*. 2018;8:9497.
187. Ashworth SL, Wean SE, Campos SB, et al. Renal ischemia induces tropomyosin dissociation-destabilizing microvilli microfilaments. *Am J Physiol Renal Physiol*. 2004;286:F988–F996.
188. Suurna MV, Ashworth SL, Hosford M, et al. Cofilin mediates ATP depletion-induced endothelial cell actin alterations. *Am J Physiol Renal Physiol*. 2006;290:F1398–F1407.
189. Wagner MC, Rhodes G, Wang E, et al. Ischemic injury to kidney induces glomerular podocyte effacement and dissociation of slit diaphragm proteins Nephl and ZO-1. *J Biol Chem*. 2008;283:35579–35589.
190. Du J, Zhang L, Yang Y, et al. ATP depletion-induced actin rearrangement reduces cell adhesion via p38 MAPK-HSP27 signaling in renal proximal tubule cells. *Cell Physiol Biochem*. 2010;25:501–510.
191. Molitoris BA, Geerdes A, McIntosh JR. Dissociation and redistribution of Na⁺,K⁺-ATPase from its surface membrane actin cytoskeletal complex during cellular ATP depletion. *J Clin Invest*. 1991;88:462–469.
192. Molitoris BA. Na⁺(+)-K⁺-ATPase that redistributes to apical membrane during ATP depletion remains functional. *Am J Physiol*. 1993;265:F693–F697.
193. Sogabe K, Roeser NF, Davis JA, et al. Calcium dependence of integrity of the actin cytoskeleton of proximal tubule cell microvilli. *Am J Physiol*. 1996;271:F292–F303.
194. Portilla D. Role of fatty acid beta-oxidation and calcium-independent phospholipase A2 in ischemic acute renal failure. *Curr Opin Nephrol Hypertens*. 1999;8:473–477.
195. Linkermann A, Green DR. Necroptosis. *N Engl J Med*. 2014;370:455–465.
196. Galluzzi L, Vitale I, Abrams JM, et al. Molecular definitions of cell death subroutines: recommendations of the Nomenclature Committee on Cell Death 2012. *Cell Death Differ*. 2012;19:107–120.
197. Tonnus W, Gembardt F, Latk M, et al. The clinical relevance of necroinflammation – spotlight on the adrenal glands. *Cell Death Differ*. 2018;In press.
198. Lieberthal W, Koh JS, Levine JS. Necrosis and apoptosis in acute renal failure. *Semin Nephrol*. 1998;18:505–518.
199. Bonogio R, Lieberthal W. Role of apoptosis in the pathogenesis of acute renal failure. *Curr Opin Nephrol Hypertens*. 2002;11:301–308.
200. Guo R, Wang Y, Minto AW, et al. Acute renal failure in endotoxemia is dependent on caspase activation. *J Am Soc Nephrol*. 2004;15:3093–3102.
201. Safirstein RL. Acute renal failure: from renal physiology to the renal transcriptome. *Kidney Int Suppl*. 2004;2004:S62–S66.
202. Nicholson DW. From bench to clinic with apoptosis-based therapeutic agents. *Nature*. 2000;407:810–816.
203. Kaufmann SH, Hengartner MO. Programmed cell death: alive and well in the new millennium. *Trends Cell Biol*. 2001;11:526–534.
204. Lee RH, Song JM, Park MY, et al. Cisplatin-induced apoptosis by translocation of endogenous Bax in mouse collecting duct cells. *Biochem Pharmacol*. 2001;62:1013–1023.
205. Peherstorfer E, Mayer B, Boehm S, et al. Effects of microinjection of synthetic Bcl-2 domain peptides on apoptosis of renal tubular epithelial cells. *Am J Physiol Renal Physiol*. 2002;283:F190–F196.
206. Kelly KJ, Plotkin Z, Vulgamott SL, et al. P53 mediates the apoptotic response to GTP depletion after renal ischemia-reperfusion: protective role of a p53 inhibitor. *J Am Soc Nephrol*. 2003;14:128–138.
207. Edelstein LC, Lagos L, Simmons M, et al. NF-kappa B-dependent assembly of an enhanceosome-like complex on the promoter region of apoptosis inhibitor Bfl-1/A1. *Mol Cell Biol*. 2003;23:2749–2761.
208. Park KM, Chen A, Bonventre JV. Prevention of kidney ischemia/reperfusion-induced functional injury and JNK. *J Biol Chem*. 2001;276(11870–11876):p38-and MAPK kinase activation by remote ischemic pretreatment.
209. Scheid MP, Schubert KM, Duronio V. Regulation of bad phosphorylation and association with Bcl-x(L) by the MAPK/Erk kinase. *J Biol Chem*. 1999;274:31108–31113.
210. Molitoris BA, Dagher PC, Sandoval RM, et al. siRNA targeted to p53 attenuates ischemic and cisplatin-induced acute kidney injury. *J Am Soc Nephrol*. 2009;20:1754–1764.
211. Lorenzen JM, Kaucsar T, Schauer C, et al. MicroRNA-24 antagonism prevents renal ischemia reperfusion injury. *J Am Soc Nephrol*. 2014;25:2717–2729.
212. Edelstein CL. Calcium-mediated proximal tubular injury-what is the role of cysteine proteases? *Nephrol Dial Transplant*. 2000;15:141–144.
213. Liu X, Van Vleet T, Schnellmann RG. The role of calpain in oncotic cell death. *Annu Rev Pharmacol Toxicol*. 2004;44:349–370.
214. Liu X, Schnellmann RG. Calpain mediates progressive plasma membrane permeability and proteolysis of cytoskeleton-associated paxillin, talin, and vinculin during renal cell death. *J Pharmacol Exp Ther*. 2003;304:63–70.
215. Devarajan P. Cellular and molecular derangements in acute tubular necrosis. *Curr Opin Pediatr*. 2005;17:193–199.

216. Sarhan M, Land WG, Tonnus W, et al. Origin and consequences of necroinflammation. *Physiol Rev*. 2018;98:727–780.
217. Holler N, Zaru R, Micheau O, et al. Fas triggers an alternative, caspase-8-independent cell death pathway using the kinase RIP as effector molecule. *Nat Immunol*. 2000;1:489–495.
218. Degterev A, Hitomi J, Gerscheid M, et al. Identification of RIP1 kinase as a specific cellular target of necrostatins. *Nat Chem Biol*. 2008;4:313–321.
219. Zhang DW, Shao J, Lin J, et al. RIP3, an energy metabolism regulator that switches TNF-induced cell death from apoptosis to necrosis. *Science*. 2009;325:332–336.
220. Sun L, Wang H, Wang Z, et al. Mixed lineage kinase domain-like protein mediates necrosis signaling downstream of RIP3 kinase. *Cell*. 2012;148:213–227.
221. Wang H, Sun L, Su L, et al. Mixed lineage kinase domain-like protein MLKL causes necrotic membrane disruption upon phosphorylation by RIP3. *Mol Cell*. 2014;54:133–146.
222. Linkermann A, Brasen JH, Darding M, et al. Two independent pathways of regulated necrosis mediate ischemia-reperfusion injury. *Proc Natl Acad Sci USA*. 2013;110:12024–12029.
223. Xu Y, Ma H, Shao J, et al. A role for tubular necroptosis in cisplatin-induced AKI. *J Am Soc Nephrol*. 2015;26:2647–2658.
224. Yang WS, Stockwell BR. Synthetic lethal screening identifies compounds activating iron-dependent, nonapoptotic cell death in oncogenic-RAS-harboring cancer cells. *Chem Biol*. 2008;15:234–245.
225. Yang WS, Stockwell BR. Ferroptosis: death by lipid peroxidation. *Trends Cell Biol*. 2016;26:165–176.
226. Kers J, Leemans JC, Linkermann A. An overview of pathways of regulated necrosis in acute kidney injury. *Semin Nephrol*. 2016;36:139–152.
227. Linkermann A, Skouta R, Himmerkus N, et al. Synchronized renal tubular cell death involves ferroptosis. *Proc Natl Acad Sci USA*. 2014;111:16836–16841.
228. Yagoda N, von Rechenberg M, Zaganjor E, et al. RAS-RAF-MEK-dependent oxidative cell death involving voltage-dependent anion channels. *Nature*. 2007;447:864–868.
229. Dolma S, Lessnick SL, Hahn WC, et al. Identification of genotype-selective antitumor agents using synthetic lethal chemical screening in engineered human tumor cells. *Cancer Cell*. 2003;3:285–296.
230. Wolpaw AJ, Shimada K, Skouta R, et al. Modulatory profiling identifies mechanisms of small molecule-induced cell death. *Proc Natl Acad Sci USA*. 2011;108:E771–E780.
231. Martin-Sanchez D, Ruiz-Andres O, Poveda J, et al. Ferroptosis, but not necroptosis, is important in nephrotoxic folic acid-induced AKI. *J Am Soc Nephrol*. 2017;28:218–229.
232. Adedoyin O, Boddu R, Traylor AM, et al. Heme oxygenase-1 mitigates ferroptosis in renal proximal tubule cells. *Am J Physiol Renal Physiol*. ajrenal.00044.02017. 2017.
233. Fink SL, Cookson BT. Apoptosis, pyroptosis, and necrosis: mechanistic description of dead and dying eukaryotic cells. *Infect Immun*. 2005;73:1907–1916.
234. Sharma D, Kanneganti TD. The cell biology of inflammasomes: mechanisms of inflammasome activation and regulation. *J Cell Biol*. 2016;213:617–629.
235. Fink SL, Cookson BT. Caspase-1-dependent pore formation during pyroptosis leads to osmotic lysis of infected host macrophages. *Cell Microbiol*. 2006;8:1812–1825.
236. Baroja-Mazo A, Martin-Sanchez F, Gomez AI, et al. The NLRP3 inflammasome is released as a particulate danger signal that amplifies the inflammatory response. *Nat Immunol*. 2014;15:738–748.
237. Havasi A, Dong Z. Autophagy and tubular cell death in the kidney. *Semin Nephrol*. 2016;36:174–188.
238. Huber TB, Edelstein CL, Hartleben B, et al. Emerging role of autophagy in kidney function, diseases and aging. *Autophagy*. 2012;8:1009–1031.
239. Ariel A, Ravichandran KS. ‘This way please’: apoptotic cells regulate phagocyte migration before and after engulfment. *Eur J Immunol*. 2016;46:1583–1586.
240. Ravichandran KS. Beginnings of a good apoptotic meal: the find-me and eat-me signaling pathways. *Immunity*. 2011;35:445–455.
241. Guan X, Qian Y, Shen Y, et al. Autophagy protects renal tubular cells against ischemia / reperfusion injury in a time-dependent manner. *Cell Physiol Biochem*. 2015;36:285–298.
242. Jiang M, Liu K, Luo J, et al. Autophagy is a renoprotective mechanism during in vitro hypoxia and in vivo ischemia-reperfusion injury. *Am J Pathol*. 2010;176:1181–1192.
243. Kimura T, Takabatake Y, Takahashi A, et al. Autophagy protects the proximal tubule from degeneration and acute ischemic injury. *J Am Soc Nephrol*. 2011;22:902–913.
244. Liu S, Hartleben B, Kretz O, et al. Autophagy plays a critical role in kidney tubule maintenance, aging and ischemia-reperfusion injury. *Autophagy*. 2012;8:826–837.
245. Livingston MJ, Ding HF, Huang S, et al. Persistent activation of autophagy in kidney tubular cells promotes renal interstitial fibrosis during unilateral ureteral obstruction. *Autophagy*. 2016;12(6):976–998.
246. Livingston MJ, Dong Z. Autophagy in acute kidney injury. *Semin Nephrol*. 2014;34:17–26.
247. Periyasamy-Thandavan S, Jiang M, Wei Q, et al. Autophagy is cytoprotective during cisplatin injury of renal proximal tubular cells. *Kidney Int*. 2008;74:631–640.
248. Mei S, Livingston M, Hao J, et al. Autophagy is activated to protect against endotoxic acute kidney injury. *Sci Rep*. 2016;6:22171.
249. Livingston MJ, Ding HF, Huang S, et al. Persistent activation of autophagy in kidney tubular cells promotes renal interstitial fibrosis during unilateral ureteral obstruction. *Autophagy*. 2016;12:976–998.
250. Braun H, Schmidt BM, Raiss M, et al. Cellular senescence limits regenerative capacity and allograft survival. *J Am Soc Nephrol*. 2012;23:1467–1473.
251. Brinkmann V, Reichard U, Goosmann C, et al. Neutrophil extracellular traps kill bacteria. *Science*. 2004;303:1532–1535.
252. Brinkmann V, Zychlinsky A. Beneficial suicide: why neutrophils die to make NETs. *Nat Rev Microbiol*. 2007;5:577–582.
253. Clark SR, Ma AC, Tavener SA, et al. Platelet TLR4 activates neutrophil extracellular traps to ensnare bacteria in septic blood. *Nat Med*. 2007;13:463–469.
254. Vanden Berghe T, Linkermann A, Jouan-Lanhouet S, et al. Regulated necrosis: the expanding network of non-apoptotic cell death pathways. *Nat Rev Mol Cell Biol*. 2014;15:135–147.
255. Nakazawa D, Kumar SV, Marschner J, et al. Histones and neutrophil extracellular traps enhance tubular necrosis and remote organ injury in ischemic AKI. *J Am Soc Nephrol*. 2017;28:1753–1768.
256. Kaissling B, Le Hir M. The renal cortical interstitium: morphological and functional aspects. *Histochem Cell Biol*. 2008;130:247–262.
257. Nelson PJ, Rees AJ, Griffin MD, et al. The renal mononuclear phagocytic system. *J Am Soc Nephrol*. 2011.
258. Ferenbach D, Hughes J. Macrophages and dendritic cells: what is the difference? *Kidney Int*. 2008;74:5–7.
259. Geissmann F, Manz MG, Jung S, et al. Development of monocytes, macrophages, and dendritic cells. *Science*. 2010;327:656–661.
260. Gordon S, Taylor PR. Monocyte and macrophage heterogeneity. *Nat Rev Immunol*. 2005;5:953–964.
261. Janeway CA Jr, Medzhitov R. Innate immune recognition. *Annu Rev Immunol*. 2002;20:197–216.
262. Matzinger P. Tolerance, danger, and the extended family. *Annu Rev Immunol*. 1994;12:991–1045.
263. Rosin DL, Okusa MD. Dangers within: DAMP responses to damage and cell death in kidney disease. *J Am Soc Nephrol*. 2011;22:416–425.
264. Soos TJ, Sims TN, Barisoni L, et al. CX3CR1+ interstitial dendritic cells form a contiguous network throughout the entire kidney. *Kidney Int*. 2006;70:591–596.
265. Anders HJ, Banas B, Schlondorff D. Signaling danger: toll-like receptors and their potential roles in kidney disease. *J Am Soc Nephrol*. 2004;15:854–867.
266. Okusa MD, Li L. Dendritic cells in acute kidney injury: cues from the microenvironment. *Trans Am Clin Climatol Assoc*. 2012;123:54–62, discussion 62–53.
267. Teteris SA, Engel DR, Kurts C. Homeostatic and pathogenic role of renal dendritic cells. *Kidney Int*. 2011;80:139–145.
268. Li L, Huang L, Vergis AL, et al. IL-17 produced by neutrophils regulates IFN-gamma-mediated neutrophil migration in mouse kidney ischemia-reperfusion injury. *J Clin Invest*. 2010;120:331–342.
269. Li L, Huang L, Sung SS, et al. The chemokine receptors CCR2 and CX3CR1 mediate monocyte/macrophage trafficking in kidney ischemia-reperfusion injury. *Kidney Int*. 2008;74:1509–1511.
270. Buch T, Heppner FL, Tertilt C, et al. A Cre-inducible diphtheria toxin receptor mediates cell lineage ablation after toxin administration. *Nat Methods*. 2005;2:419–426.
271. Li L, Okusa MD. Macrophages, dendritic cells, and kidney ischemia-reperfusion injury. *Semin Nephrol*. 2010;30:268–277.

272. Bajwa A, Huang L, Ye H, et al. Dendritic cell sphingosine 1-phosphate receptor-3 regulates Th1-Th2 polarity in kidney ischemia-reperfusion injury. *J Immunol*. 2012;189:2584–2596.
273. Park SW, Kim M, Brown KM, et al. Inhibition of sphingosine 1-phosphate receptor 2 protects against renal ischemia-reperfusion injury. *J Am Soc Nephrol*. 2012;23:266–280.
274. Dong X, Swaminathan S, Bachman LA, et al. Resident dendritic cells are the predominant TNF-secreting cell in early renal ischemia-reperfusion injury. *Kidney Int*. 2007;71:619–628.
275. Kim MG, Boo CS, Ko YS, et al. Depletion of kidney CD11c+ F4/80+ cells impairs the recovery process in ischaemia/reperfusion-induced acute kidney injury. *Nephrol Dial Transplant*. 2010;25:2908–2921.
276. Morelli AE, Thomson AW. Tolerogenic dendritic cells and the quest for transplant tolerance. *Nat Rev Immunol*. 2007;7:610–621.
277. Hu J, Wan Y. Tolerogenic dendritic cells and their potential applications. *Immunology*. 2011;132:307–314.
278. Li L, Huang L, Ye H, et al. Dendritic cells tolerized with adenosine A(2)AR agonist attenuate acute kidney injury. *J Clin Invest*. 2012;122:3931–3942.
279. Gilboa E. DC-based cancer vaccines. *J Clin Invest*. 2007;117:1195–1203.
280. Tadagavadi RK, Reeves WB. Renal dendritic cells ameliorate nephrotoxic acute kidney injury. *J Am Soc Nephrol*. 2010;21:53–63.
281. Dong X, Swaminathan S, Bachman LA, et al. Antigen presentation by dendritic cells in renal lymph nodes is linked to systemic and local injury to the kidney. *Kidney Int*. 2005;68:1096–1108.
282. Kawakami T, Lichtnekert J, Thompson LJ, et al. Resident renal mononuclear phagocytes comprise five discrete populations with distinct phenotypes and functions. *J Immunol*. 2013;191:3358–3372.
283. Huen SC, Cantley LG. Macrophages in renal injury and repair. *Annu Rev Physiol*. 2017;79:449–469.
284. Geissmann F, Jung S, Littman DR. Blood monocytes consist of two principal subsets with distinct migratory properties. *Immunity*. 2003;19:71–82.
285. Wilson HM, Walbaum D, Rees AJ. Macrophages and the kidney. *Curr Opin Nephrol Hypertens*. 2004;13:285–290.
286. Fogg DK, Sibon C, Miled C, et al. A clonogenic bone marrow progenitor specific for macrophages and dendritic cells. *Science*. 2006;311:83–87.
287. Sunderkotter C, Nikolic T, Dillon MJ, et al. Subpopulations of mouse blood monocytes differ in maturation stage and inflammatory response. *J Immunol*. 2004;172:4410–4417.
288. Dauer M, Obermaier B, Herten J, et al. Mature dendritic cells derived from human monocytes within 48 hours: a novel strategy for dendritic cell differentiation from blood precursors. *J Immunol*. 2003;170:4069–4076.
289. Mohamadzadeh M, Berard F, Essert G, et al. Interleukin 15 skews monocyte differentiation into dendritic cells with features of Langerhans cells. *J Exp Med*. 2001;194:1013–1020.
290. Chomarat P, Dantin C, Bennett L, et al. TNF skews monocyte differentiation from macrophages to dendritic cells. *J Immunol*. 2003;171:2262–2269.
291. Delneste Y, Charbonnier P, Herbault N, et al. Interferon-gamma switches monocyte differentiation from dendritic cells to macrophages. *Blood*. 2003;101:143–150.
292. Chomarat P, Banchereau J, Davoust J, et al. IL-6 switches the differentiation of monocytes from dendritic cells to macrophages. *Nat Immunol*. 2000;1:510–514.
293. Furuichi K, Wada T, Iwata Y, et al. CCR2 signaling contributes to ischemia-reperfusion injury in kidney. *J Am Soc Nephrol*. 2003;14:2503–2515.
294. Martinez FO, Gordon S. The M1 and M2 paradigm of macrophage activation: time for reassessment. *F1000Prime Rep*. 2014;6:13.
295. Martinez FO, Helming L, Gordon S. Alternative activation of macrophages: an immunologic functional perspective. *Annu Rev Immunol*. 2009;27:451–483.
296. van Rooijen N, Sanders A, van den Berg TK. Apoptosis of macrophages induced by liposome-mediated intracellular delivery of clodronate and propamidine. *J Immunol Methods*. 1996;193:93–99.
297. Day YJ, Huang L, Ye H, et al. Renal ischemia-reperfusion injury and adenosine 2A receptor-mediated tissue protection: role of macrophages. *Am J Physiol Renal Physiol*. 2005;288:F722–F731.
298. Ricardo SD, van Goor H, Eddy AA. Macrophage diversity in renal injury and repair. *J Clin Invest*. 2008;118:3522–3530.
299. Lin SL, Li B, Rao S, et al. Macrophage Wnt7b is critical for kidney repair and regeneration. *Proc Natl Acad Sci USA*. 2010;107:4194–4199.
300. Jang HR, Rabb H. Immune cells in experimental acute kidney injury. *Nat Rev Nephrol*. 2015;11:88–101.
301. Li L, Huang L, Sung SJ, et al. NKT cell activation mediates neutrophil IFN-gamma production and renal ischemia-reperfusion injury. *J Immunol*. 2007;178:5899–5911.
302. Sica A, Mantovani A. Macrophage plasticity and polarization: in vivo veritas. *J Clin Invest*. 2012;122:787–795.
303. Lee S, Huen S, Nishio H, et al. Distinct macrophage phenotypes contribute to kidney injury and repair. *J Am Soc Nephrol*. 2011;22:317–326.
304. Hato T, Zollman A, Plotkin Z, et al. Endotoxin preconditioning reprograms S1 tubules and macrophages to protect the kidney. *J Am Soc Nephrol*. 2018;29:104–117.
305. Awad AS, Rouse M, Huang L, et al. Compartmentalization of neutrophils in the kidney and lung following acute ischemic kidney injury. *Kidney Int*. 2009;75:689–698.
306. Okusa MD, Linden J, Huang L, et al. A2a-Adenosine receptor mediated inhibition of renal injury and neutrophil adhesion. *Am J Physiol*. 2000;279:F809–F818.
307. Akcay A, Nguyen Q, Edelstein CL. Mediators of inflammation in acute kidney injury. *Mediators Inflamm*. 2009;2009:137072.
308. Gluba A, Banach M, Hannam S, et al. The role of Toll-like receptors in renal diseases. *Nat Rev Nephrol*. 2010;6:224–235.
309. Wu H, Chen G, Wyburn KR, et al. TLR4 activation mediates kidney ischemia/reperfusion injury. *J Clin Invest*. 2007;117:2847–2859.
310. Chaturvedi S, Yuen DA, Bajwa A, et al. Slit2 prevents neutrophil recruitment and renal ischemia-reperfusion injury. *J Am Soc Nephrol*. 2013;24:1274–1287.
311. Kelly KJ, Williams WW, Colvin RB, et al. Intercellular adhesion molecule-1-deficient mice are protected against ischemic renal injury. *J Clin Invest*. 1996;97:1056–1063.
312. Burne-Taney MJ, Yokota-Ikeda N, Rabb H. Effects of combined T- and B-cell deficiency on murine ischemia reperfusion injury. *Am J Transplant*. 2005;5:1186–1193.
313. Salmi M, Jalkanen S. VAP-1: an adhesin and an enzyme. *Trends Immunol*. 2001;22:211–216.
314. Pannecoeck R, Serruys D, Benmeridja L, et al. Vascular adhesion protein-1: role in human pathology and application as a biomarker. *Crit Rev Clin Lab Sci*. 2015;52:284–300.
315. Tanaka S, Tanaka T, Kawakami T, et al. Vascular adhesion protein-1 enhances neutrophil infiltration by generation of hydrogen peroxide in renal ischemia/reperfusion injury. *Kidney Int*. 2017;92:154–164.
316. Burne-Taney MJ, Ascon DB, Daniels F, et al. B cell deficiency confers protection from renal ischemia reperfusion injury. *J Immunol*. 2003;171:3210–3215.
317. Kinsey GR, Li L, Okusa MD. Inflammation in acute kidney injury. *Nephron Exp Nephrol*. 2008;109:e102–e107.
318. Kinsey GR, Okusa MD. Expanding role of T cells in acute kidney injury. *Curr Opin Nephrol Hypertens*. 2014;23:9–16.
319. Kinsey GR, Sharma R, Okusa MD. Regulatory T cells in AKI. *J Am Soc Nephrol*. 2013;24:1720–1726.
320. Tsuboi N, Yoshikai Y, Matsuo S, et al. Roles of toll-like receptors in C-C chemokine production by renal tubular epithelial cells. *J Immunol*. 2002;169:2026–2033.
321. El-Achkar TM, Huang X, Plotkin Z, et al. Sepsis induces changes in the expression and distribution of Toll-like receptor 4 in the rat kidney. *Am J Physiol Renal Physiol*. 2006;290:F1034–F1043.
322. Rusai K, Sollinger D, Baumann M, et al. Toll-like receptors 2 and 4 in renal ischemia/reperfusion injury. *Pediatr Nephrol*. 2010;25:853–860.
323. Zhang B, Ramesh G, Uematsu S, et al. TLR4 signaling mediates inflammation and tissue injury in nephrotoxicity. *J Am Soc Nephrol*. 2008;19:923–932.
324. Burne MJ, Daniels F, El Ghandour A, et al. Identification of the CD4+ T cell as a major pathogenic factor in ischemic acute renal failure. *J Clin Invest*. 2001;108:1283–1290.
325. Burne-Taney MJ, Rabb H. The role of adhesion molecules and T cells in ischemic renal injury. *Curr Opin Nephrol Hypertens*. 2003;12:85–90.
326. Seino K, Motohashi S, Fujisawa T, et al. Natural killer T cell-mediated antitumor immune responses and their clinical applications. *Cancer Sci*. 2006;97:807–812.
327. Kinjo Y, Kronenberg M. Valpha14i NKT cells are innate lymphocytes that participate in the immune response to diverse microbes. *J Clin Immunol*. 2005;25:522–533.

328. Kinjo Y, Tupin E, Wu D, et al. Natural killer T cells recognize diacylglycerol antigens from pathogenic bacteria. *Nat Immunol.* 2006;7:978–986.
329. Miyake S, Yamamura T. Therapeutic potential of glycolipid ligands for natural killer (NK) T cells in the suppression of autoimmune diseases. *Curr Drug Targets Immune Endocr Metabol Disord.* 2005;5:315–322.
330. Yu KO, Porcelli SA. The diverse functions of CD1d-restricted NKT cells and their potential for immunotherapy. *Immunol Lett.* 2005;100:42–55.
331. Mercer JC, Ragin MJ, August A. Natural killer T cells: rapid responders controlling immunity and disease. *Int J Biochem Cell Biol.* 2005;37:1337–1343.
332. Van Kaer L. Regulation of immune responses by CD1d-restricted natural killer T cells. *Immunol Res.* 2004;30:139–153.
333. Zhang C, Zhang J, Tian Z. The regulatory effect of natural killer cells: do “NK-reg cells” exist? *Cell Mol Immunol.* 2006;3:241–254.
334. Bendelac A, Savage PB, Teyton L. The biology of NKT cells. *Annu Rev Immunol.* 2006.
335. Montoya CJ, Jie HB, Al-Harthi L, et al. Activation of plasmacytoid dendritic cells with TLR9 agonists initiates invariant NKT cell-mediated cross-talk with myeloid dendritic cells. *J Immunol.* 2006;177:1028–1039.
336. Nagaraj S, Ziske C, Strehl J, et al. Dendritic cells pulsed with alpha-galactosylceramide induce anti-tumor immunity against pancreatic cancer in vivo. *Int Immunol.* 2006;18:1279–1283.
337. Toyofuku A, Yasunami Y, Nabeyama K, et al. Natural killer T-cells participate in rejection of islet allografts in the liver of mice. *Diabetes.* 2006;55:34–39.
338. Yasunami Y, Kojo S, Kitamura H, et al. Valpha14 NK T cell-triggered IFN-gamma production by Gr-1+CD11b+ cells mediates early graft loss of syngeneic transplanted islets. *J Exp Med.* 2005;202:913–918.
339. Day YJ, Huang L, McDuffie MJ, et al. Renal protection from ischemia mediated by A2A adenosine receptors on bone marrow-derived cells. *J Clin Invest.* 2003;112:883–891.
340. Day YJ, Huang L, Ye H, et al. Renal ischemia-reperfusion and adenosine 2A receptor-mediated tissue protection: the role of CD4+ T cells and interferon gamma. *J Immunol.* 2006;176:3108–3114.
341. Sharma R, Kinsey GR. Regulatory T cells in acute and chronic kidney diseases. *Am J Physiol Renal Physiol.* 2018;314:F679–F698.
342. Liu M, Chien CC, Grigoryev DN, et al. Effect of T cells on vascular permeability in early ischemic acute kidney injury in mice. *Microvasc Res.* 2009;77:340–347.
343. Kinsey GR, Sharma R, Huang L, et al. Regulatory T cells suppress innate immunity in kidney ischemia-reperfusion injury. *J Am Soc Nephrol.* 2009;20:1744–1753.
344. Francisco LM, Sage PT, Sharpe AH. The PD-1 pathway in tolerance and autoimmunity. *Immunol Rev.* 2010;236:219–242.
345. Riella LV, Paterson AM, Sharpe AH, et al. Role of the PD-1 pathway in the immune response. *Am J Transplant.* 2012;12:2575–2587.
346. Jaworska K, Ratajczak J, Huang L, et al. Both PD-1 ligands protect the kidney from ischemia reperfusion injury. *J Immunol.* 2015;194:325–333.
347. Strembska ME, Jose S, Sabapathy V, et al. IL233, a novel IL-2 and IL-33 hybrid cytokine, ameliorates renal injury. *J Am Soc Nephrol.* 2017;28:2681–2693.
348. Licon-Limon P, Kim LK, Palm NW, et al. TH2, allergy and group 2 innate lymphoid cells. *Nat Immunol.* 2013;14:536–542.
349. Renner B, Strassheim D, Amura CR, et al. B cell subsets contribute to renal injury and renal protection after ischemia/reperfusion. *J Immunol.* 2010;185:4393–4400.
350. Lobo PI, Schlegel KH, Bajwa A, et al. Natural IgM switches the function of lipopolysaccharide-activated murine bone marrow-derived dendritic cells to a regulatory dendritic cell that suppresses innate inflammation. *J Immunol.* 2015;195:5215–5226.
351. Joao C, Ogle BM, Gay-Rabinstein C, et al. B cell-dependent TCR diversification. *J Immunol.* 2004;172:4709–4716.
352. Ray A, Basu S, Williams CB, et al. A novel IL-10-independent regulatory role for B cells in suppressing autoimmunity by maintenance of regulatory T cells via GITR ligand. *J Immunol.* 2012;188:3188–3198.
353. Zhang M, Austen WG Jr, Chiu I, et al. Identification of a specific self-reactive IgM antibody that initiates intestinal ischemia/reperfusion injury. *Proc Natl Acad Sci USA.* 2004;101:3886–3891.
354. Zhang M, Michael LH, Grosjean SA, et al. The role of natural IgM in myocardial ischemia-reperfusion injury. *J Mol Cell Cardiol.* 2006;41:62–67.
355. Lobo PI, Bajwa A, Schlegel KH, et al. Natural IgM anti-leukocyte autoantibodies attenuate excess inflammation mediated by innate and adaptive immune mechanisms involving Th-17. *J Immunol.* 2012;188:1675–1685.
356. Savransky V, Molls RR, Burne-Taney MJ, et al. Role of T cell receptor in kidney ischemia reperfusion injury. *Kidney Int.* 2006.
357. Ashworth SL, Sandoval RM, Tanner GA, et al. Two-photon microscopy: visualization of kidney dynamics. *Kidney Int.* 2007;72:416–421.
358. Thurman A, Jo SK, Ye H, et al. Activation of sphingosine-1-phosphate 1 receptor in the proximal tubule protects against ischemia-reperfusion injury. *J Am Soc Nephrol.* 2010;21:955–965.
359. Thurman HM, Huang L, Ye H, et al. Endothelial sphingosine 1-phosphate receptor1 mediates protection and recovery from acute kidney injury. *J Am Soc Nephrol.* 2016;27:3383–3393.
360. Chen J, Matzuk MM, Zhou XJ, et al. Endothelial pentraxin 3 contributes to murine ischemic acute kidney injury. *Kidney Int.* 2012;82:1195–1207.
361. Xu C, Chang A, Hack BK, et al. TNF-mediated damage to glomerular endothelium is an important determinant of acute kidney injury in sepsis. *Kidney Int.* 2013.
362. Hato T, El-Achkar TM, Dagher PC. Sisters in arms: myeloid and tubular epithelial cells shape renal innate immunity. *Am J Physiol Renal Physiol.* 2013;304:F1243–F1251.
363. Kalakeche R, Hato T, Rhodes G, et al. Endotoxin uptake by S1 proximal tubular segment causes oxidative stress in the downstream S2 segment. *J Am Soc Nephrol.* 2011;22:1505–1516.
364. Leemans JC, Stokman G, Claessen N, et al. Renal-associated TLR2 mediates ischemia/reperfusion injury in the kidney. *J Clin Invest.* 2005;115:2894–2903.
365. Choong FX, Sandoval RM, Molitoris BA, et al. Multiphoton microscopy applied for real-time intravital imaging of bacterial infections in vivo. *Methods Enzymol.* 2012;506:35–61.
366. Mansson LE, Melican K, Boekel J, et al. Real-time studies of the progression of bacterial infections and immediate tissue responses in live animals. *Cell Microbiol.* 2007;9:413–424.
367. Melican K, Boekel J, Mansson LE, et al. Bacterial infection-mediated mucosal signalling induces local renal ischaemia as a defence against sepsis. *Cell Microbiol.* 2008;10:1987–1998.
368. El-Achkar TM, Wu XR. Uromodulin in kidney injury: an instigator, bystander, or protector? *Am J Kidney Dis.* 2012;59:452–461.
369. Serafini-Cessi F, Malagolini N, Cavallone D. Tamm-Horsfall glycoprotein: biology and clinical relevance. *Am J Kidney Dis.* 2003;42:658–676.
370. El-Achkar TM, McCracken R, Rauchman M, et al. Tamm-Horsfall protein-deficient thick ascending limbs promote injury to neighboring S3 segments in an MIP-2-dependent mechanism. *Am J Physiol Renal Physiol.* 2011;300:F999–F1007.
371. El Achkar TM, Wu XR, Rauchman M, et al. Tamm-Horsfall protein protects the kidney from ischemic injury by decreasing inflammation and altering TLR4 expression. *Am J Physiol Renal Physiol.* 2008;295:F534–F544.
372. El-Achkar TM, McCracken R, Liu Y, et al. Tamm-Horsfall protein translocates to the basolateral domain of thick ascending limbs, interstitium, and circulation during recovery from acute kidney injury. *Am J Physiol Renal Physiol.* 2013;304:F1066–F1075.
373. Micanovic R, Khan S, Janosevic D, et al. Tamm-Horsfall protein regulates mononuclear phagocytes in the kidney. *J Am Soc Nephrol.* 2018;29:841–856.
374. Ricklin D, Lambris JD. Preformed mediators of defense-Gatekeepers enter the spotlight. *Immunol Rev.* 2016;274:5–8.
375. Ricklin D, Mastellos DC, Reis ES, et al. The renaissance of complement therapeutics. *Nat Rev Nephrol.* 2018;14:26–47.
376. Freely S, Kemper C, Le Fric G. The “ins and outs” of complement-driven immune responses. *Immunol Rev.* 2016;274:16–32.
377. Thurman JM, Nester CM. All things complement. *Clin J Am Soc Nephrol.* 2016;11:1856–1866.
378. Mather DR, Heeger PS. Molecules great and small: the complement system. *Clin J Am Soc Nephrol.* 2015;10:1636–1650.
379. Riedemann NC, Guo RF, Ward PA. The enigma of sepsis. *J Clin Invest.* 2003;112:460–467.
380. Thurman JM, Ljubanovic D, Edelstein CL, et al. Lack of a functional alternative complement pathway ameliorates ischemic acute renal failure in mice. *J Immunol.* 2003;170:1517–1523.
381. Xavier S, Sahu RK, Landes SG, et al. Pericytes and immune cells contribute to complement activation in tubulointerstitial fibrosis. *Am J Physiol Renal Physiol.* 2017;312:F516–F532.

382. Nezu M, Suzuki N, Yamamoto M. Targeting the KEAP1-NRF2 system to prevent kidney disease progression. *Am J Nephrol*. 2017;45:473–483.
383. Galli F, Piroddi M, Annetti C, et al. Oxidative stress and reactive oxygen species. *Contrib Nephrol*. 2005;149:240–260.
384. Pinsky MR. Pathophysiology of sepsis and multiple organ failure: pro- versus anti-inflammatory aspects. *Contrib Nephrol*. 2004;144:31–43.
385. Kelly KJ. Stress response proteins and renal ischemia. *Minerva Urol Nephrol*. 2002;54:81–91.
386. Paller MS, Weber K, Patten M. Nitric oxide-mediated renal epithelial cell injury during hypoxia and reoxygenation. *Ren Fail*. 1998;20:459–469.
387. Bidmon B, Endemann M, Muller T, et al. HSP-25 and HSP-90 stabilize Na,K-ATPase in cytoskeletal fractions of ischemic rat renal cortex. *Kidney Int*. 2002;62:1620–1627.
388. Barrera-Chimal J, Perez-Villalva R, Ortega JA, et al. Intra-renal transfection of heat shock protein 90 alpha or beta (Hsp90alpha or Hsp90beta) protects against ischemia/reperfusion injury. *Nephrol Dial Transplant*. 2014;29:301–312.
389. Nayak B, Xie P, Akagi S, et al. Modulation of renal-specific oxidoreductase/myo-inositol oxygenase by high-glucose ambience. *Proc Natl Acad Sci USA*. 2005;102:17952–17957.
390. Dutta RK, Kondeti VK, Sharma I, et al. Beneficial effects of myo-inositol oxygenase deficiency in cisplatin-induced AKI. *J Am Soc Nephrol*. 2017;28:1421–1436.
391. Liu M, Grigoryev DN, Crow MT, et al. Transcription factor Nrf2 is protective during ischemic and nephrotoxic acute kidney injury in mice. *Kidney Int*. 2009;76:277–285.
392. Wu QQ, Wang Y, Senitko M, et al. Bardoxolone methyl (BARD) ameliorates ischemic AKI and increases expression of protective genes Nrf2, PPARgamma, and HO-1. *Am J Physiol Renal Physiol*. 2011;300:F1180–F1192.
393. Pergola PE, Raskin P, Toto RD, et al. Bardoxolone methyl and kidney function in CKD with type 2 diabetes. *N Engl J Med*. 2011;365:327–336.
394. Tan RJ, Chartoumpekis DV, Rush BM, et al. Keap1 hypomorphism protects against ischemic and obstructive kidney disease. *Sci Rep*. 2016;6:36185.
395. Swaminathan S. Iron, hormesis, and protection in acute kidney injury. *Kidney Int*. 2016;90:16–17.
396. Baliga R, Zhang Z, Baliga M, et al. In vitro and in vivo evidence suggesting a role for iron in cisplatin-induced nephrotoxicity. *Kidney Int*. 1998;53:394–401.
397. Winterbourn CC. Toxicity of iron and hydrogen peroxide: the Fenton reaction. *Toxicol Lett*. 1995;82-83:969–974.
398. Shah SV, Rajapurkar MM, Baliga R. The role of catalytic iron in acute kidney injury. *Clin J Am Soc Nephrol*. 2011;6:2329–2331.
399. Baliga R, Ueda N, Walker PD, et al. Oxidant mechanisms in toxic acute renal failure. [Review] [84 refs]. *Drug Metab Rev*. 1999;31:971–997.
400. Baliga R, Ueda N, Walker PD, et al. Oxidant mechanisms in toxic acute renal failure. *Drug Metab Rev*. 1999;31:971–997.
401. Shah SV, Walker PD. Evidence suggesting a role for hydroxyl radical in glycerol-induced acute renal failure. *Am J Physiol*. 1988;255:F438–F443.
402. Paller MS. Hemoglobin- and myoglobin-induced acute renal failure in rats: role of iron in nephrotoxicity. *Am J Physiol*. 1988;255:F539–F544.
403. Walker VJ, Agarwal A. Targeting iron homeostasis in acute kidney injury. *Semin Nephrol*. 2016;36:62–70.
404. Scindia Y, Dey P, Thirunagari A, et al. Hepcidin mitigates renal ischemia-reperfusion injury by modulating systemic iron homeostasis. *J Am Soc Nephrol*. 2015;26:2800–2814.
405. Baliga R, Zhang Z, Baliga M, et al. Evidence for cytochrome P-450 as a source of catalytic iron in myoglobinuric acute renal failure. *Kidney Int*. 1996;49:362–369.
406. Baliga R, Ueda N, Shah SV. Increase in bleomycin-detectable iron in ischaemia/reperfusion injury to rat kidneys. *Biochem J*. 1993;291(Pt 3):901–905.
407. van Swelm RP, Wetzels JF, Verweij VG, et al. Renal handling of circulating and renal-synthesized hepcidin and its protective effects against hemoglobin-mediated kidney injury. *J Am Soc Nephrol*. 2016;27:2720–2732.
408. Bolisetty S, Zarjou A, Agarwal A. Heme oxygenase 1 as a therapeutic target in acute kidney injury. *Am J Kidney Dis*. 2017;69:531–545.
409. Nath KA. Heme oxygenase-1 and acute kidney injury. *Curr Opin Nephrol Hypertens*. 2014;23:17–24.
410. Agarwal A, Bolisetty S. Adaptive responses to tissue injury: role of heme oxygenase-1. *Trans Am Clin Climatol Assoc*. 2013;124:111–122.
411. Nath KA. Heme oxygenase-1: a provenance for cytoprotective pathways in the kidney and other tissues. *Kidney Int*. 2006;70:432–443.
412. Bolisetty S, Traylor A, Joseph R, et al. Proximal tubule-targeted heme oxygenase-1 in cisplatin-induced acute kidney injury. *Am J Physiol Renal Physiol*. 2016;310:F385–F394.
413. Shiraishi F, Curtis LM, Truong L, et al. Heme oxygenase-1 gene ablation or expression modulates cisplatin-induced renal tubular apoptosis. *Am J Physiol Renal Physiol*. 2000;278:F726–F736.
414. Nath KA, Haggard JJ, Croatt AJ, et al. The indispensability of heme oxygenase-1 in protecting against acute heme protein-induced toxicity in vivo. *Am J Pathol*. 2000;156:1527–1535.
415. Inguaggiato P, Gonzalez-Michaca L, Croatt AJ, et al. Cellular overexpression of heme oxygenase-1 up-regulates p21 and confers resistance to apoptosis. *Kidney Int*. 2001;60:2181–2191.
416. Adedoyin O, Boddu R, Traylor A, et al. Heme oxygenase-1 mitigates ferroptosis in renal proximal tubule cells. *Am J Physiol Renal Physiol*. 2018;314:F702–F714.
417. Kapturczak MH, Wasserfall C, Brusko T, et al. Heme oxygenase-1 modulates early inflammatory responses: evidence from the heme oxygenase-1-deficient mouse. *Am J Pathol*. 2004;165:1045–1053.
418. Ferenbach DA, Bonventre JV. Mechanisms of maladaptive repair after AKI leading to accelerated kidney ageing and CKD. *Nat Rev Nephrol*. 2015;11:264–276.
419. Gupta S, Verfaillie C, Chmielewski D, et al. Isolation and characterization of kidney-derived stem cells. *J Am Soc Nephrol*. 2006;17:3028–3040.
420. Bussolati B, Bruno S, Grange C, et al. Isolation of renal progenitor cells from adult human kidney. *Am J Pathol*. 2005;166:545–555.
421. De Broe ME. Tubular regeneration and the role of bone marrow cells: 'stem cell therapy'—a panacea? *Nephrol Dial Transplant*. 2005;20:2318–2320.
422. Fang TC, Alison MR, Cook HT, et al. Proliferation of bone marrow-derived cells contributes to regeneration after folic acid-induced acute tubular injury. *J Am Soc Nephrol*. 2005;16:1723–1732.
423. Poulsom R, Forbes SJ, Hodivala-Dilke K, et al. Bone marrow contributes to renal parenchymal turnover and regeneration. *J Pathol*. 2001;195:229–235.
424. Duffield JS, Park KM, Hsiao LL, et al. Restoration of tubular epithelial cells during repair of the postischemic kidney occurs independently of bone marrow-derived stem cells. *J Clin Invest*. 2005;115:1743–1755.
425. Chang-Panesso M, Humphreys BD. Cellular plasticity in kidney injury and repair. *Nat Rev Nephrol*. 2017;13:39–46.
426. Humphreys BD, Valerius MT, Kobayashi A, et al. Intrinsic epithelial cells repair the kidney after injury. *Cell Stem Cell*. 2008;2:284–291.
427. Humphreys BD, Czerniak S, DiRocco DP, et al. Repair of injured proximal tubule does not involve specialized progenitors. *Proc Natl Acad Sci USA*. 2011;108:9226–9231.
428. Kusaba T, Lalli M, Kramann R, et al. Differentiated kidney epithelial cells repair injured proximal tubule. *Proc Natl Acad Sci USA*. 2014;111:1527–1532.
429. Berger K, Bangen JM, Hammerich L, et al. Origin of regenerating tubular cells after acute kidney injury. *Proc Natl Acad Sci USA*. 2014;111:1533–1538.
430. Lindgren D, Bostrom AK, Nilsson K, et al. Isolation and characterization of progenitor-like cells from human renal proximal tubules. *Am J Pathol*. 2011;178:828–837.
431. Smeets B, Boor P, Dijkman H, et al. Proximal tubular cells contain a phenotypically distinct, scattered cell population involved in tubular regeneration. *J Pathol*. 2013;229:645–659.
432. Hansson J, Hulténby K, Cramnert C, et al. Evidence for a morphologically distinct and functionally robust cell type in the proximal tubules of human kidney. *Hum Pathol*. 2014;45:382–393.
433. Lazzeri E, Angelotti ML, Peired A, et al. Endocycle-related tubular cell hypertrophy and progenitor proliferation recover renal function after acute kidney injury. *Nat Commun*. 2018;9:1344.
434. Hammerman MR. Growth factors and apoptosis in acute renal injury. *Curr Opin Nephrol Hypertens*. 1998;7:419–424.
435. Fiaschi-Taesch NM, Santos S, Reddy V, et al. Prevention of acute ischemic renal failure by targeted delivery of growth factors to the proximal tubule in transgenic mice: the efficacy of parathyroid hormone-related protein and hepatocyte growth factor. *J Am Soc Nephrol*. 2004;15:112–125.

436. Matsumoto M, Makino Y, Tanaka T, et al. Induction of renoprotective gene expression by cobalt ameliorates ischemic injury of the kidney in rats. *J Am Soc Nephrol*. 2003;14:1825–1832.
437. Vannay A, Fekete A, Adori C, et al. Divergence of renal vascular endothelial growth factor mRNA expression and protein level in post-ischaemic rat kidneys. *Exp Physiol*. 2004;89:435–444.
438. Hladunewich MA, Corrigan G, Derby GC, et al. A randomized, placebo-controlled trial of IGF-1 for delayed graft function: a human model to study postischemic ARF. *Kidney Int*. 2003;64:593–602.
439. Hirschberg R, Kopple J, Lipsett P, et al. Multicenter clinical trial of recombinant human insulin-like growth factor I in patients with acute renal failure. *Kidney Int*. 1999;55:2423–2432.
440. Bruno S, Tapparo M, Collino F, et al. Renal regenerative potential of different extracellular vesicle populations derived from bone marrow mesenchymal stromal cells. *Tissue Eng Part A*. 2017.
441. Erdbrugger U, Le TH. Extracellular vesicles in renal diseases: more than novel biomarkers? *J Am Soc Nephrol*. 2016;27:12–26.
442. Bruno S, Grange C, Derogibus MC, et al. Mesenchymal stem cell-derived microvesicles protect against acute tubular injury. *J Am Soc Nephrol*. 2009;20:1053–1067.
443. Molitoris BA, Sutton TA. Endothelial injury and dysfunction: role in the extension phase of acute renal failure. *Kidney Int*. 2004;66:496–499.
444. Conger JD, Schrier RW. Renal hemodynamics in acute renal failure. *Annu Rev Physiol*. 1980;42:603–614.
445. Conger JD, Robinette JB, Hammond WS. Differences in vascular reactivity in models of ischemic acute renal failure. *Kidney Int*. 1991;39:1087–1097.
446. Hellberg POA, Källskog Ö, Wolgast M. Red cell trapping and postischemic renal blood flow. Differences between the cortex, outer and inner medulla. *Kidney Int*. 1991;40:625–631.
447. Noiri E, Nakao A, Uchida K, et al. Oxidative and nitrosative stress in acute renal ischemia. *Am J Physiol Renal Physiol*. 2001;281:F948–F957.
448. Ling H, Edelstein C, Gengaro P, et al. Attenuation of renal ischemia-reperfusion injury in inducible nitric oxide synthase knockout mice. *Am J Physiol*. 1999;277:F383–F390.
449. Goligorsky MS, Noiri E. Duality of nitric oxide in acute renal injury. *Semin Nephrol*. 1999;19:263–271.
450. Goligorsky MS, Brodsky SV, Noiri E. NO bioavailability, endothelial dysfunction, and acute renal failure: new insights into pathophysiology. *Semin Nephrol*. 2004;24:316–323.
451. Mattson DL, Wu F. Control of arterial blood pressure and renal sodium excretion by nitric oxide synthase in the renal medulla. *Acta Physiol Scand*. 2000;168:149–154.
452. Chander V, Chopra K. Renal protective effect of molsidomine and L-arginine in ischemia-reperfusion induced injury in rats. *J Surg Res*. 2005;128:132–139.
453. Langenberg C, Bellomo R, May C, et al. Renal blood flow in sepsis. *Crit Care*. 2005;9:R363–R374.
454. Wan L, Langenberg C, Bellomo R, et al. Angiotensin II in experimental hyperdynamic sepsis. *Crit Care*. 2009;13:R190.
455. Khanna A, English SW, Wang XS, et al. Angiotensin II for the Treatment of Vasodilatory Shock. *N Engl J Med*. 2017.
456. Bogatcheva NV, Verin AD. The role of cytoskeleton in the regulation of vascular endothelial barrier function. *Microvasc Res*. 2008;76:202–207.
457. Kramann R, Humphreys BD. Kidney pericytes: roles in regeneration and fibrosis. *Semin Nephrol*. 2014;34:374–383.
458. Kramann R, Wongboonsin J, Chang-Panesso M, et al. Gli1+ pericyte loss induces capillary rarefaction and proximal tubular injury. *J Am Soc Nephrol*. 2017;28:776–784.
459. Lemos DR, Marsh G, Huang A, et al. Maintenance of vascular integrity by pericytes is essential for normal kidney function. *Am J Physiol Renal Physiol*. 2016;311:F1230–F1242.
460. Chelazzi C, Villa G, Mancinelli P, et al. Glycocalyx and sepsis-induced alterations in vascular permeability. *Crit Care*. 2015;19:26.
461. Woodcock TE, Woodcock TM. Revised Starling equation and the glycocalyx model of transvascular fluid exchange: an improved paradigm for prescribing intravenous fluid therapy. *Br J Anaesth*. 2012;108:384–394.
462. Schmidt EP, Lee WL, Zemans RL, et al. On, around, and through: neutrophil-endothelial interactions in innate immunity. *Physiology*. 2011;26:334–347.
463. Sutton TA, Kelly KJ, Mang HE, et al. Minocycline reduces renal microvascular leakage in a rat model of ischemic renal injury. *Am J Physiol Renal Physiol*. 2004.
464. Lee SY, Horbelt M, Mang HE, et al. MMP-9 gene deletion mitigates microvascular loss in a model of ischemic acute kidney injury. *Am J Physiol Renal Physiol*. 2011;301:F101–F109.
465. Horbelt M, Lee SY, Mang HE, et al. Acute and chronic microvascular alterations in a mouse model of ischemic acute kidney injury. *Am J Physiol Renal Physiol*. 2007;293:F688–F695.
466. Gupta A, Rhodes GJ, Berg DT, et al. Activated protein C ameliorates LPS-induced acute kidney injury and downregulates renal iNOS and angiotensin 2. *Am J Physiol Renal Physiol*. 2007;293:F245–F254.
467. Sharfuddin AA, Sandoval RM, Berg DT, et al. Soluble thrombomodulin protects ischemic kidneys. *J Am Soc Nephrol*. 2009;20:524–534.
468. Singbartl K, Ley K. Leukocyte recruitment and acute renal failure. *J Mol Med*. 2004;82:91–101.
469. Burne MJ, Rabb H. Pathophysiological contributions of fucosyltransferases in renal ischemia reperfusion injury. *J Immunol*. 2002;169:2648–2652.
470. Matsukawa A, Lukacs NW, Hogaboam CM, et al. Mice genetically lacking endothelial selectins are resistant to the lethality in septic peritonitis. *Exp Mol Pathol*. 2002;72:68–76.
471. Dehnadi A, Benedict Cosimi A, Neal Smith R, et al. Prophylactic orthosteric inhibition of leukocyte integrin CD11b/CD18 prevents long-term fibrotic kidney failure in cynomolgus monkeys. *Nat Commun*. 2017;8:13899.
472. Souza AC, Yuen PS, Star RA. Microparticles: markers and mediators of sepsis-induced microvascular dysfunction, immunosuppression, and AKI. *Kidney Int*. 2015;87:1100–1108.
473. Zafrani L, Gerotziakas G, Byrnes C, et al. Calpastatin controls polymicrobial sepsis by limiting procoagulant microparticle release. *Am J Respir Crit Care Med*. 2012;185:744–755.
474. Chawla LS, Eggers PW, Star RA, et al. Acute kidney injury and chronic kidney disease as interconnected syndromes. *N Engl J Med*. 2014;371:58–66.
475. Venkatachalam MA, Griffin KA, Lan R, et al. Acute kidney injury: a springboard for progression in chronic kidney disease. *Am J Physiol Renal Physiol*. 2010.
476. Cerda J, Lameire N, Eggers P, et al. Epidemiology of acute kidney injury. *Clin J Am Soc Nephrol*. 2008;3:881–886.
477. Hsu CY, Chertow GM, McCulloch CE, et al. Nonrecovery of kidney function and death after acute on chronic renal failure. *Clin J Am Soc Nephrol*. 2009;4:891–898.
478. Hsu CY. Linking the population epidemiology of acute renal failure, chronic kidney disease and end-stage renal disease. *Curr Opin Nephrol Hypertens*. 2007;16:221–226.
479. Grgic I, Campanholle G, Bijol V, et al. Targeted proximal tubule injury triggers interstitial fibrosis and glomerulosclerosis. *Kidney Int*. 2012;82:172–183.
480. Schrimpf C, Duffield JS. Mechanisms of fibrosis: the role of the pericyte. *Curr Opin Nephrol Hypertens*. 2011;20:297–305.
481. Liu BC, Tang TT, Lv LL, et al. Renal tubule injury: a driving force toward chronic kidney disease. *Kidney Int*. 2018.
482. Choi YJ, Chakraborty S, Nguyen V, et al. Peritubular capillary loss is associated with chronic tubulointerstitial injury in human kidney: altered expression of vascular endothelial growth factor. *Hum Pathol*. 2000;31:1491–1497.
483. Basile DP. The endothelial cell in ischemic acute kidney injury: implications for acute and chronic function. *Kidney Int*. 2007;72:151–156.
484. Ehling J, Babickova J, Gremse F, et al. Quantitative micro-computed tomography imaging of vascular dysfunction in progressive kidney diseases. *J Am Soc Nephrol*. 2016;27:520–532.
485. Kawakami T, Mimura I, Shoji K, et al. Hypoxia and fibrosis in chronic kidney disease: crossing at pericytes. *Kidney Int Suppl* (2011). 2014;4:107–112.
486. Tanaka S, Tanaka T, Nangaku M. Hypoxia as a key player in the AKI-to-CKD transition. *Am J Physiol Renal Physiol*. 2014;307:F1187–F1195.
487. Basile DP, Bonventre JV, Mehta R, et al. Progression after AKI: understanding maladaptive repair processes to predict and identify therapeutic treatments. *J Am Soc Nephrol*. 2016;27:687–697.
488. Leonard EC, Friedrich JL, Basile DP. VEGF-121 preserves renal microvessel structure and ameliorates secondary renal disease following acute kidney injury. *Am J Physiol Renal Physiol*. 2008;295:F1648–F1657.
489. Zager RA, Johnson AC. Renal ischemia-reperfusion injury upregulates histone-modifying enzyme systems and alters histone expression at proinflammatory/profibrotic genes. *Am J Physiol Renal Physiol*. 2009;296:F1032–F1041.

490. Shahrzad S, Bertrand K, Minhas K, et al. Induction of DNA hypomethylation by tumor hypoxia. *Epigenetics*. 2007;2:119–125.
491. Tang J, Zhuang S. Epigenetics in acute kidney injury. *Curr Opin Nephrol Hypertens*. 2015;24:351–358.
492. Nangaku M, Hirakawa Y, Mimura I, et al. Epigenetic changes in the acute kidney injury-to-chronic kidney disease transition. *Nephron*. 2017;137:256–259.
493. Basile DP, Leonard EC, Beal AG, et al. Persistent oxidative stress following renal ischemia-reperfusion injury increases ANG II hemodynamic and fibrotic activity. *Am J Physiol Renal Physiol*. 2012;302:F1494–F1502.
494. Yang L, Brooks CR, Xiao S, et al. KIM-1-mediated phagocytosis reduces acute injury to the kidney. *J Clin Invest*. 2015;125:1620–1636.
495. Humphreys BD, Xu F, Sabbisetti V, et al. Chronic epithelial kidney injury molecule-1 expression causes murine kidney fibrosis. *J Clin Invest*. 2013;123:4023–4035.
496. Krishnamurthy J, Torrice C, Ramsey MR, et al. Ink4a/Arf expression is a biomarker of aging. *J Clin Invest*. 2004;114:1299–1307.
497. Humphreys BD. Mechanisms of renal fibrosis. *Annu Rev Physiol*. 2018;80:309–326.
498. Cianciollo Cosentino C, Skrypnik NI, Brilli LL, et al. Histone deacetylase inhibitor enhances recovery after AKI. *J Am Soc Nephrol*. 2013;24:943–953.
499. Tang J, Liu N, Tolbert E, et al. Sustained activation of EGFR triggers renal fibrogenesis after acute kidney injury. *Am J Pathol*. 2013;183:160–172.
500. Huen SC, Huynh L, Marlier A, et al. GM-CSF promotes macrophage alternative activation after renal ischemia/reperfusion injury. *J Am Soc Nephrol*. 2015;26:1334–1345.
501. Zhang MZ, Yao B, Yang S, et al. CSF-1 signaling mediates recovery from acute kidney injury. *J Clin Invest*. 2012;122:4519–4532.
502. Clements M, Gershenovich M, Chaber C, et al. Differential Ly6C expression after renal ischemia-reperfusion identifies unique macrophage populations. *J Am Soc Nephrol*. 2016;27:159–170.
503. Lin SL, Castano AP, Nowlin BT, et al. Bone marrow Ly6Chigh monocytes are selectively recruited to injured kidney and differentiate into functionally distinct populations. *J Immunol*. 2009;183:6733–6743.
504. Cao Q, Harris DC, Wang Y. Macrophages in kidney injury, inflammation, and fibrosis. *Physiology*. 2015;30:183–194.
505. Kalluri R, Zeisberg M. Fibroblasts in cancer. *Nat Rev Cancer*. 2006;6:392–401.
506. Gabbiani G. The myofibroblast in wound healing and fibrocontractile diseases. *J Pathol*. 2003;200:500–503.
507. Haase VH. Pathophysiological consequences of HIF activation: HIF as a modulator of fibrosis. *Ann N Y Acad Sci*. 2009;1177:57–65.
508. Wick G, Backovic A, Rabensteiner E, et al. The immunology of fibrosis: innate and adaptive responses. *Trends Immunol*. 2010;31:110–119.
509. Yang L, Besschetnova TY, Brooks CR, et al. Epithelial cell cycle arrest in G2/M mediates kidney fibrosis after injury. *Nat Med*. 2010;16:535–543, 531p following 143.
510. Bechtel W, McGoochan S, Zeisberg EM, et al. Methylation determines fibroblast activation and fibrogenesis in the kidney. *Nat Med*. 2010;16:544–550.
511. Wynn TA. Fibrosis under arrest. *Nat Med*. 2010;16:523–525.
512. LeBleu VS, Taduri G, O'Connell J, et al. Origin and function of myofibroblasts in kidney fibrosis. *Nat Med*. 2013;19:1047–1053.
513. Humphreys BD, Lin SL, Kobayashi A, et al. Fate tracing reveals the pericyte and not epithelial origin of myofibroblasts in kidney fibrosis. *Am J Pathol*. 2010;176:85–97.
514. Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. *Nat Med*. 2012;18:1028–1040.
515. Asada N, Takase M, Nakamura J, et al. Dysfunction of fibroblasts of extrarenal origin induces renal fibrosis and renal anemia in mice. *J Clin Invest*. 2011;121:3981–3990.
516. Lin SL, Kisseleva T, Brenner DA, et al. Pericytes and perivascular fibroblasts are the primary source of collagen-producing cells in obstructive fibrosis of the kidney. *Am J Pathol*. 2008;173:1617–1627.
517. Zeisberg EM, Potenta SE, Sugimoto H, et al. Fibroblasts in kidney fibrosis emerge via endothelial-to-mesenchymal transition. *J Am Soc Nephrol*. 2008;19:2282–2287.
518. Zeisberg EM, Tarnavski O, Zeisberg M, et al. Endothelial-to-mesenchymal transition contributes to cardiac fibrosis. *Nat Med*. 2007;13:952–961.
519. Boyer B, Tucker GC, Valles AM, et al. Reversible transition towards a fibroblastic phenotype in a rat carcinoma cell line. *Int J Cancer Suppl*. 1989;4:69–75.
520. Miettinen PJ, Ebner R, Lopez AR, et al. TGF-beta induced transdifferentiation of mammary epithelial cells to mesenchymal cells: involvement of type I receptors. *J Cell Biol*. 1994;127:2021–2036.
521. Okada H, Danoff TM, Kalluri R, et al. Early role of Fsp1 in epithelial-mesenchymal transformation. *Am J Physiol*. 1997;273:F563–F574.
522. Boutet A, De Frutos CA, Maxwell PH, et al. Snail activation disrupts tissue homeostasis and induces fibrosis in the adult kidney. *EMBO J*. 2006;25:5603–5613.
523. Lamouille S, Xu J, Derynck R. Molecular mechanisms of epithelial-mesenchymal transition. *Nat Rev Mol Cell Biol*. 2014;15:178–196.
524. Lovisa S, LeBleu VS, Tampe B, et al. Epithelial-to-mesenchymal transition induces cell cycle arrest and parenchymal damage in renal fibrosis. *Nat Med*. 2015;21:998–1009.
525. Grande MT, Sanchez-Laorden B, Lopez-Blau C, et al. Snail1-induced partial epithelial-to-mesenchymal transition drives renal fibrosis in mice and can be targeted to reverse established disease. *Nat Med*. 2015;21:989–997.
526. Ovadya Y, Krizhanovsky V. A new Twist in kidney fibrosis. *Nat Med*. 2015;21:975–977.
527. Maarouf OH, Aravamudhan A, Rangarajan D, et al. Paracrine Wnt1 drives interstitial fibrosis without inflammation by tubulointerstitial cross-talk. *J Am Soc Nephrol*. 2016;27:781–790.
528. Maarouf OH, Ikeda Y, Humphreys BD. Wnt signaling in kidney tubulointerstitium during disease. *Histol Histopathol*. 2015;30:163–171.
529. Zhou D, Tan RJ, Zhou L, et al. Kidney tubular beta-catenin signaling controls interstitial fibroblast fate via epithelial-mesenchymal communication. *Sci Rep*. 1878;3:2013.
530. Zhou D, Fu H, Zhang L, et al. Tubule-derived Wnts are required for fibroblast activation and kidney fibrosis. *J Am Soc Nephrol*. 2017.
531. English D, Welch Z, Kovala AT, et al. Sphingosine 1-phosphate released from platelets during clotting accounts for the potent endothelial cell chemotactic activity of blood serum and provides a novel link between hemostasis and angiogenesis. *FASEB J*. 2000;14:2255–2265.
532. An S, Zheng Y, Bleu T. Sphingosine 1-phosphate-induced cell proliferation, survival, and related signaling events mediated by G protein-coupled receptors Edg3 and Edg5. *J Biol Chem*. 2000;275:288–296.
533. Donati C, Cencetti F, Nincheri P, et al. Sphingosine 1-phosphate mediates proliferation and survival of mesoangioblasts. *Stem Cells*. 2007;25:1713–1719.
534. Goetzl EJ, Lee H, Dolezalova H, et al. Mechanisms of lysolipid phosphate effects on cellular survival and proliferation. *Ann N Y Acad Sci*. 2000;905:177–187.
535. Karliner JS, Honbo N, Summers K, et al. The lysophospholipids sphingosine-1-phosphate and lysophosphatidic acid enhance survival during hypoxia in neonatal rat cardiac myocytes. *J Mol Cell Cardiol*. 2001;33:1713–1717.
536. Maceyka M, Payne SG, Miltien S, et al. Sphingosine kinase, sphingosine-1-phosphate, and apoptosis. *Biochim Biophys Acta*. 2002;1585:193–201.
537. Hait NC, Allegood J, Maceyka M, et al. Regulation of histone acetylation in the nucleus by sphingosine-1-phosphate. *Science*. 2009;325:1254–1257.
538. Bajwa A, Huang L, Kurmaeva E, et al. Sphingosine kinase 2 deficiency attenuates kidney fibrosis via IFN-gamma. *J Am Soc Nephrol*. 2016;28:1145–1161.
539. Schwalm S, Beyer S, Frey H, et al. Sphingosine Kinase-2 deficiency ameliorates kidney fibrosis by up-regulating Smad7 in a mouse model of unilateral ureteral obstruction. *Am J Pathol*. 2017;187:2413–2429.
540. Wynn TA. Cellular and molecular mechanisms of fibrosis. *J Pathol*. 2008;214:199–210.
541. Keane MP, Belperio JA, Burdick MD, et al. IL-12 attenuates bleomycin-induced pulmonary fibrosis. *Am J Physiol Lung Cell Mol Physiol*. 2001;281:L92–L97.
542. Gurujeyalakshmi G, Giri SN. Molecular mechanisms of antifibrotic effect of interferon gamma in bleomycin-mouse model of lung fibrosis: downregulation of TGF-beta and procollagen I and III gene expression. *Exp Lung Res*. 1995;21:791–808.
543. Oldroyd SD, Thomas GL, Gabbiani G, et al. Interferon-gamma inhibits experimental renal fibrosis. *Kidney Int*. 1999;56:2116–2127.
544. Schrimpf C, Xin C, Campanholle G, et al. Pericyte TIMP3 and ADAMTS1 modulate vascular stability after kidney injury. *J Am Soc Nephrol*. 2012;23:868–883.
545. Handsley MM, Edwards DR. Metalloproteinases and their inhibitors in tumor angiogenesis. *Int J Cancer*. 2005;115:849–860.

546. Wu CF, Chiang WC, Lai CF, et al. Transforming growth factor beta-1 stimulates profibrotic epithelial signaling to activate pericyte-myofibroblast transition in obstructive kidney fibrosis. *Am J Pathol*. 2013;182:118–131.
547. Lin SL, Chang FC, Schrimpf C, et al. Targeting endothelium-pericyte cross talk by inhibiting VEGF receptor signaling attenuates kidney microvascular rarefaction and fibrosis. *Am J Pathol*. 2011;178:911–923.
548. Kang HM, Ahn SH, Choi P, et al. Defective fatty acid oxidation in renal tubular epithelial cells has a key role in kidney fibrosis development. *Nat Med*. 2015;21:37–46.
549. Kramann R, DiRocco DP, Humphreys BD. Understanding the origin, activation and regulation of matrix-producing myofibroblasts for treatment of fibrotic disease. *J Pathol*. 2013;231:273–289.
550. Decleves AE, Sharma K. Novel targets of antifibrotic and anti-inflammatory treatment in CKD. *Nat Rev Nephrol*. 2014;10:257–267.
551. Wick G, Grundtman C, Mayerl C, et al. The immunology of fibrosis. *Annu Rev Immunol*. 2013;31:107–135.
552. Tampe D, Zeisberg M. Potential approaches to reverse or repair renal fibrosis. *Nat Rev Nephrol*. 2014;10:226–237.
553. Boor P, Ostendorf T, Floege J. Renal fibrosis: novel insights into mechanisms and therapeutic targets. *Nat Rev Nephrol*. 2010;6:643–656.
554. de Seigneux S, Martin PY. Preventing the Progression of AKI to CKD: the Role of Mitochondria. *J Am Soc Nephrol*. 2017.
555. Birk AV, Liu S, Soong Y, et al. The mitochondrial-targeted compound SS-31 re-energizes ischemic mitochondria by interacting with cardiolipin. *J Am Soc Nephrol*. 2013;24:1250–1261.
556. Eirin A, Ebrahimi B, Kwon SH, et al. Restoration of mitochondrial cardiolipin attenuates cardiac damage in swine renovascular hypertension. *J Am Heart Assoc*. 2016;5.
557. Sabbah HN, Gupta RC, Kohli S, et al. Chronic therapy with elamipretide (MTP-131), a novel mitochondria-targeting peptide, improves left ventricular and mitochondrial function in dogs with advanced heart failure. *Circ Heart Fail*. 2016;9:e002206.
558. Sweetwyne MT, Pippin JW, Eng DG, et al. The mitochondrial-targeted peptide, SS-31, improves glomerular architecture in mice of advanced age. *Kidney Int*. 2017.
559. Perry HM, Huang L, Wilson RJ, et al. Dynamin-related protein 1 deficiency promotes recovery from AKI. *J Am Soc Nephrol*. 2018; 29:194–206.
560. Uchino S, Kellum JA, Bellomo R, et al. Acute renal failure in critically ill patients: a multinational, multicenter study. *JAMA*. 2005;294:813–818.
561. Doi K, Rabb H. Impact of acute kidney injury on distant organ function: recent findings and potential therapeutic targets. *Kidney Int*. 2016;89:555–564.
562. Okusa MD, Rosin DL, Tracey KJ. Targeting neural reflex circuits in immunity to treat kidney disease. *Nat Rev Nephrol*. 2017;13:669–680.
563. Inoue T, Abe C, Sung SS, et al. Vagus nerve stimulation mediates protection from kidney ischemia-reperfusion injury through alpha7nAChR+ splenocytes. *J Clin Invest*. 2016;126:1939–1952.
564. Abe C, Inoue T, Inglis MA, et al. C1 neurons mediate a stress-induced anti-inflammatory reflex in mice. *Nat Neurosci*. 2017;20:700–707.
565. Gigliotti JC, Okusa MD. The spleen: the forgotten organ in acute kidney injury of critical illness. *Nephron Clin Pract*. 2014;127: 153–157.
566. Gigliotti JC, Huang L, Ye H, et al. Ultrasound prevents renal ischemia-reperfusion injury by stimulating the splenic cholinergic anti-inflammatory pathway. *J Am Soc Nephrol*. 2013;24:1451–1460.
567. Rabb H, Wang Z, Nemoto T, et al. Acute renal failure leads to dysregulation of lung salt and water channels. *Kidney Int*. 2003;63:600–606.
568. Liu M, Liang Y, Chigurupati S, et al. Acute kidney injury leads to inflammation and functional changes in the brain. *J Am Soc Nephrol*. 2008;19:1360–1370.
569. Basu RK, Wheeler DS. Kidney-lung cross-talk and acute kidney injury. *Pediatr Nephrol*. 2013;28:2239–2248.
570. Imai Y, Parodo J, Kajikawa O, et al. Injurious mechanical ventilation and end-organ epithelial cell apoptosis and organ dysfunction in an experimental model of acute respiratory distress syndrome. *JAMA*. 2003;289:2104–2112.
571. Pratschke J, Wilhelm MJ, Laskowski I, et al. Influence of donor brain death on chronic rejection of renal transplants in rats. *J Am Soc Nephrol*. 2001;12:2474–2481.
572. de Vries DK, Lindeman JH, Ringers J, et al. Donor brain death predisposes human kidney grafts to a proinflammatory reaction after transplantation. *Am J Transplant*. 2011;11:1064–1070.
573. Pavlov VA, Tracey KJ. Neural regulation of immunity: molecular mechanisms and clinical translation. *Nature neuroscience*. 2017.
574. Tracey KJ. Reflex control of immunity. *Nat Rev Immunol*. 2009;9: 418–428.
575. Tracey KJ. The inflammatory reflex. *Nature*. 2002;420:853–859.
576. Tracey KJ. Physiology and immunology of the cholinergic anti-inflammatory pathway. *J Clin Invest*. 2007;117:289–296.
577. Borovikova LV, Ivanova S, Zhang M, et al. Vagus nerve stimulation attenuates the systemic inflammatory response to endotoxin. *Nature*. 2000;405:458–462.
578. Nijima A, Hori T, Katafuchi T, et al. The effect of interleukin-1 beta on the efferent activity of the vagus nerve to the thymus. *J Auton Nerv Syst*. 1995;54:137–144.
579. Nijima A. The afferent discharges from sensors for interleukin 1 beta in the hepatoportal system in the anesthetized rat. *J Auton Nerv Syst*. 1996;61:287–291.
580. Saeed RW, Varma S, Peng-Nemeroff T, et al. Cholinergic stimulation blocks endothelial cell activation and leukocyte recruitment during inflammation. *J Exp Med*. 2005;201:1113–1123.
581. McAllen RM, Cook AD, Khiew HW, et al. The interface between cholinergic pathways and the immune system and its relevance to arthritis. *Arthritis Res Ther*. 2015;17:87.
582. Govere G, Stakenborg M, Matteoli G. The intestinal cholinergic anti-inflammatory pathway. *J Physiol*. 2016;594:5771–5780.
583. Gigliotti JC, Huang L, Bajwa A, et al. Ultrasound modulates the splenic neuroimmune axis in attenuating AKI. *J Am Soc Nephrol*. 2015;26:2407–2481.
584. Andres-Hernando A, Altmann C, Ahuja N, et al. Splenectomy exacerbates lung injury after ischemic acute kidney injury in mice. *Am J Physiol Renal Physiol*. 2011;301:F907–F916.
585. Basile DP, Leonard EC, Tonade D, et al. Distinct effects on long-term function of injured and contralateral kidneys following unilateral renal ischemia-reperfusion. *Am J Physiol Renal Physiol*. 2012;302:F625–F635.

BOARD REVIEW QUESTIONS

- Which of the following are true statements regarding the pathogenesis of acute kidney injury?
 - Following acute tubule necrosis, remaining renal epithelial cells are thought to undergo cell division and repopulate the tubule epithelium.
 - Following AKI, scattered progenitors cells are thought to repopulate tubule epithelium.
 - During early phase of acute tubular necrosis, epithelial cell polarity is lost.
 - None of the above
 - All of the above

Answer: c

Rationale: In AKI there is loss of epithelial polarity due to the disruption of the actin cytoskeleton and the loss of tight junctions and adherens junctions. These junctional complexes actively participate in numerous functions, including paracellular transport, cell polarity, and cellular shape. Early ischemic injury results in “opening” of these tight junctions, leading to increased paracellular permeability causing further back leak of the glomerular filtrate into the interstitium. Following the acute insult, repair process are initiated. There are 2 major hypotheses to explain recovery of renal function by tubule regeneration. The first hypothesis suggests that tubules regenerate from any surviving tubular epithelial cells. Genetic fate mapping studies demonstrated that regeneration following injury was achieved primarily by surviving epithelial cells. The second hypothesis suggests that tubules regenerate from a specific tubular cell subpopulation with high regenerative potential or so called “scattered tubular cells”.

- Which of the following statements are true regarding the pathogenesis of acute kidney injury?
 - Renal medulla is the most susceptible region for injury in ischemia-reperfusion injury.
 - The oxygen tension (PO_2) is lowest in the medulla.
 - Following ischemia-reperfusion injury, RBC flow decreases to the greatest degree in the medulla.
 - None of the above
 - All of the above

Answer: c

Rationale: The renal medulla is particular susceptible to AKI due to the low oxygen tension, a condition which is further exacerbated by endothelial cell activation facilitating leukocyte adherence as a consequence of expression of endothelial cell adhesion molecules. There is a decrease in RBC flow, leading to medullary circulatory congestion and reduced oxygen and nutrient supply to the outer medulla.

- Which of the following are mechanisms of hypoxia in the kidney of chronic kidney disease?
 - Loss of peritubular capillaries
 - stagnation of peritubular capillary blood flow
 - decreased oxygen delivery as a result of anemia.
 - All of the above
 - None of the above

Answer: d

Rationale: Recent studies have begun to elucidate the importance of the endothelium in the initiation and progression of fibrosis. Studies on human biopsies have associated microvascular rarefaction with progressive renal failure. Renal endothelial injury can result in a reduction of capillary density, called capillary rarefaction, occurring several weeks after AKI which may be due to hypoxia and a lack of upregulation of VEGF in response to injury. Capillary rarefaction may in turn affect tubular epithelial cells, (myo)fibroblasts, and inflammatory cells, leading to tubulointerstitial fibrosis. There is evidence that epigenetic changes are closely related to renal hypoxia and CKD progression. Lastly the anemia associated with CKD, in part, due to decreased erythropoietin production leads to a decrease in oxygen delivery.

- Which of the following contributes to fibrosis in AKI to CKD transition?
 - Persistent inflammation
 - Pericyte transformation into myofibroblasts
 - Direct proximal tubule damage
 - All of the above
 - None of the above

Answer: d

Rationale: A key feature of progression of AKI to CKD is the activation of extracellular matrix-producing myofibroblasts. Other factors important in CKD progression include proximal tubule cell, endothelial cell and vascular cell damage in AKI, increase in tubular oxygen consumption, increased oxidative stress that can affect the expression of Hypoxia-Inducible factors in renal cells. In addition, innate and adaptive immunity, cell cycle arrest and epigenetic mechanisms also contribute. While some injured tubules may undergo repair and regeneration, injury may also be accompanied by persistent inflammation, maturation and proliferation of fibroblasts, and extracellular matrix deposition as part of the process of fibrosis. The source of fibroblasts in the injured kidney - fibrocytes, epithelial cells through EMT, intrinsic fibroblasts and pericytes - remains controversial. Recent studies also suggest that in the absence of fibrosis persistent regulated epithelial cell necrosis may represent a pathophysiological mechanism of transition from AKI to CKD.